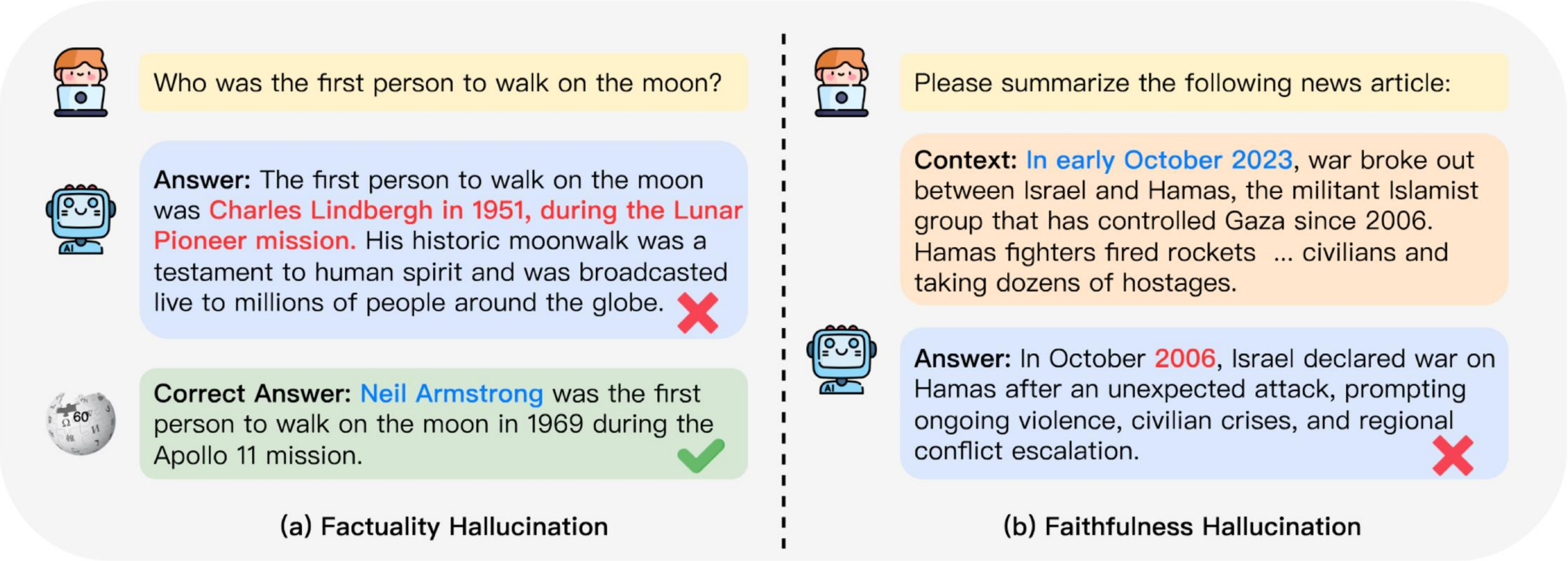
Natural Language Processing

Retrieval-Augmented Generation 檢索強化生成

Hallucination of LLM

- It is discovered that NLG models often generate text that is nonsensical, or unfaithful to the provided input. Such undesirable generation is referred to Hallucination (Ji et al., 2023).



Ji et al. "Survey of hallucination in natural language generation." ACM Computing Surveys 55.12 (2023): 1-38.
Figure source: Munkhdalai, Tsendsuren, Manaal Faruqui, and Siddharth Gopal. "Leave no context behind: Efficient infinite context transformers with infini-attention." arXiv preprint arXiv:2404.07143 (2024).

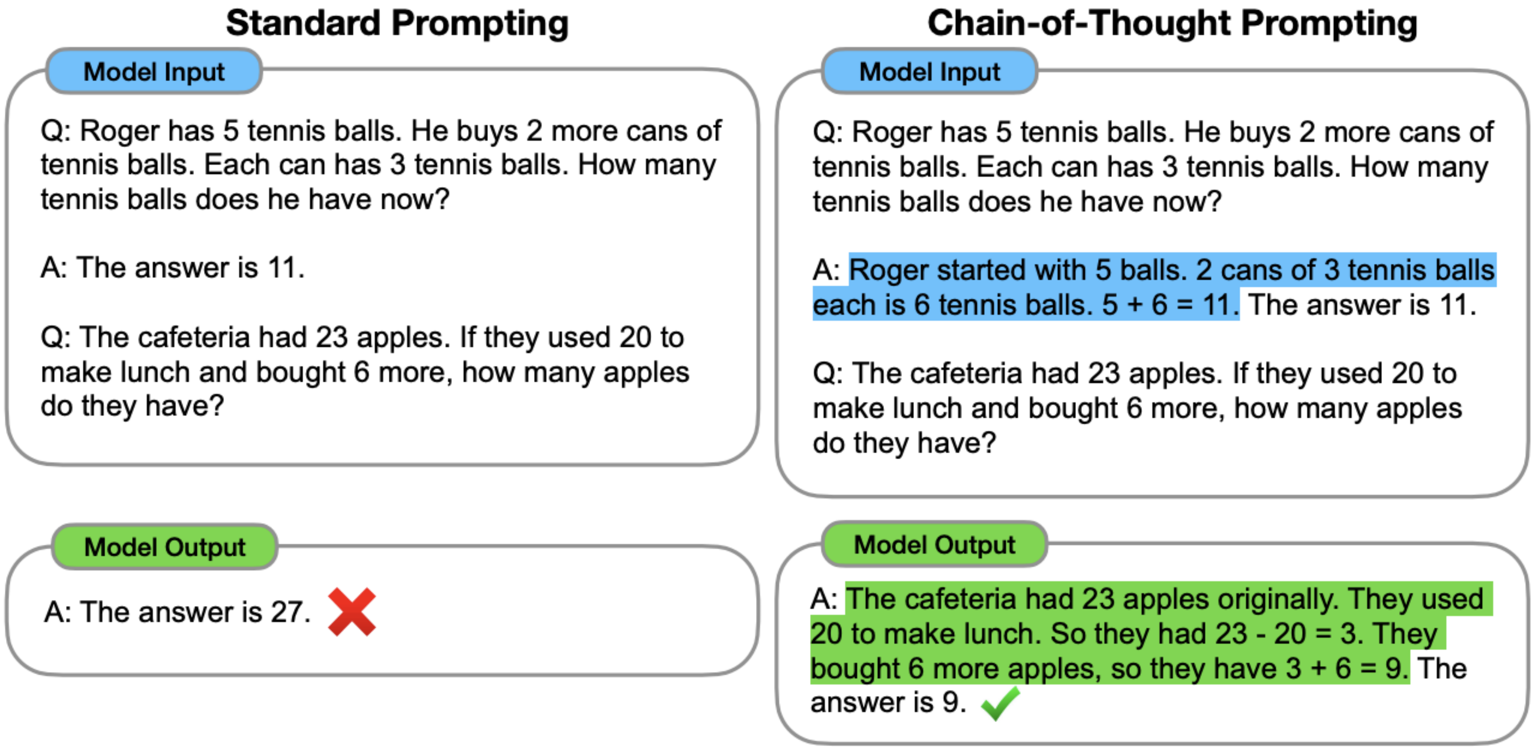
Solutions to Mitigating Hallucinations

解決幻覺問題

- Chain-of-Thought Prompting (CoT) 提供解題順序，讓decoder依照順序進行
- Retrieval-Augmented Generation (RAG)
- …

Chain-of-Thought Prompting (CoT)

- Wei et al., Chain-of-Thought Prompting Elicits Reasoning in Large Language Models. NeurIPS 2022. By Google Brain. (43 pages)



提供問題解析/思索步驟，可以提升回答準確度以及詳細程度

Chain-of-Thought Prompting (CoT)

- Wei et al., Chain-of-Thought Prompting Elicits Reasoning in Large Language Models. NeurIPS 2022. By Google Brain. (43 pages)

Prompt: Few-shot examples (Rationales written by human)

Question: Sammy wanted to go to where the people were. Where might he go?
Answer Choices: (a) populated areas (b) race track (c) desert (d) apartment (e) roadblock

Answer: The answer must be a place with a lot of people. Of the above choices, only populated areas have a lot of people. So the answer is (a) populated areas.

xN

Similar in task

+

A gentleman is carrying equipment for golf, what is he likely to have?
Answer Choices: (a) supermarket (b) park (c) pub (d) school (e) club

Answer: The answer must be something that is used for golf. Of the above choices, only clubs are used for golf. So the answer is (e) club.

Rationale
Answer

Retrieval-Augmented Generation (RAG)

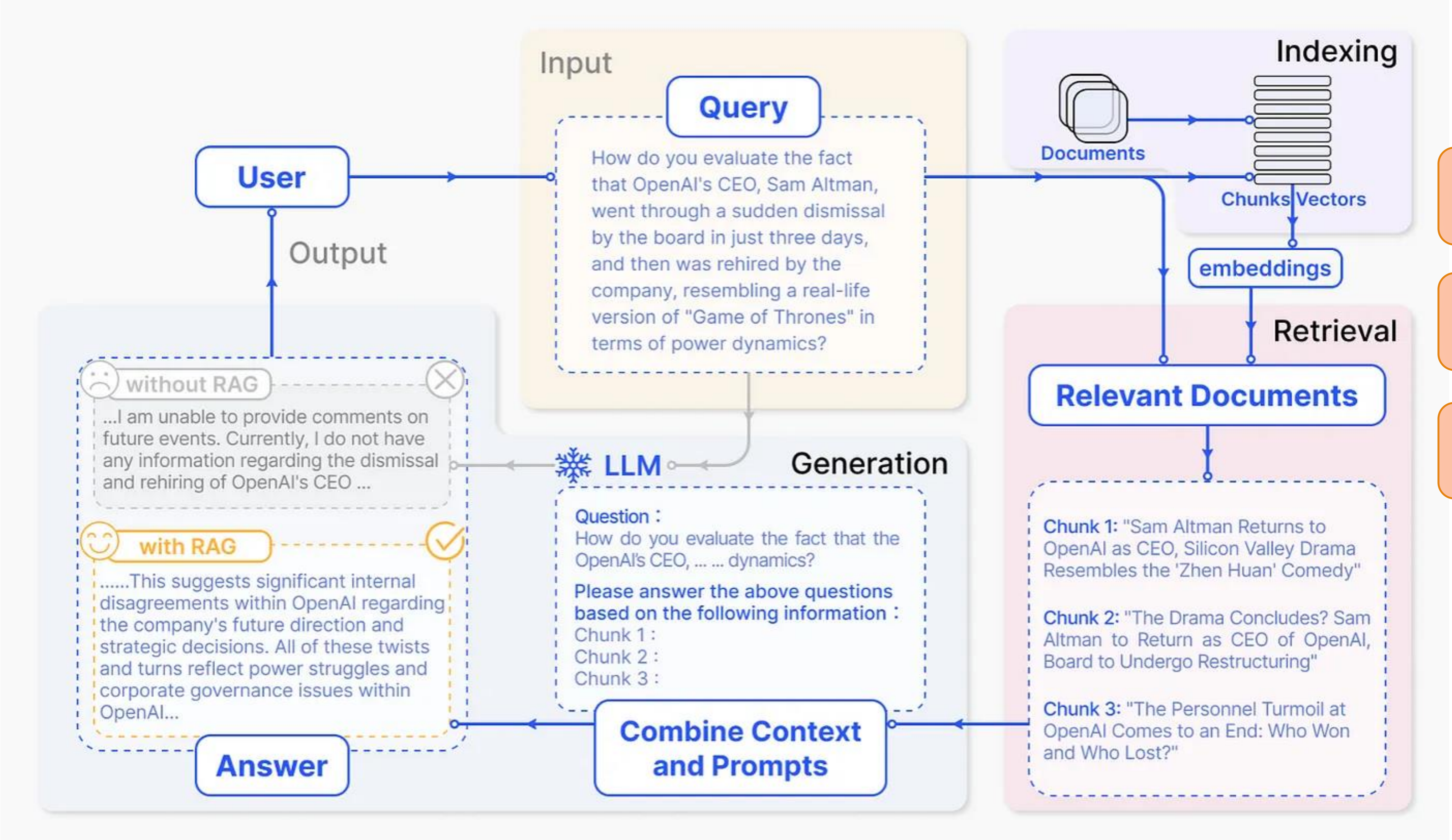
Information Retrieval

檢索

- Retrieval: get relevant information from a pool (like a search engine)

- Retrieval-Augmented Generation (RAG): Perform generation with additional retrieved information

Gao, Yunfan, et al. "Retrieval-augmented generation for large language models: A survey." (arXiv 2023)

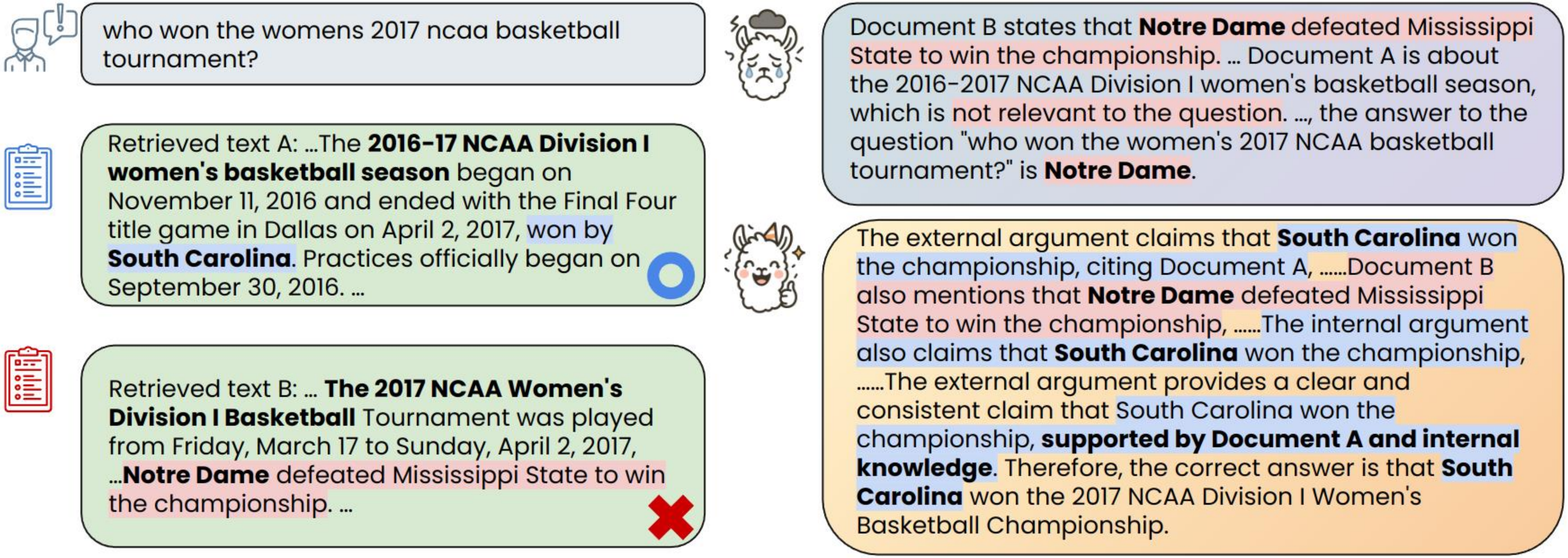


What Has Been Improved?

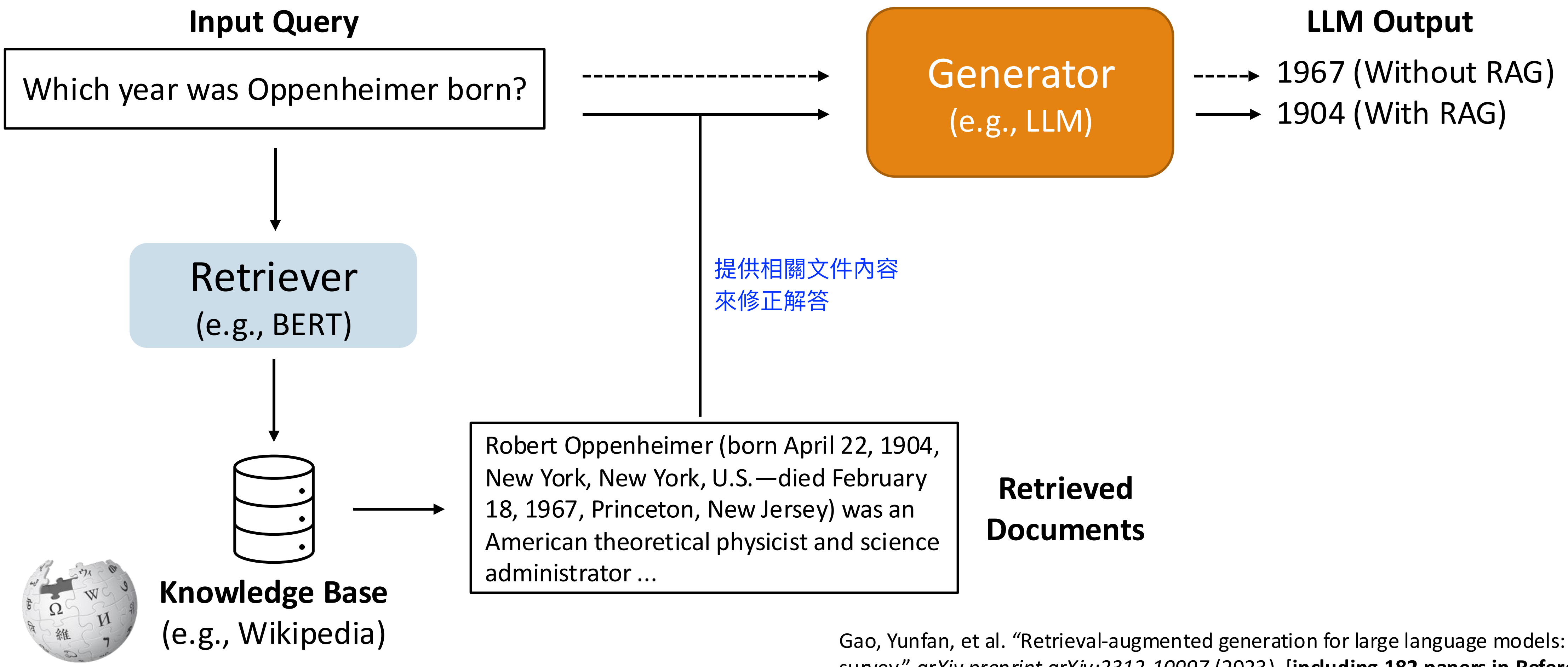
What Problems Remain Unresolved?

What New Problems Have Emerged?

An example of RAG with noise chunks



Retrieval-Augmented Generation (RAG)



Gao, Yunfan, et al. "Retrieval-augmented generation for large language models: A survey." *arXiv preprint arXiv:2312.10997* (2023). **[including 182 papers in Reference]**

Gao, Yunfan, et al. "Retrieval-augmented generation for large language models: A survey." *arXiv preprint arXiv:2312.10997* (2023). [**including 182 papers in Reference]**

Retrieval-Augmented Generation (RAG)

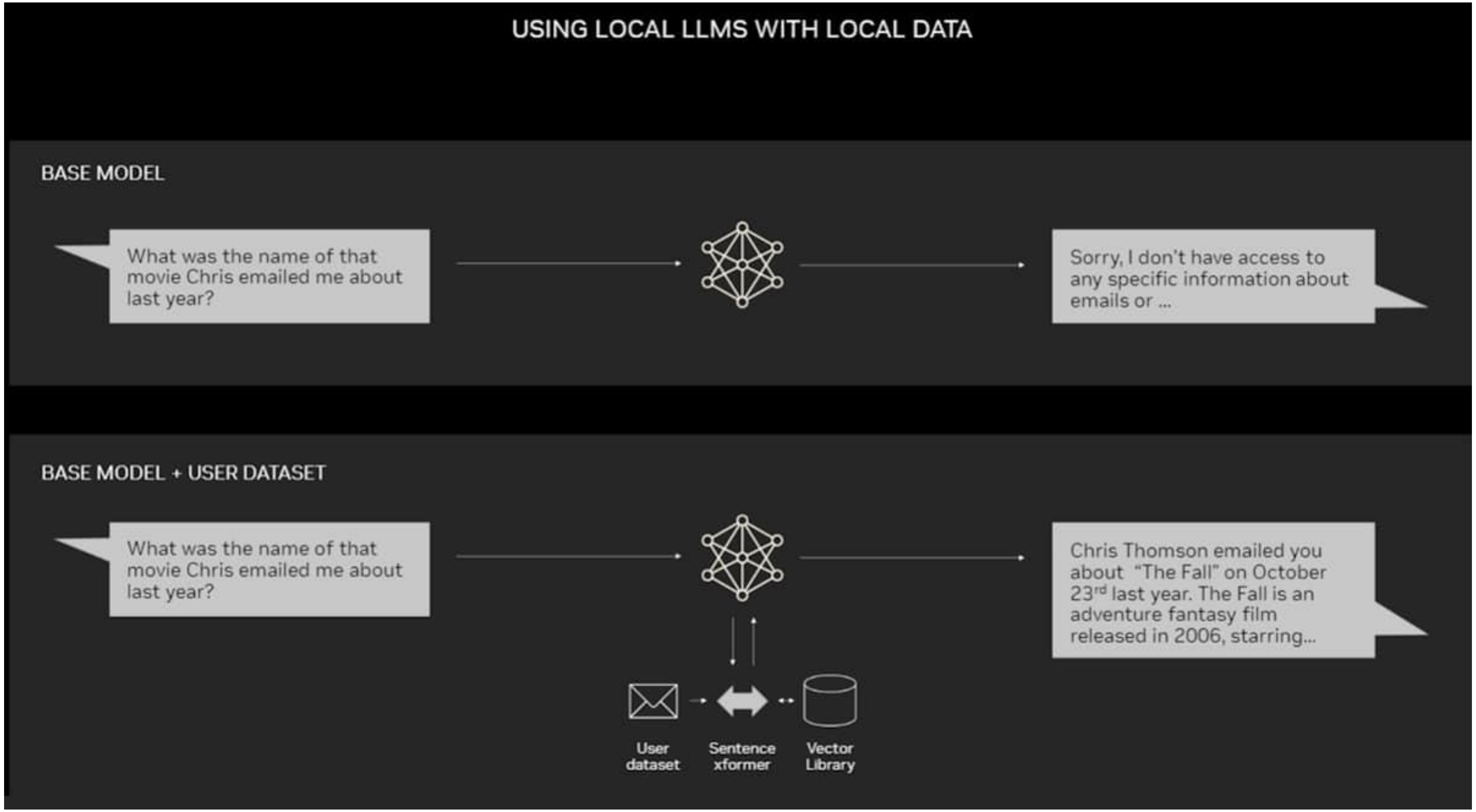
TABLE I
SUMMARY OF RAG METHODS

Retrieval Source	Retrieval Data Type	Retrieval Granularity	Augmentation Stage	Retrieval process
Wikipedia	Text	Phase	Pre-training	Once
FactoidWiki	Tabular text	Sentence	Inference	Iterative
Dataset-base	KG	Chunk	Tuning	Recursive
Search Engine	Graph	Doc		Adaptive
Synthesized dataset		Entity		Multi-time
Common Crawl		Triplet		
Pre-training Corpus		Sub-graph		
BEIR				
MSMARCO				
Arxiv, Online database				
LLM				

Why do we need RAG?

- LLMs have profound parameterized knowledge that makes them useful in responding to general prompts.
- However, LLMs are error-prone due to a lack of domain knowledge or outdated information.
- Standalone LLMs do not serve users who want a deeper dive into a current or more specific topic.

Why do we need RAG?



Retrievers for RAG

- A retriever is aimed at searching relevant documents based on an input query.
- A retriever plays an important role in enhancing the performance of an LLM. Therefore, a good retriever is needed.
- Usually, a retriever produces outputs by computing similarities between query embeddings and document embeddings, which come from other encoder models or the retriever itself.

Embedding Types for Retrieval

- **Sparse Embeddings** (sparse vectors)
  - E.g., TF-IDF, BM25
- **Dense Embeddings** (dense vectors)
  - E.g., BERT, Sentence-BERT, DPR (Dense Passage Retrieval)

稀疏向量檢索
Ex:
1.”I like cats.”
2.”I love Kittens.”
–>Sparse向量會完全不同，因為cat不等於kittens
向量間的cosine similarity會很低

稠密向量檢索
Ex:.
I like cats ~= I love kittens
–>Dense向量距離會很近，因為語意相似
向量間的cosine similarity會很高

Sparse Embeddings
概念：大多數位置都是 0，只有少數幾個詞的權重是非零的。
代表方法：
TF–IDF (Term Frequency – Inverse Document Frequency)
→ 衡量一個詞在文件中出現的頻率，並根據它在整體語料中的稀有度加權。
BM25 (Best Match 25)
→ 改進版的 TF–IDF，常用於搜尋引擎，如 Elasticsearch。
特點：
維度非常高每個詞都是一個維度、強調詞的出現次數和重要性，而非語意。
優點： 可解釋性高、計算快、缺點： 無法捕捉語意相似\例如「cat」與「kitten」的距離不近

Dense Embeddings
概念：透過深度學習模型，將整個句子轉換成低維度、連續值的向量。
每個維度都蘊含語意資訊。
代表方法：
BERT：可提取 contextual embeddings。
Sentence–BERT：針對句子相似度優化的 BERT 變體。
DPR (Dense Passage Retrieval)：專門用於問答檢索任務，將問題與段落都轉成dense 向量。
特點：
維度較低（例如 768 維）、向量之間的距離,如 cosine similarity反映語意相似。
優點： 能理解語意關係、跨語言/同義詞、缺點： 訓練成本高、不易解釋。

Training Retrievers for RAG

- Task-oriented training:
  - *Retrieval for open-domain question answering (ODQA)*
    - Usually used in RAG because it is more common to search relevant documents.
  - *Sentence embeddings for semantic similarity tasks*

Retrieval for open-domain question answering

Query (a question) | Relevant doc | Irrelevant doc | … | Irrelevant doc → Model → Finding a relevant document

Sentence embeddings for semantic similarity tasks

Sentence | Relevant Sentence | Irrelevant sentence | … | Irrelevant sentence → Model → Finding a semantically similar sentence

- Both approaches are suitable for retrieval (depends on your task for an LLM).

Sentence embeddings for semantic similarity tasks

Sparse Vectors

通常一篇文章塞30000維度的向量進行比較

- In information retrieval, sparse vectors are vectors with most elements set to zero.
- The advantage of sparse vectors is computational efficiency because most elements are zero and can be ignored.

For an example:
We have a small vocabulary with five words: ["cat", "dog", "fish", "bird", "snake"].
We have a document that only contains the words "cat" and "dog".
The sparse vector for this document would look like this: [1, 1, 0, 0, 0]

The elements represented to "fish", "bird", and "snake" are 0s

Sparse Embeddings: Bag-of-words

one hug=> 有:1, 沒有:0

```
texts = [
    "This is a book",
    "These are pens and my pen is here"
]
```

- The Bag-of-words approach creates document embeddings.
- The embedding size is equal to the vocabulary size.
- Each value of an embedding is based on frequency counts.

Transform via frequency

Vocabulary size

	a	and	are	book	here	is	my	pen	these	this
sent_0	1	0	0	1	0	1	0	0	0	1
sent_1	0	1	1	0	1	1	1	2	1	0

Since the outputs contain many zeros, this approach is called a sparse embedding method.

Sparse Embeddings: TF-IDF

- TF (Term Frequency)
- IDF (Inverse Document Frequency)

```
texts = [
    "This is a book",
    "These are pens and my pen is here"
]
```

Transform via TF-IDF

- The **TF-IDF** approach also creates document embeddings.
- The embedding size is equal to the vocabulary size.
- Each value of an embedding is based on **TF x IDF**.

將文章裡面出現過的次數和位置考慮進去
=>增加權重概念、有小數點0–1之間(顆粒度較小的表示方式

Vocabulary size

	a	and	are	book	here	is	my	pen	these	this
sent_0	0.534046	0.000000	0.000000	0.534046	0.000000	0.379978	0.000000	0.000000	0.000000	0.534046
sent_1	0.000000	0.324336	0.324336	0.000000	0.324336	0.230768	0.324336	0.648673	0.324336	0.000000

Since the outputs contain many zeros, this approach is called a sparse embedding method.

https://github.com/tsmatz/nlp-tutorials/blob/master/01_sparse_vector.ipynb

TF-IDF word的重要程度從在文件的出現頻率和整個資料集裡面的稀有度評估有關

The mathematical representation of TF-IDF:

所有Document的frame大小

$$TF-IDF = TF \times IDF$$ where $$TF_{i,j} = \frac{n_{i,j}}{\sum_k n_{k,j}} \quad IDF_i = lg\frac{|D|}{|\{j : t_i \in d_j\}|}$$

- Where $n_{i,j}$ is the i-th word in j-th text in the dataset.

TF (Term Frequency) 該字在文件裡面的出現頻率

- Represents the "frequency" of a term appearing in a text.

IDF (Inverse Document Frequency) 越少文件提到該字，表示越具辨識度、IDF越高越重要

- Aims for terms to have higher specificity, meaning the fewer texts in the dataset contain the term, the better.

BM25 (an improved version of TF-IDF)

解決TF–IDF問題：
1.TF比例增長不合理
2.文件長度造成bias
3.TF–IDF沒有參數可以調整

Best Matching

相似度比對，找到最佳的結果=> Probabilistic IR Model基於機率資訊檢索的rank演算法
=>主要應用在計算某篇文件對某個query的相關性分數

$$score = \frac{(k_1 + 1)TF}{TF + k_1 * (1 - b + b * \frac{|D|}{avgD})} * IDF, b \in [0,1]$$

query term
在文件中的次數

文件長度/資料集的平均文件長度

控制TF飽和(saturation)
=>TF對機率的邊際效應

- **k_1 :** A term **frequency saturation** hyper-parameter. For best performance, the value of k_1 should be between **0 and 3.**
  - This **reduces the effect of high-frequency terms** so that they don't overpower the score excessively.
- **b :** A document length normalization parameter, this hyper-parameter controls the influence of sequence length.
  - This allows shorter and longer documents to compete more equally in retrieval relevance.

控制文件長度正規化
=>b=0不管文章長度
=>b=1完全依據長度修正

k_1=0, b=0 → term frequency is ignored

k_1 increases → the weight of term frequency increases

在TF–IDF中的TF為線性
=>某字一直出現，分數可以推很高
BM25引入saturation
=>TF出現頻率高但是最大影響力有限
(模擬機率飽和狀態)

Robertson, Stephen, and Hugo Zaragoza. "The probabilistic relevance framework: BM25 and beyond." *Foundations and Trends® in Information Retrieval* 3.4 (2009): 333-389.

Dense Vectors

- In NLP, dense vectors comprise compact numerical values representing semantic features of text.
- “Dense” is concept contrary to “sparse.”
- Word2vec is also an approach for creating dense embeddings.

0	1	0	0	0	1	0	1

Sparse vector

0.12	1.56	0.48	1.21	0.08	1.08

Dense vector

Approach for Dense Vectors: Dual Encoder

用specific的資料retrain–encoder

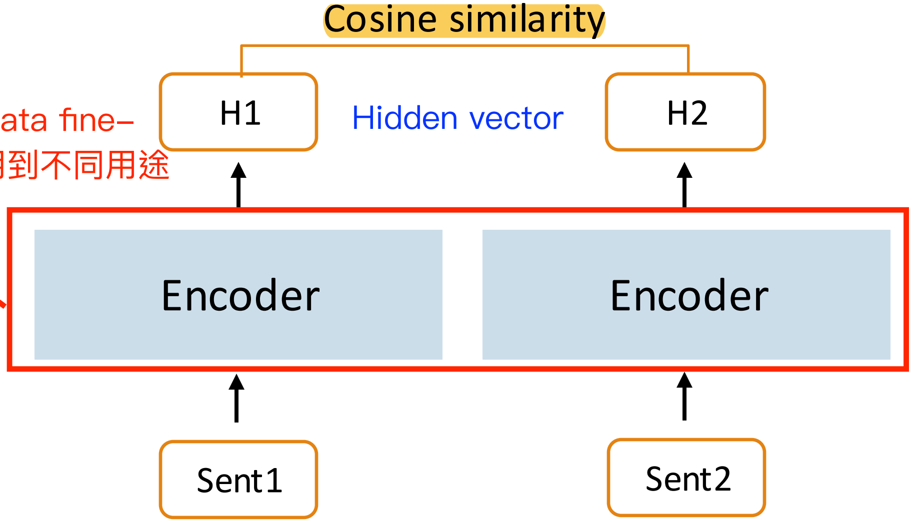
學生網路

- Also called **bi-encoder**, **Siamese network.**
- Structure:
  - Two identical/similar encoders
  - Processes two inputs independently
  - Outputs separate vectors for each input
- Tasks:

建造相似度高的網路
=>常遇到overlap
=>因此在透過supervise data fine–tune到multi–model，應用到不同用途

Task	Inputs
Information Retrieval	Document and query
Semantic similarity (or any sentence pair classification)	Two sentences

判斷相似度，算出loss,接著在更新Encoder

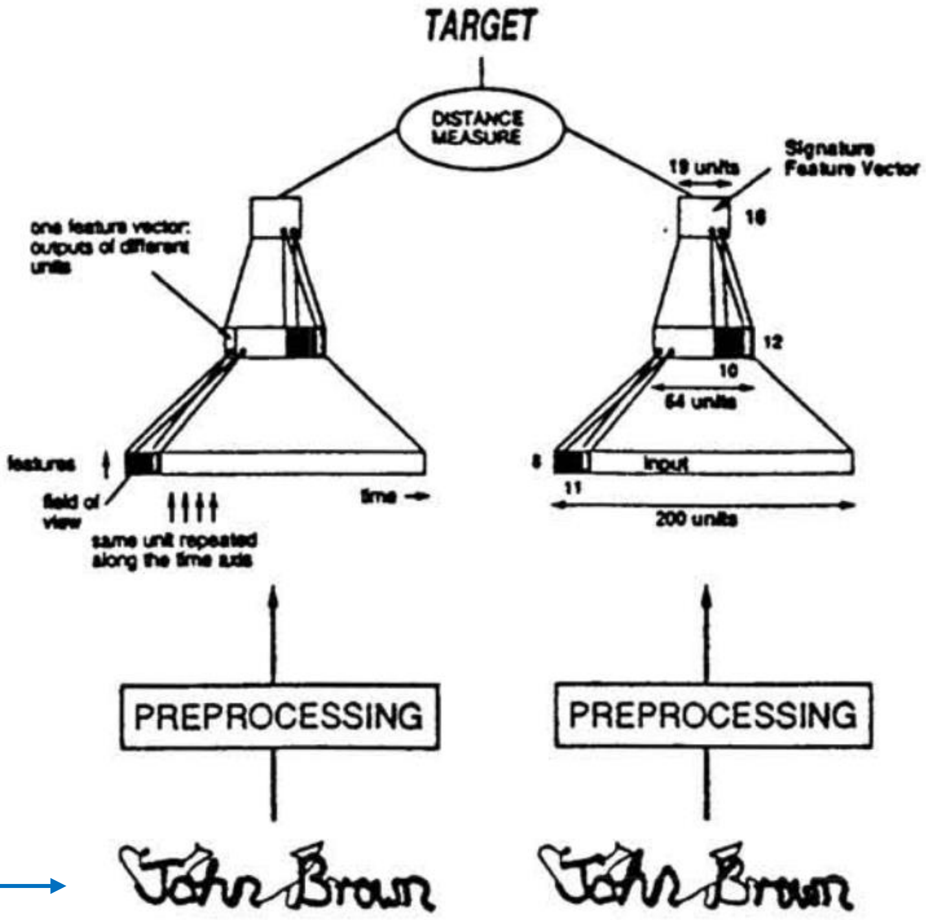


The First Siamese Network (類似對比學習)

- Bromley, J., Guyon, I., LeCun, Y., Säckinger, E., & Shah, R. (1993). Signature verification using a" siamese" time delay neural network. NeurIPS.
- Siamese" neural network consists of **two identical** sub-networks joined at their outputs.

流程：
同一人寫的兩種版本Input >>放入兩種不同的Encode>>產出的相似度應該要接近

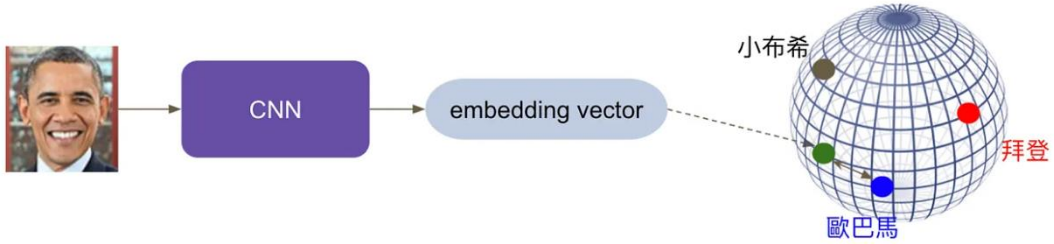
Signature Verification

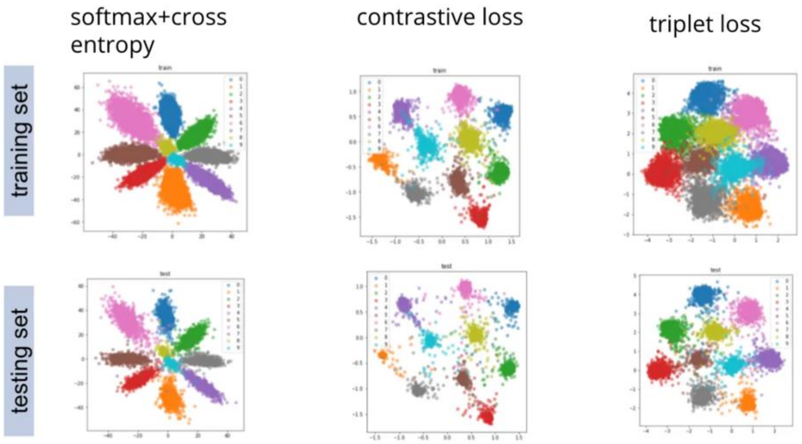


(Figure source: Bromley et al., 1993)

A Brief History of Siamese Networks

- Signature Verification (Bromley et al., 1993)[1]
- Face Verification (Chopra et al., 2005)[2]
- Image similarity (Koch et al., 2015)[3]
- Text Similarity (Neculoiu et al., 2016)[4]





[1] Bromley, Jane, et al. "Signature verification using a" siamese" time delay neural network." NeurIPS 1993.
[2] Chopra, Sumit, Raia Hadsell, and Yann LeCun. "Learning a similarity metric discriminatively, with application to face verification." 2005 IEEE computer society conference on computer vision and pattern recognition. CVPR 2005.
[3] Koch, Gregory, Richard Zemel, and Ruslan Salakhutdinov. "Siamese neural networks for one-shot image recognition." *ICML deep learning workshop*. 2015.
[4] Neculoiu, Paul, Maarten Versteegh, and Mihai Rotaru. "Learning text similarity with siamese recurrent networks." Proceedings of the 1st Workshop on Representation Learning for NLP. 2016.

Sentence-BERT

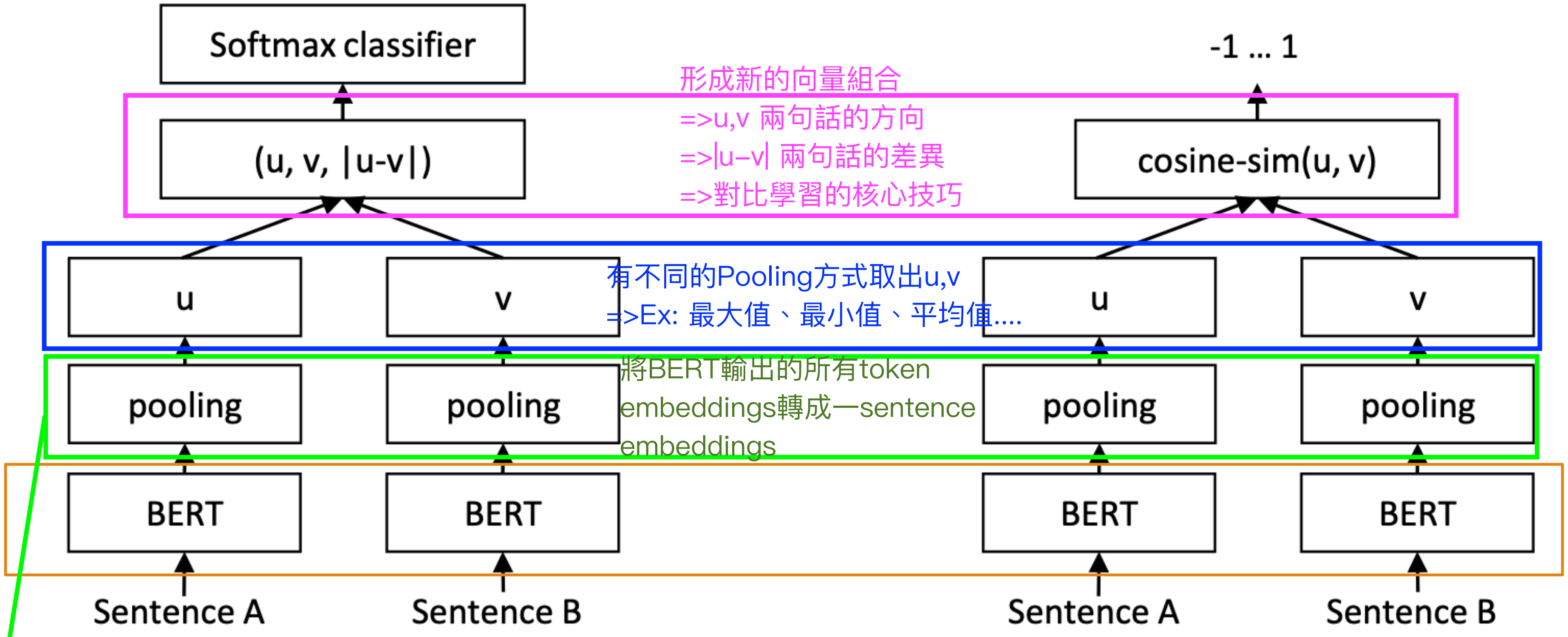
將sentence 編碼成可比較的vector(sentence embeddings)

=>句子A/B語意越接近則向量越接近 =>可用cosine similarity比較相似度

```
# Pseudo code for Dual Encoder
query_vector = encoder(query)          # [0.1, 0.2, 0.3]
document_vector = encoder(document)       # [0.2, 0.2, 0.4]
similarity = cosine_similarity(query_vector, document_vector)
```

對比學習 >> 處理分類問題

Siamese Network
(**weights shared**)



訓練階段 **Training time**

用 Siamese BERT + pooling + softmax 讓模型學習句子相似度

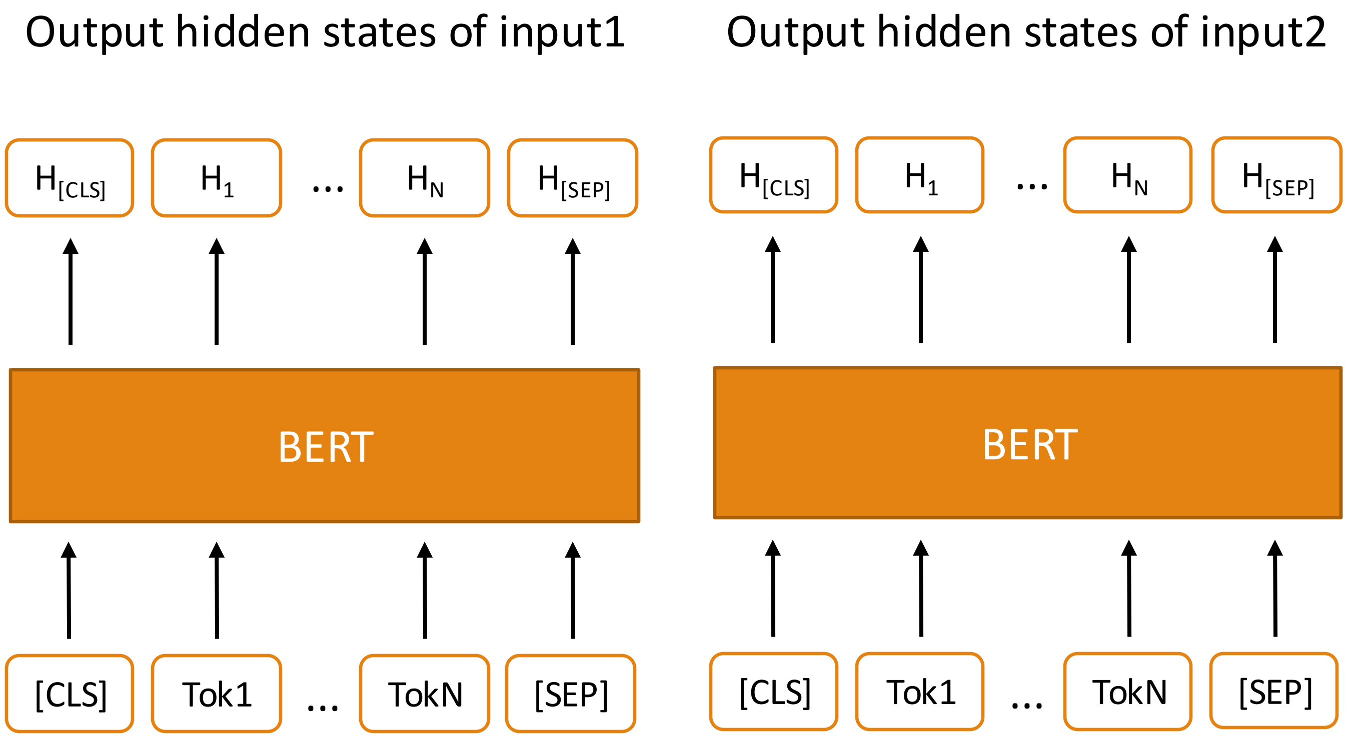
推論/使用階段 **Inference time**

將句子編碼成向量 → 用 cosine similarity 比對

Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. *(EMNLP-IJCNLP 2019)*

Why does Sentence-BERT need pooling?

- BERT produces embeddings (hidden states from the final layer) for each token.
- We need a single fixed-size vector for the entire sentence.



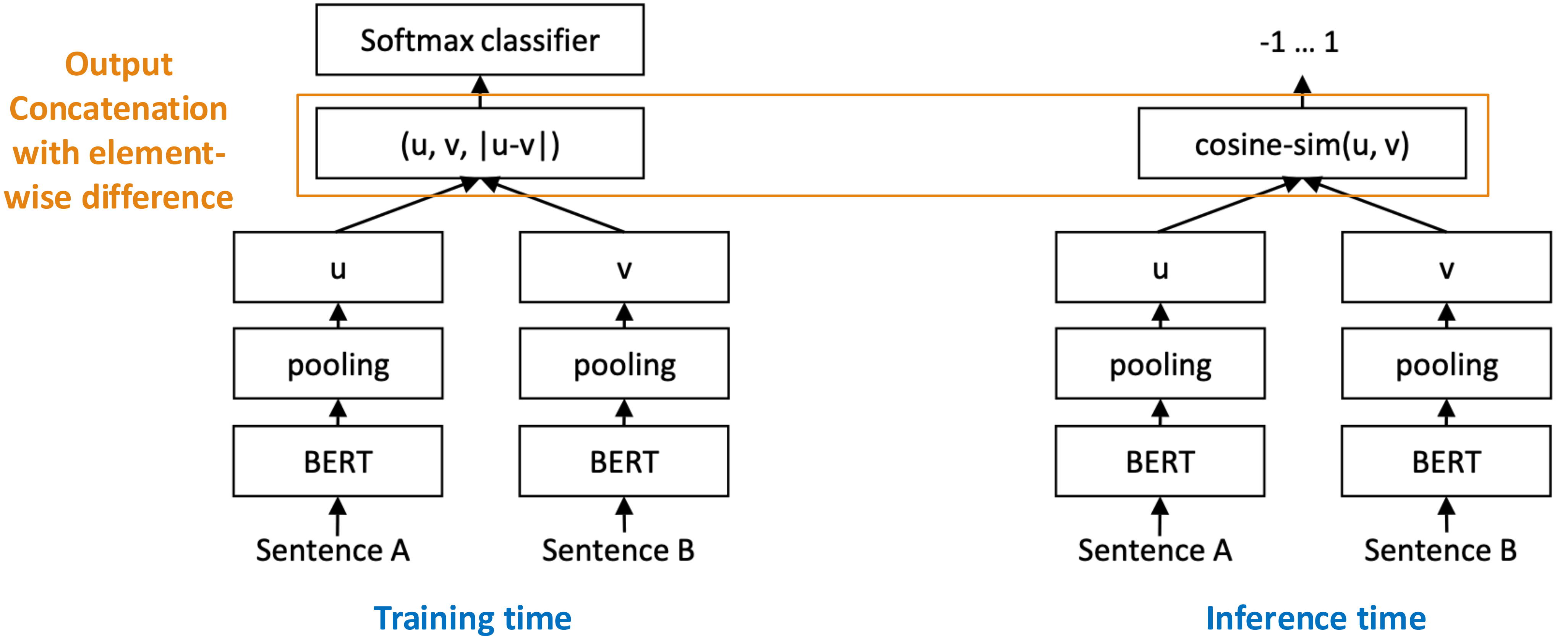
Pooling of Sentence-BERT 常用的不同pooling方法

We need a single fixed-size vector for the entire sentence.

- **CLS: Use the [CLS] token**
  - This is the default setting in original BERT.
- **MEAN: the mean of all output vectors**
  - Averages all token embeddings.
- **MAX: max-over-time of the output vectors**
  - Takes maximum value across each dimension.

Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. *(EMNLP-IJCNLP 2019)*

Sentence-BERT(Dual encoder)



Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. *(EMNLP-IJCNLP 2019)*

Performance comparison of Pooling and Concatenation

- Indeed, [CLS] token can directly be used for representing the entire sentence.
- But pooling may bring better performance.

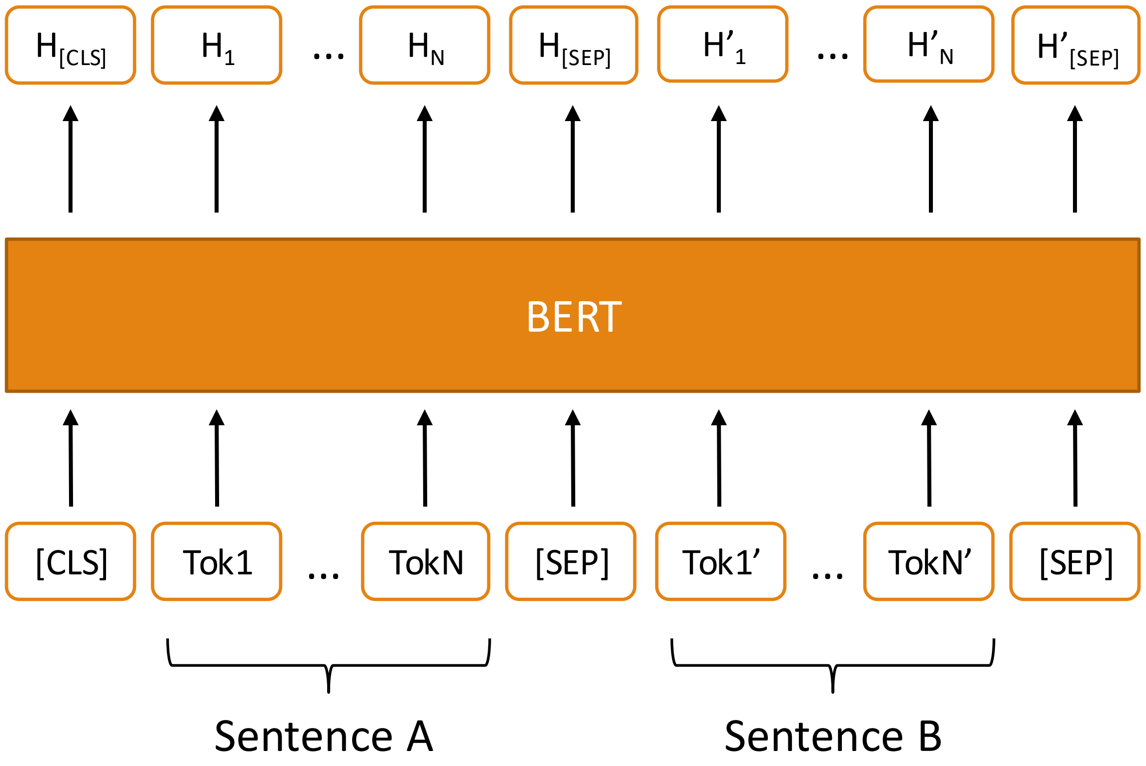
- **Experiment** shows using element-wise difference is better than the other settings.
- Note that the concatenation mode is only used for training.

	NLI	**STSb**
Pooling Strategy		
MEAN	**80.78**	**87.44**
MAX	79.07	69.92
CLS	79.80	86.62
Concatenation		
(u, v)	66.04	-
$(\|u - v\|)$ 線性關係	69.78	-
$(u * v)$	70.54	-
$(\|u - v\|, u * v)$	78.37	-
$(u, v, u * v)$	77.44	-
$(u, v, \|u - v\|)$	**80.78**	-
$(u, v, \|u - v\|, u * v)$	80.44	-

Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. *(EMNLP-IJCNLP 2019)*

BERT as a Cross Encoder 根據下游任務，fine–tune BERT

- For a cross encoder, representations of two input sentences are attended with each other.
- The hidden state of [CLS] represents the relationship between the two input sentences.



Devlin, J., Chang, M.W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. NAACL 2019.

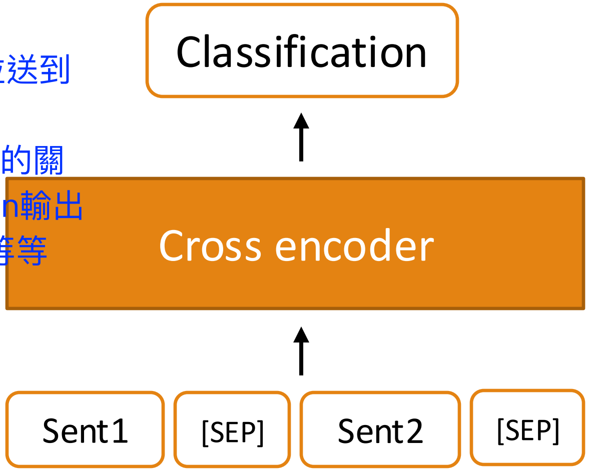
Computation time for Bi-encoders and Cross encoders

- For 10,000 sentence pairs:

每次比較：Sent1 [SEP] Sent2, Sent2 [SEP] Sent3....
所以總次數為10,000 x2,才能算出兩兩句子的相似度

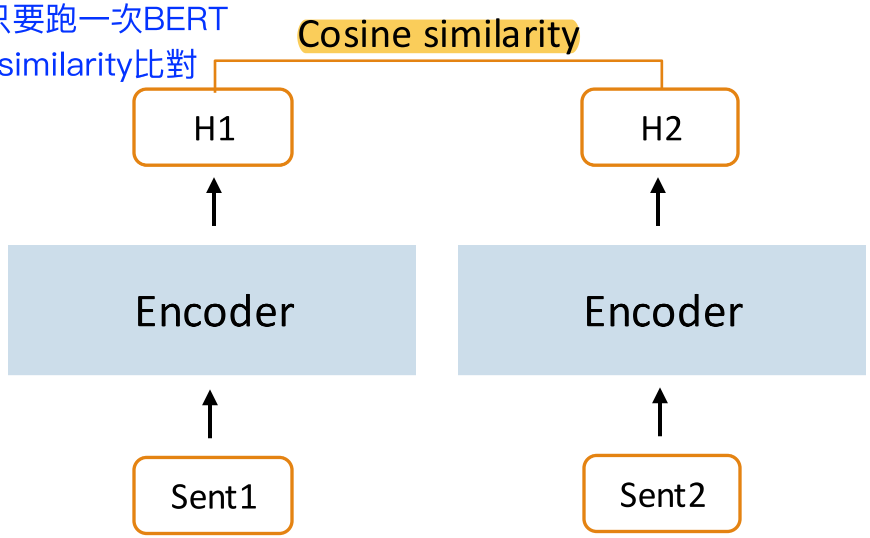
  - Cross encoders: n·(n−1)/2 = 49,995,000 inference times
  - Bi-encoders: 10,000 * 2 inference times (can be parallel) with cosine similarity calculation

Cross Encoder:
1.將兩句子拼接再一起並送到同一個BERT
2.讓BERT一起讀兩句話的關係，最後用classificatoin輸出
ex: 相似/不相似、0–5等等



=>每比較一對句子就要跑一次BERT

Bi encoder:
1.每句話都只要跑一次BERT
2.用cosine similarity比對



=>每個encoder都可以平行運算，且都只使用cosine similarity比較

Performance comparison for Bi-encoders and Cross encoders

Model	Spearman
Not trained for STS	
Avg. GloVe embeddings	58.02
Avg. BERT embeddings	46.35
InferSent - GloVe	68.03
Universal Sentence Encoder	74.92
SBERT-NLI-base	77.03
SBERT-NLI-large	79.23
Trained on STS benchmark dataset	
BERT-STSb-base	84.30 ± 0.76
SBERT-STSb-base	84.67 ± 0.19
SRoBERTa-STSb-base	**84.92** ± 0.34
BERT-STSb-large	**85.64** ± 0.81
SBERT-STSb-large	84.45 ± 0.43
SRoBERTa-STSb-large	85.02 ± 0.76
Trained on NLI data + STS benchmark data	
BERT-NLI-STSb-base	**88.33** ± 0.19
SBERT-NLI-STSb-base	85.35 ± 0.17
SRoBERTa-NLI-STSb-base	84.79 ± 0.38
BERT-NLI-STSb-large	**88.77** ± 0.46
SBERT-NLI-STSb-large	86.10 ± 0.13
SRoBERTa-NLI-STSb-large	86.15 ± 0.35

BERT: Cross encoder

SBERT / SRoBERTa: Bi-encoders

- Generally, the difference in performance between bi-encoders and cross encoders is not large.
- However, bi-encoders are much faster with respect to computation time.
    - If >1M documents in a database, the difference in computation time will be extremely huge.

SimCSE (Dual encoder)

對比學習框架

- SimCSE: a simple contrastive sentence embedding framework =>可以自動產生正樣本的方法，不需要人工標注資料
- Both unsupervised and supervised training approaches were proposed in SimCSE:
  - **Unsupervised** training of SimCSE
    - Relying on **Dropout**

      同一句話經過BERT 2次，因dropout不同，可以得到兩個不同的向量(但是語意相同)
      =>可以避免model 對data overfitting
      =>同一句話當作positive pair/不同句子當作negative pair >>因此不需要人工標注資料
  - **Supervised** training of SimCSE
    - Relying on **labels in a dataset**

      使用人工標註，種類包含entailment, neutral, contradiction
      =>entailment當作positive、contradiction當作negative
      =>樣本品質會比unsupervised高、但是耗費成本高

Gao, Tianyu, Xingcheng Yao, and Danqi Chen. "SimCSE: Simple Contrastive Learning of Sentence Embeddings." EMNLP 2021.

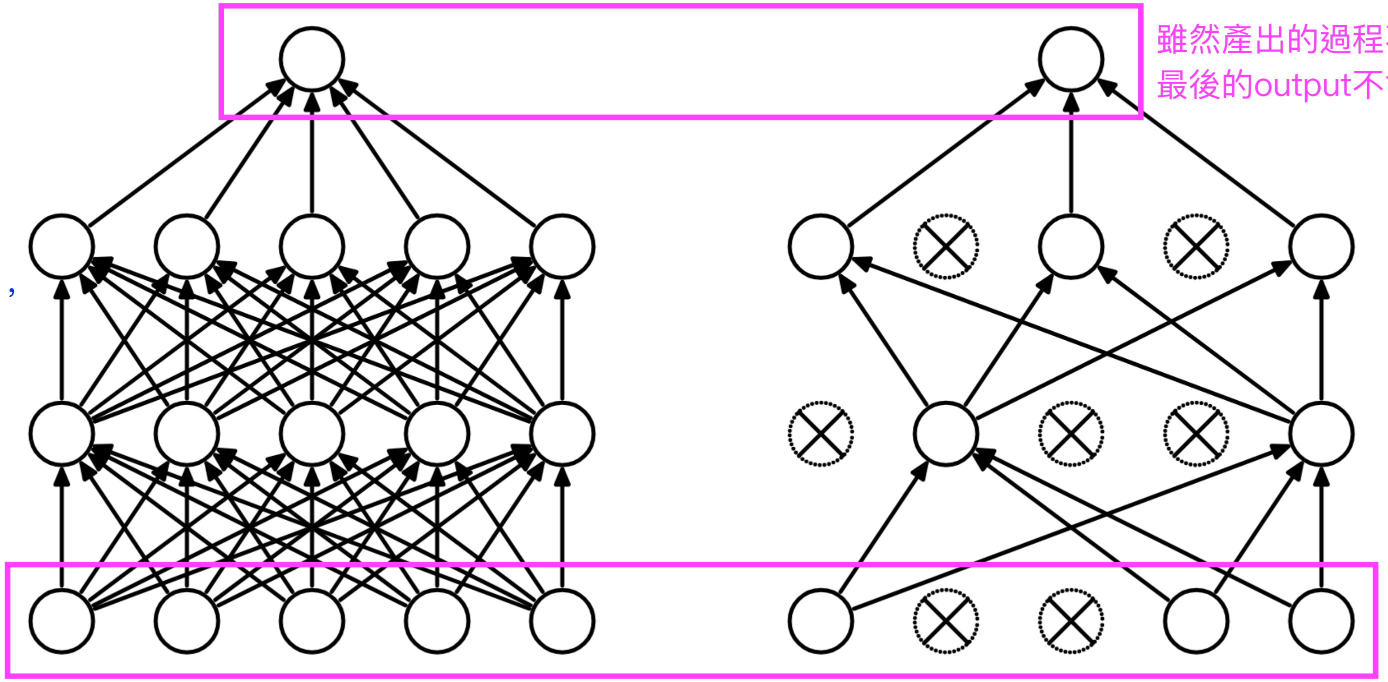
Dropout

讓positive pair的向量更接近、negative pair的向量更遙遠
=>達到data augmentation效果

- Dropout randomly drop units (along with their connections) from the neural network during training. This approach usually brings regularization and reduces overfitting.

雖然產出的過程不一樣，但是在input素材相同的條件下
最後的output不會相差太多(通常可以視為一樣的結果)

以往Neural Network訓練方式，
可能會造成each node的data
overfitting



(a) Standard Neural Net (b) After applying dropout.

透過隨機拔掉一些node,
雖然效果會比原本的training data差
=>但是可以讓model有更好的泛化性
(意指有更多的空間可以塞不同的data)
=>通常在train階段實行

Srivastava, Nitish, et al. "Dropout: a simple way to prevent neural networks from overfitting." The journal of machine learning research 15.1 (2014): 1929-1958.

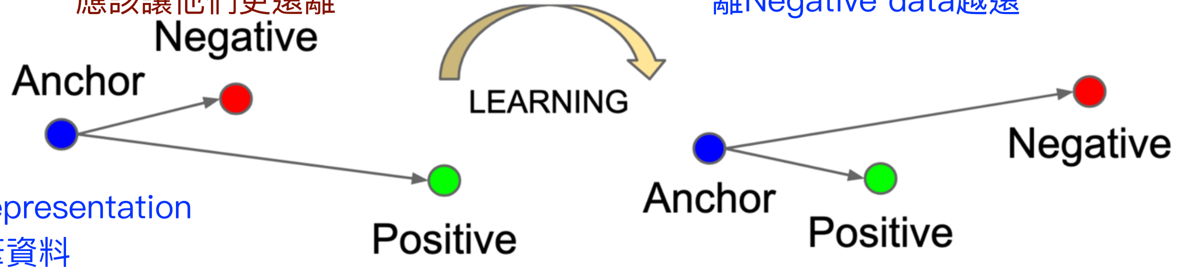
Contrastive Learning

透過對比哪些是相似、哪些不相似來學representation的方法

f (·): encoder network
g(·): projection head
z: output logits

和Anchor不同語意的資料
=>model learning完之後，
應該讓他們更遠離

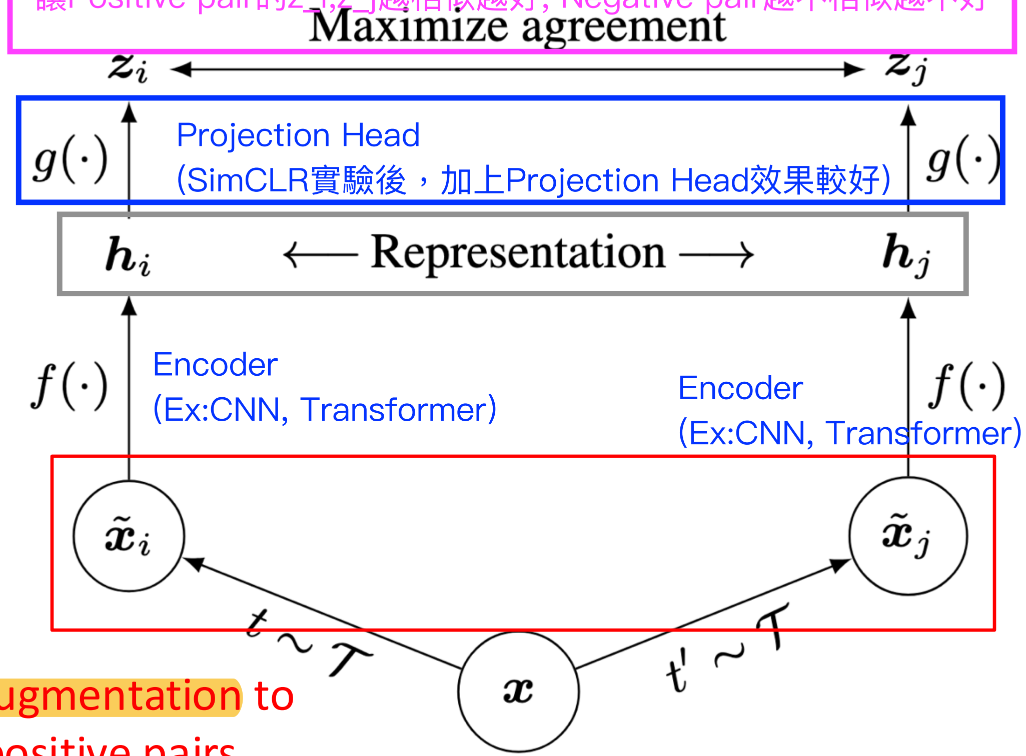
學習目標：
Anchor離positive data越近
離Negative data越遠



要學representation
的那筆資料

和Anchor同一語意的資料
=> model learning完之後，
應該讓他們更靠近

讓Positive pair的z_i,z_j越相似越好, Negative pair越不相似越不好



Projection Head
(SimCLR實驗後，加上Projection Head效果較好)

Encoder
(Ex:CNN, Transformer)

Encoder
(Ex:CNN, Transformer)

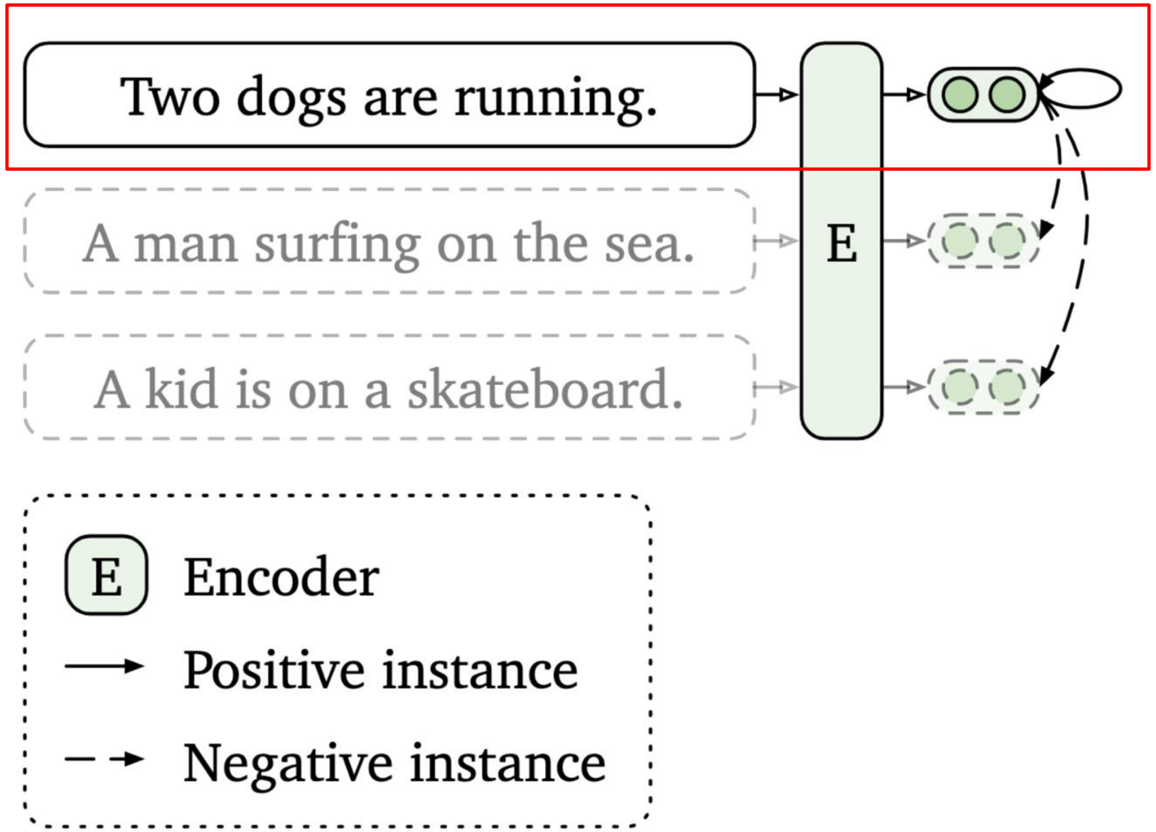
Use data augmentation to
create positive pairs

同一張input data photo並使用不同的增強策略調整成2版本

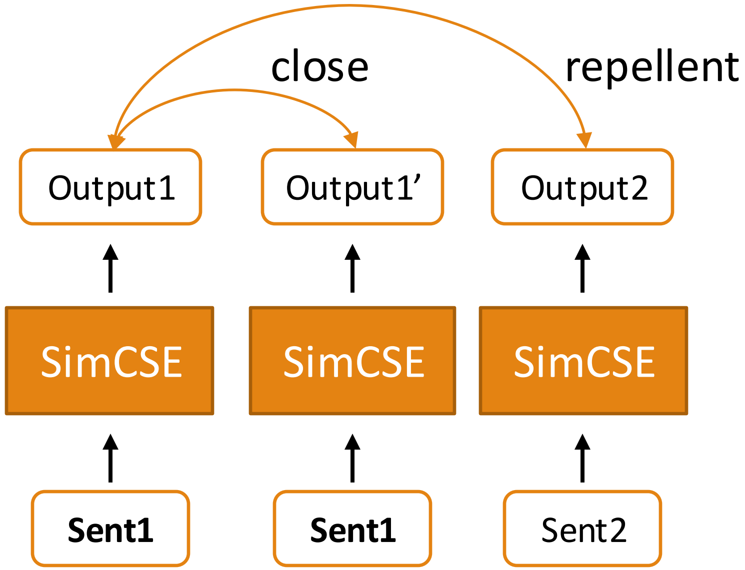
Left Figure source: Schroff, Florian, Dmitry Kalenichenko, and James Philbin. "Facenet: A unified embedding for face recognition and clustering." CVPR 2015.

Right Figure source: Chen, Ting, et al. "A simple framework for contrastive learning of visual representations." ICLR 2020.

Unsupervised training of SimCSE



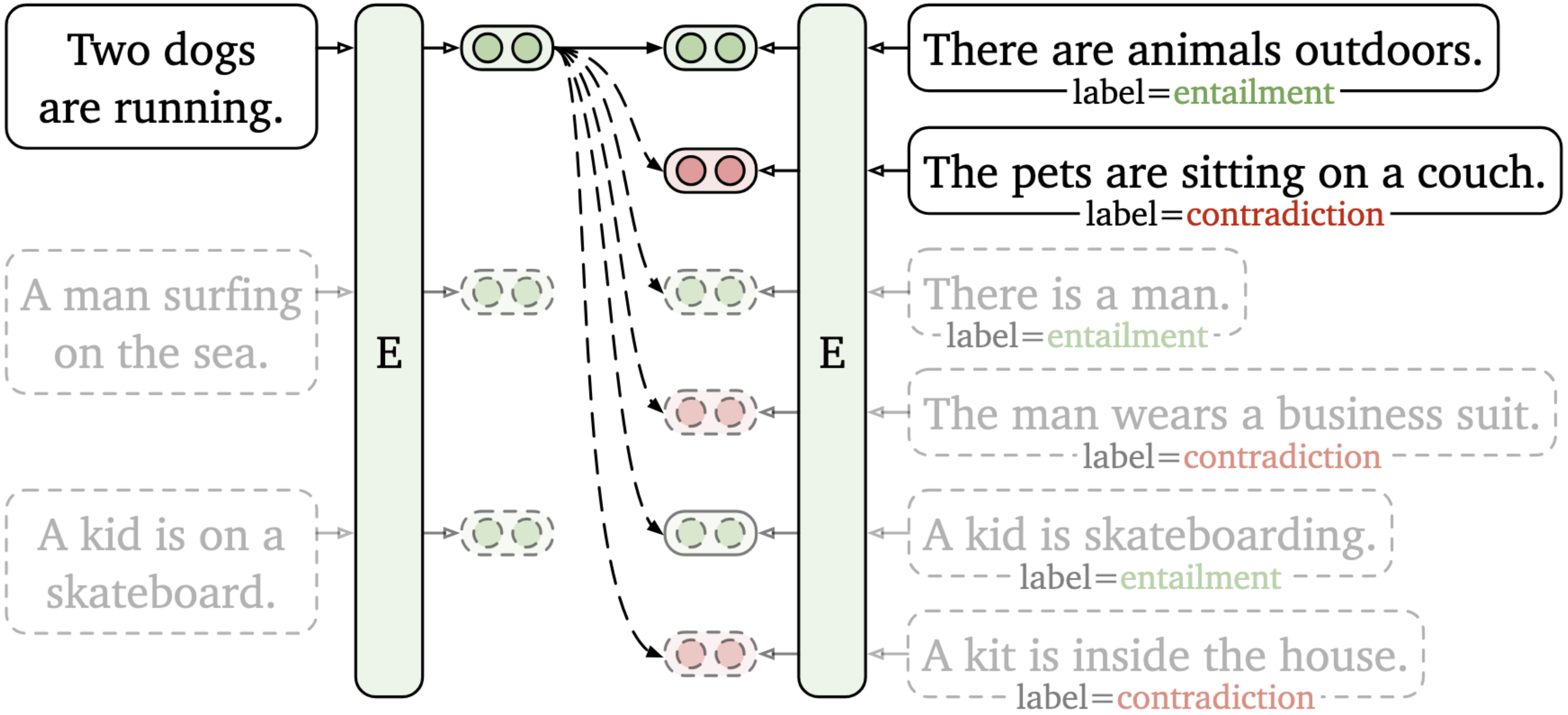
Input a sentence twice with a dropout mask
(Dropout as data augmentation)
同樣的句子放到BERT兩次，取得不同的版本



Gao, Tianyu, Xingcheng Yao, and Danqi Chen. "SimCSE: Simple Contrastive Learning of Sentence Embeddings." EMNLP 2021.

Supervised training of SimCSE

- Supervised training of SimCSE relies on labels in a dataset to define positives and negatives.



Gao, Tianyu, Xingcheng Yao, and Danqi Chen. "SimCSE: Simple Contrastive Learning of Sentence Embeddings." EMNLP 2021.

SimCSE outperforms Sentence-BERT

- SimCSE outperforms Sentence-BERT on semantic similarity tasks.

Model	STS12	STS13	STS14	STS15	STS16	STS-B	SICK-R	Avg.
Supervised models								
$\text{SRoBERTa}_{\texttt{base}}$ ♣	71.54	72.49	70.80	78.74	73.69	77.77	74.46	74.21
$\text{SRoBERTa}_{\texttt{base}}$-whitening	70.46	77.07	74.46	81.64	76.43	79.49	76.65	76.60
$*$ $\text{SimCSE-RoBERTa}_{\texttt{base}}$	**76.53**	**85.21**	**80.95**	**86.03**	**82.57**	**85.83**	**80.50**	**82.52**
$*$ $\text{SimCSE-RoBERTa}_{\texttt{large}}$	**77.46**	**87.27**	**82.36**	**86.66**	**83.93**	**86.70**	**81.95**	**83.76**

Retrieval for open-domain question answering (ODQA)

檢索出跟問題最相似的答案，而不是檢索出跟問題一樣的關鍵字
=>嘗試讓model去理解問題本身，並且去檢索出相關性最高的答案

Open-domain question answering (ODQA)

ODQA: 模型必須從知識庫自己找答案，pre-train資料不包含答案
SQuAD:模型在pre-train階段已經看過答案
=>ODQA-從開發資料找答案、SQuAD-從封閉資料找答案

- Given a question x such as "What is the currency of the UK?", a model must output the correct answer string y, "pound".
- The "open" part of ODQA refers to the fact that the model does not receive a pre-identified document that is known to contain the answer.
- ODQA is like Reading comprehension (RC) tasks, such as SQuAD, but no relevant articles provided.

(Example of SQuAD)

In meteorology, precipitation is any product of the condensation of atmospheric water vapor that falls under **gravity**. The main forms of precipitation include drizzle, rain, sleet, snow, **graupel** and hail... Precipitation forms as smaller droplets coalesce via collision with other rain drops or ice crystals **within a cloud**. Short, intense periods of rain in scattered locations are called "showers".

What causes precipitation to fall?
gravity

主要是回答檔案的位置

Guu, Kelvin, et al. "Retrieval augmented language model pre-training." ICML 2020.
Rajpurkar, P., Zhang, J., Lopyrev, K., & Liang, P. (2016). SQuAD: 100,000+ Questions for Machine Comprehension of Text. EMNLP 2016.

https://aclanthology.org/2020.emnlp-main.550/

Dense Passage Retrieval (DPR)

把問題和段落都轉成 dense vectors（高維向量），用cosine similarity找出語意最接近的段落
=> BM25–字詞重合lexical match、DPR–semantic similarity

- Dense retrieval focuses on **semantic similarity**
- Passages and questions are embedded into dense vectors
- Dense vectors enable better matching for related words or phrases

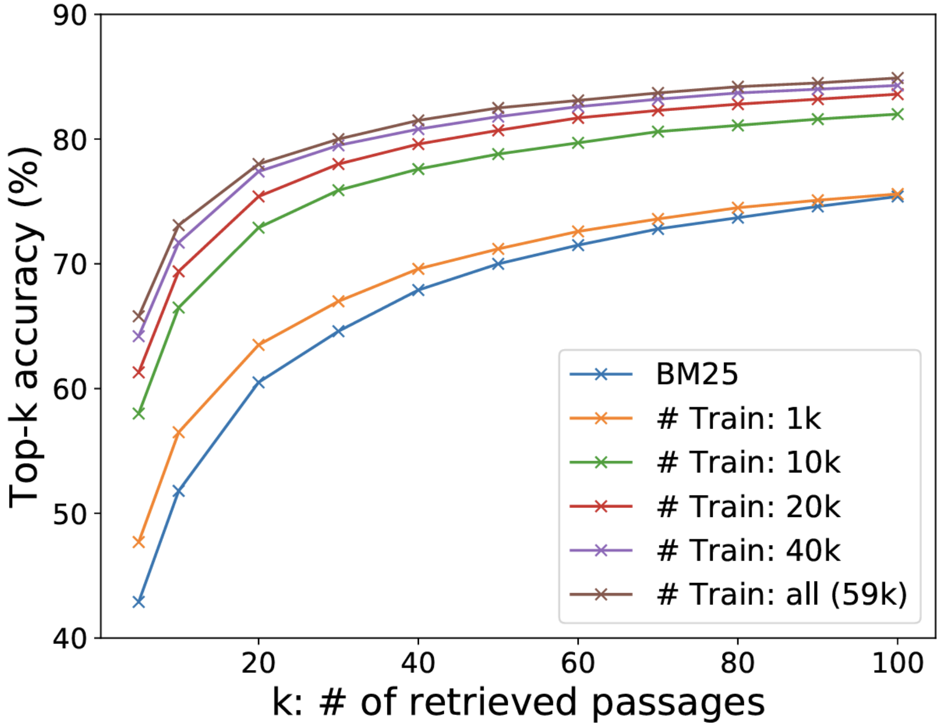
(e.g., "the body of water" matched with "sea")

Question	Passage received by BM25	Passage retrieved by DPR
What is the body of water between England and Ireland?	Title:British Cycling . . . **England** is not recognised as a region by the UCI, and there is no English cycling team outside the Commonwealth Games. For those occasions, British Cycling selects and supports the **England** team. Cycling is represented on the Isle of Man by the Isle of Man Cycling Association. Cycling in Northern **Ireland** is organised under Cycling Ulster, part of the all-Ireland governing **body** Cycling **Ireland**. Until 2006, a rival governing **body** existed, . . .	Title: Irish Sea . . . Annual traffic between Great Britain and **Ireland** amounts to over 12 million passengers and of traded goods. **The Irish Sea** is connected to the North Atlantic at both its northern and southern ends. To the north, the connection is through the North Channel between Scotland and Northern **Ireland** and the Malin Sea. The southern end is linked to the Atlantic through the St George's Channel between Ireland and Pembrokeshire, and the Celtic Sea. . . .

Karpukhin et al., 2020. Dense Passage Retrieval for Open-Domain Question Answering, EMNLP 2020.

Dense Passage Retrieval (DPR)

- Outperforms **BM25** using only 1000 training data!



Dense Passage Retrieval (DPR)

- After training, two **BERT-based encoders** can **independently** encode question (**q**) and passage (**p**) into dense vectors.
- **Similarity** between question and passage = **dot product** between their embeddings

$$\mathrm{sim}(q,p) = E_Q(q)^\intercal E_P(p).$$

q : question text
p : passage text
E_Q : BERT model that outputs question representation
E_p : BERT model that outputs passage representation

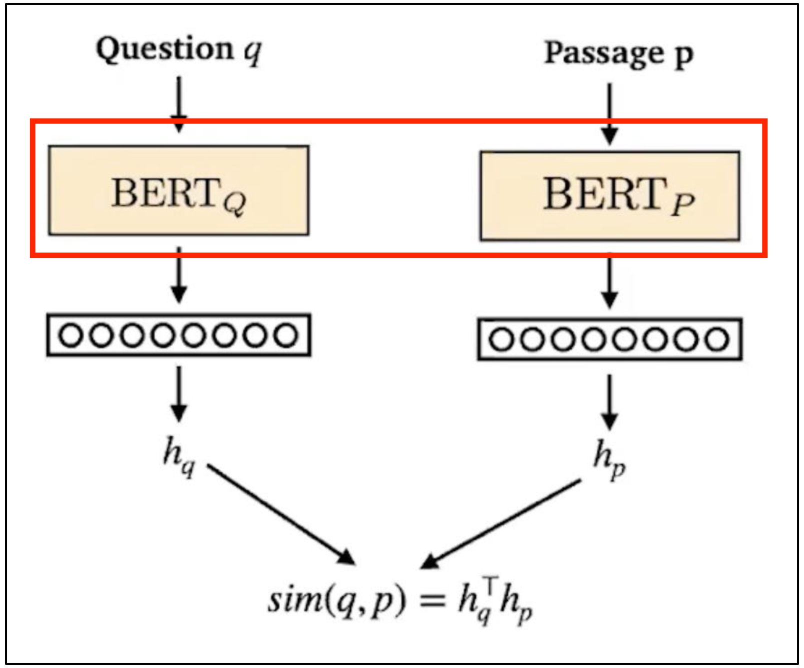


Image source: Stanford CS224N Lecture 12 - Question Answering

Karpukhin et al., 2020. Dense Passage Retrieval for Open-Domain Question Answering

Dense Passage Retrieval (DPR)

Training the encoders

- Goal: **Relevant** pairs of questions and passages will have **smaller distance** than the irrelevant ones

- Training data

m training instances

$$\mathcal{D} = \{\langle \boxed{q_i}, \boxed{p_i^+}, \boxed{p_{i,1}^-, \cdots, p_{i,n}^-} \rangle\}_{i=1}^{\boxed{m}}$$

Question

Relevant Passage

n Irrelevant Passages

Karpukhin et al., 2020. Dense Passage Retrieval for Open-Domain Question Answering

Dense Passage Retrieval (DPR)

Training the encoders

- Base model: bert-base-uncased

- Loss function: Negative log-likelihood of the positive passage

$$
\begin{aligned}
& L(q_i, p_i^+, p_{i,1}^-, \cdots, p_{i,n}^-) \\
= \quad & -\log \frac{e^{\mathrm{sim}(q_i, p_i^+)}}{e^{\mathrm{sim}(q_i, p_i^+)} + \sum_{j=1}^{n} e^{\mathrm{sim}(q_i, p_{i,j}^-)}}.
\end{aligned}
$$

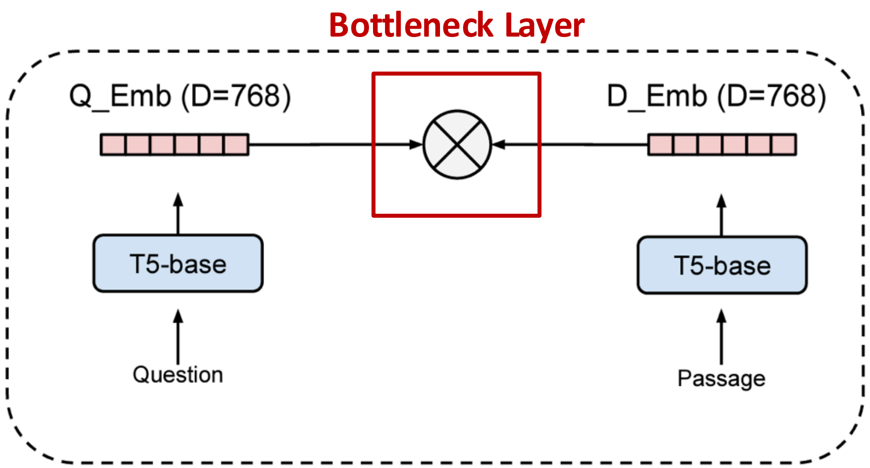
  - **Maximize** the similarity between $\mathbf{q_i}$ and $\mathbf{p_i^+}$
  - **Minimize** the similarity between non-relevant pairs ($\mathbf{q_i}$ and $\mathbf{p_{i,j}^-}$)

Karpukhin et al., 2020. Dense Passage Retrieval for Open-Domain Question Answering

https://aclanthology.org/2022.emnlp-main.669/

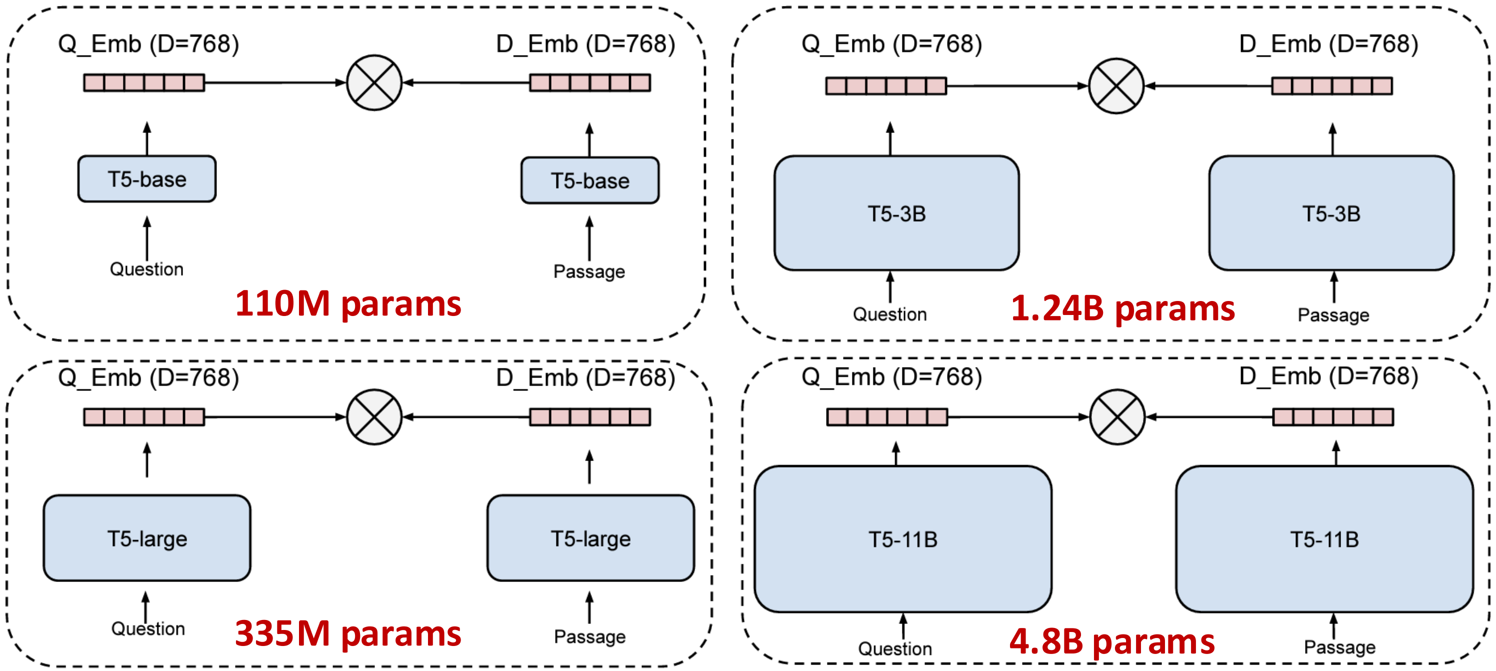
Limitations of Dual Encoders

- Often fail to generalize to other domains for retrieval tasks
- **Bottleneck layer** of dual encoders (simple dot-product or cosine similarity) *might not be powerful enough* to **capture semantic relevance**?



Ni et al., 2022. Large Dual Encoders Are Generalizable Retrievers, EMNLP 2022.

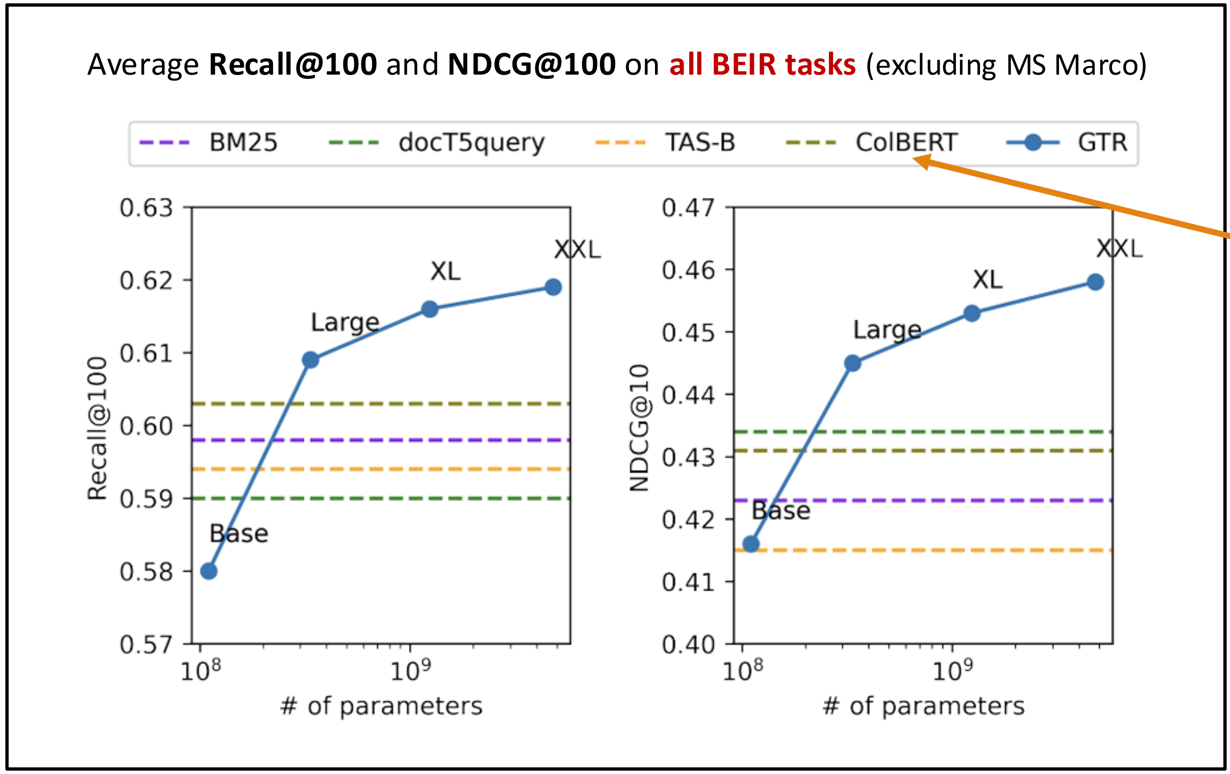
Generalizable T5-based Retriever (GTR)



Can **scaling up** dual encoder model size *improve the retrieval performance*, while keeping the bottleneck layers **fixed**?

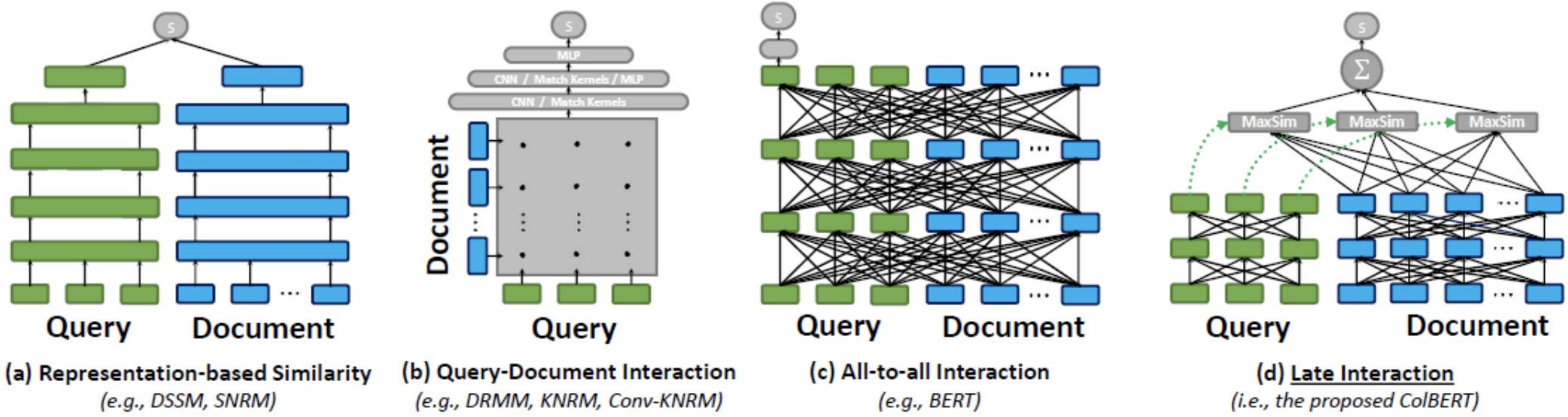
Generalizable T5-based Retriever (GTR)

- Scaling up *consistently improves* dual encoders' **out-of-domain** performance.



ColBERT: Efficient and Effective Passage Search via Contextualized Late Interaction over BERT, SIGIR 2020

Interactions of Query and Document



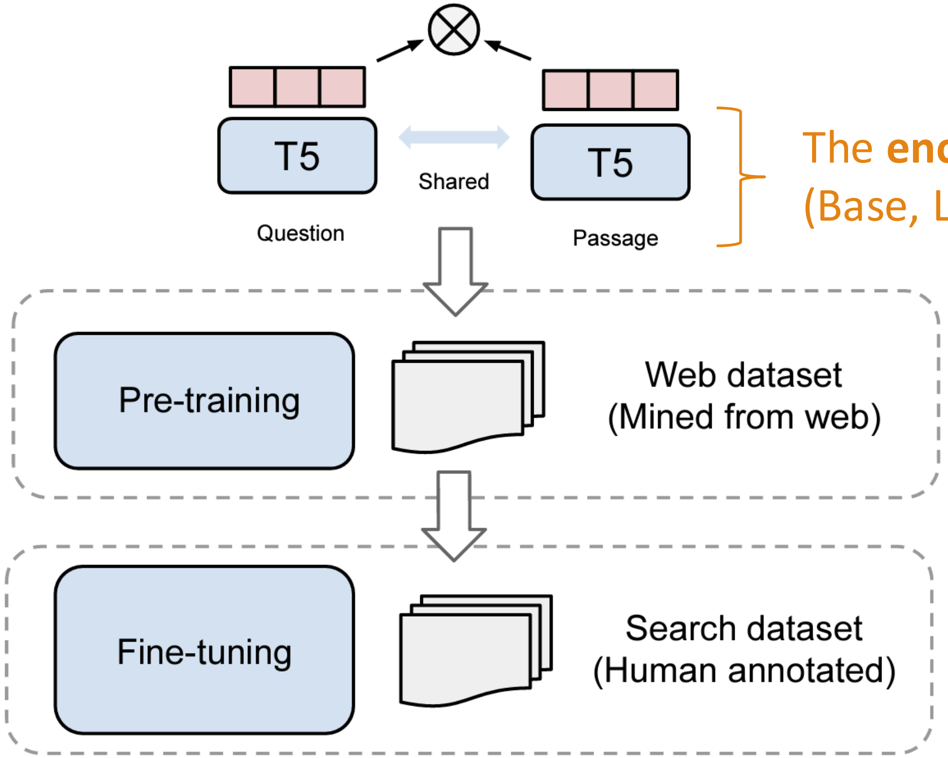
ColBERT: Efficient and Effective Passage Search via Contextualized Late Interaction over BERT, SIGIR 2020

Generalizable T5-based Retriever (GTR)

Multi-stage training for GTR

Stage 1:

Stage 2:



The **encoder** part from pretrained T5 models (Base, Large, XL, XXL)

2 billion **QA pairs** from:
- Reddit
- Stack-Overflow

Dataset:
- MS Marco
- NaturalQuestions

https://arxiv.org/pdf/1611.09268

MS MARCO: A Human Generated MAchine Reading COmprehension Dataset

- Machine Reading Comprehension (MRC) dataset
  - by Microsoft 2016, which was adapted in 2018 for retrieval.
- 8.8 M passages from the Web to 1M real-world queries.
- Each query is associated with sparse relevance judgments of one (or very few) documents marked as relevant and no documents explicitly indicated as irrelevant.
- three different tasks
  - (i) predict if a question is answerable given a set of context passages, and extract and synthesize the answer as a human would
  - (ii) generate a well-formed answer (if possible) based on the context passages that can be understood with the question and passage context
  - (iii) rank a set of retrieved passages given a question.

Generalizable T5-based Retriever (GTR)

- **Data efficiency**: Only needs **10%** of MS Marco *supervised data* to achieve the best **out-of-domain** performance!

*GTR **w/o** Pre-training *GTR w/ Pre-training + Fine-tuning

	GTR-FT		GTR		
Ratio of data	Large	XL	Large	XL	XXL
	NDCG@10 on MS Marco *in-domain				
10%	0.402	0.397	0.428	0.426	-
100%	0.415	0.418	0.430	0.439	0.430
	Zero-shot average NDCG@10 w/o MS Marco *out-of-domain				
10%	**0.413**	0.418	**0.452**	**0.462**	**0.465**
100%	0.412	**0.433**	0.445	0.453	0.458

Summary Parameter Sharing

SimCSE:
和Sentence-BERT一樣採用兩個共享權重的 Encoder
但差異在weight sharing使用drop out

DPR:
用於 Dense Retrieval
將 Query 和 Passage 都
轉成向量後比相似度

	BERT (Devlin et al., NAACL 2019)	Sentence-BERT (Reimers et al., EMNLP 2019)	SimCSE (Gao et al., EMNLP 2021)	DPR (Karpukhin et al., EMNLP 2020)	GTR (Ni et al., EMNLP 2022)
Encoder Type	Cross	Dual	Dual	Dual	Dual
Weight Sharing	Nan	Yes	Yes	No (Separate Query Encoder and Passage Encoder)	Yes

BERT:
將兩個句子一起丟進同一個 BERT
格式是：[CLS] Sent1 [SEP] Sent2 [SEP]
模型同時讀取兩句話之間的交互關係
適合分類、NLI、QA，但 不適合大規模搜尋

Sentence-BERT:
Encoder Type：Dual Encoder（Siamese Network）
句子 A → Encoder
句子 B → Encoder
用 cosine similarity 做比對

Query/Passage Encoder不共用DPR?
=>語言分佈不同，需要專門的embedding空間
=>訓練model可以善用passage來回應Query

https://arxiv.org/pdf/2104.08663

BEIR: A Heterogeneous Benchmark for Zero-shot Evaluation of Information Retrieval Models

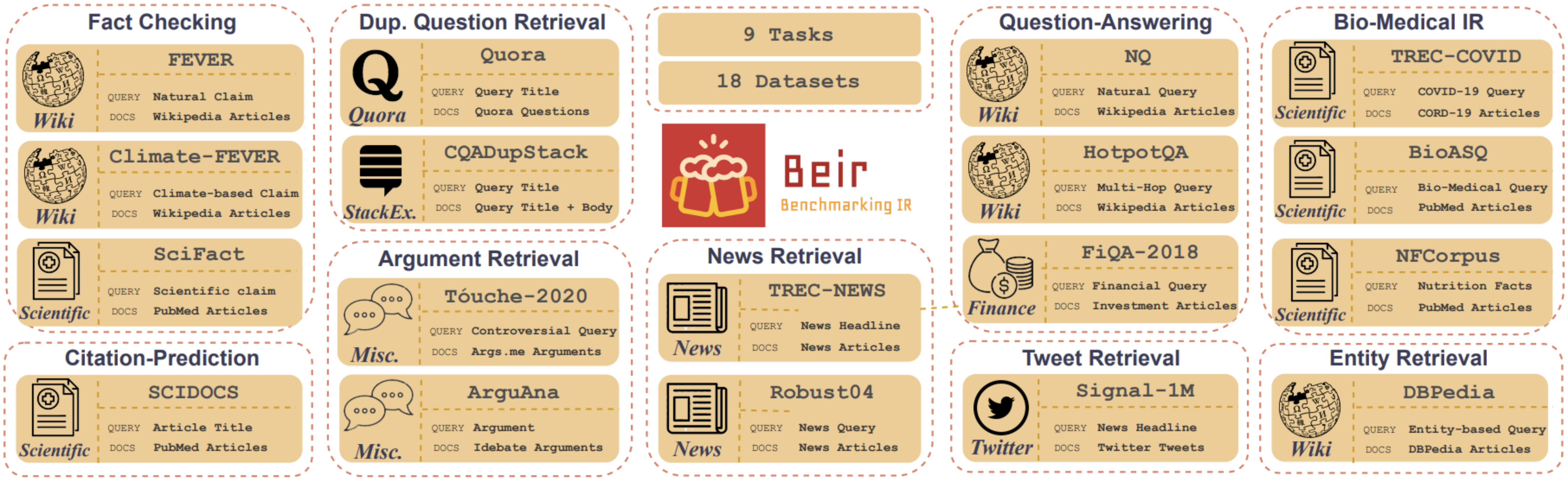


Figure 1: An overview of the diverse tasks and datasets in BEIR benchmark.

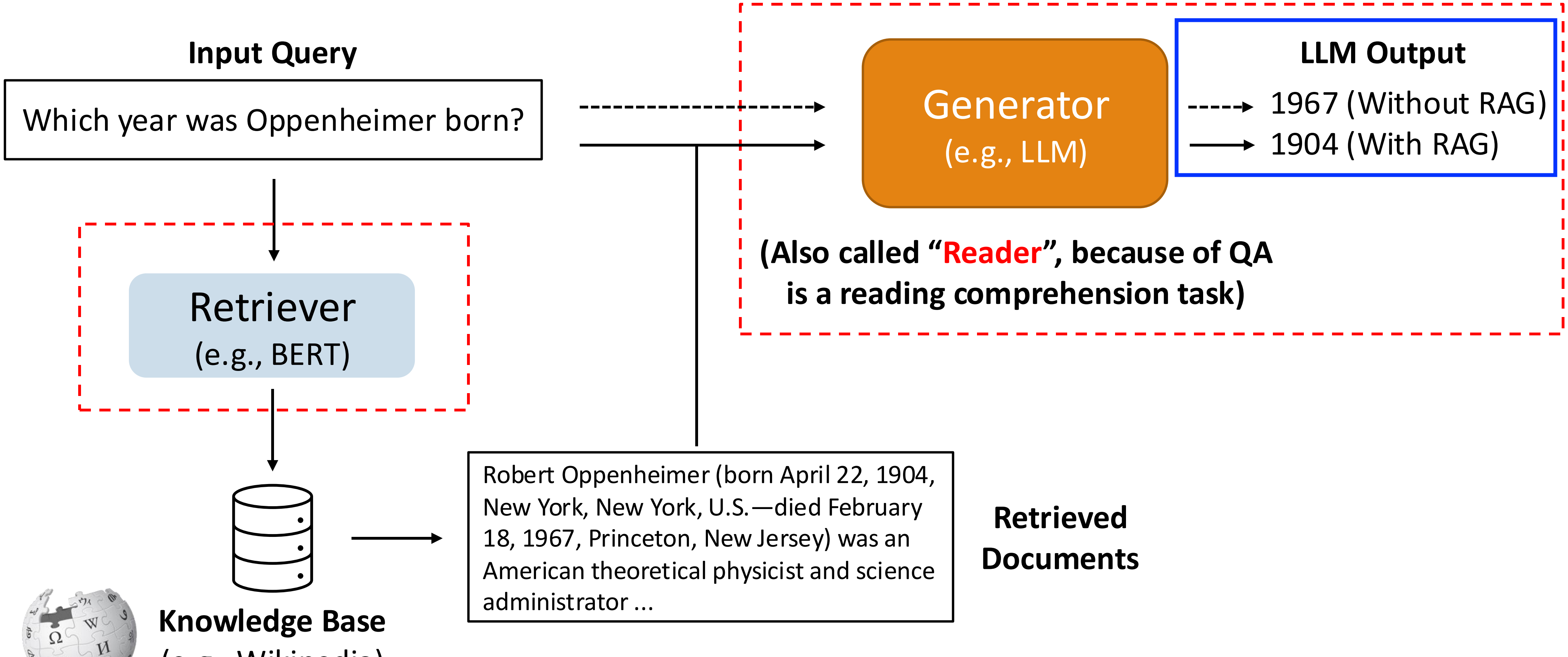
https://arxiv.org/pdf/2104.08663

BEIR: A Heterogeneous Benchmark for Zero-shot Evaluation of Information Retrieval Models

Model (→)	Lexical	Sparse			Dense				Late-Interaction	Re-ranking
Dataset (↓)	**BM25**	**DeepCT**	**SPARTA**	**docT5query**	**DPR**	**ANCE**	**TAS-B**	**GenQ**	**ColBERT**	**BM25+CE**
MS MARCO	0.228	0.296‡	0.351‡	0.338‡	0.177	0.388‡	0.408‡	0.408‡	0.401‡	**0.413**‡
TREC-COVID	0.656	0.406	0.538	0.713	0.332	0.654	0.481	0.619	0.677	**0.757**
BioASQ	0.465	0.407	0.351	0.431	0.127	0.306	0.383	0.398	0.474	**0.523**
NFCorpus	0.325	0.283	0.301	0.328	0.189	0.237	0.319	0.319	0.305	**0.350**
NQ	0.329	0.188	0.398	0.399	0.474‡	0.446	0.463	0.358	0.524	**0.533**
HotpotQA	0.603	0.503	0.492	0.580	0.391	0.456	0.584	0.534	0.593	**0.707**
FiQA-2018	0.236	0.191	0.198	0.291	0.112	0.295	0.300	0.308	0.317	**0.347**
Signal-1M (RT)	0.330	0.269	0.252	0.307	0.155	0.249	0.289	0.281	0.274	**0.338**
TREC-NEWS	0.398	0.220	0.258	0.420	0.161	0.382	0.377	0.396	0.393	**0.431**
Robust04	0.408	0.287	0.276	0.437	0.252	0.392	0.427	0.362	0.391	**0.475**
ArguAna	0.315	0.309	0.279	0.349	0.175	0.415	0.429	**0.493**	0.233	0.311
Touché-2020	**0.367**	0.156	0.175	0.347	0.131	0.240	0.162	0.182	0.202	0.271
CQADupStack	0.299	0.268	0.257	0.325	0.153	0.296	0.314	0.347	0.350	**0.370**
Quora	0.789	0.691	0.630	0.802	0.248	0.852	0.835	0.830	**0.854**	0.825
DBPedia	0.313	0.177	0.314	0.331	0.263	0.281	0.384	0.328	0.392	**0.409**
SCIDOCS	0.158	0.124	0.126	0.162	0.077	0.122	0.149	0.143	0.145	**0.166**
FEVER	0.753	0.353	0.596	0.714	0.562	0.669	0.700	0.669	0.771	**0.819**
Climate-FEVER	0.213	0.066	0.082	0.201	0.148	0.198	0.228	0.175	0.184	**0.253**
SciFact	0.665	0.630	0.582	0.675	0.318	0.507	0.643	0.644	0.671	**0.688**
Avg. Performance vs. BM25		**- 27.9%**	**- 20.3%**	**+ 1.6%**	**- 47.7%**	**- 7.4%**	**- 2.8%**	**- 3.6%**	**+ 2.5%**	**+ 11%**

From Retrievers to QA

From now we focus on the full QA pipeline.



Gao, Yunfan, et al. "Retrieval-augmented generation for large language models: A survey." *arXiv preprint arXiv:2312.10997* (2023).

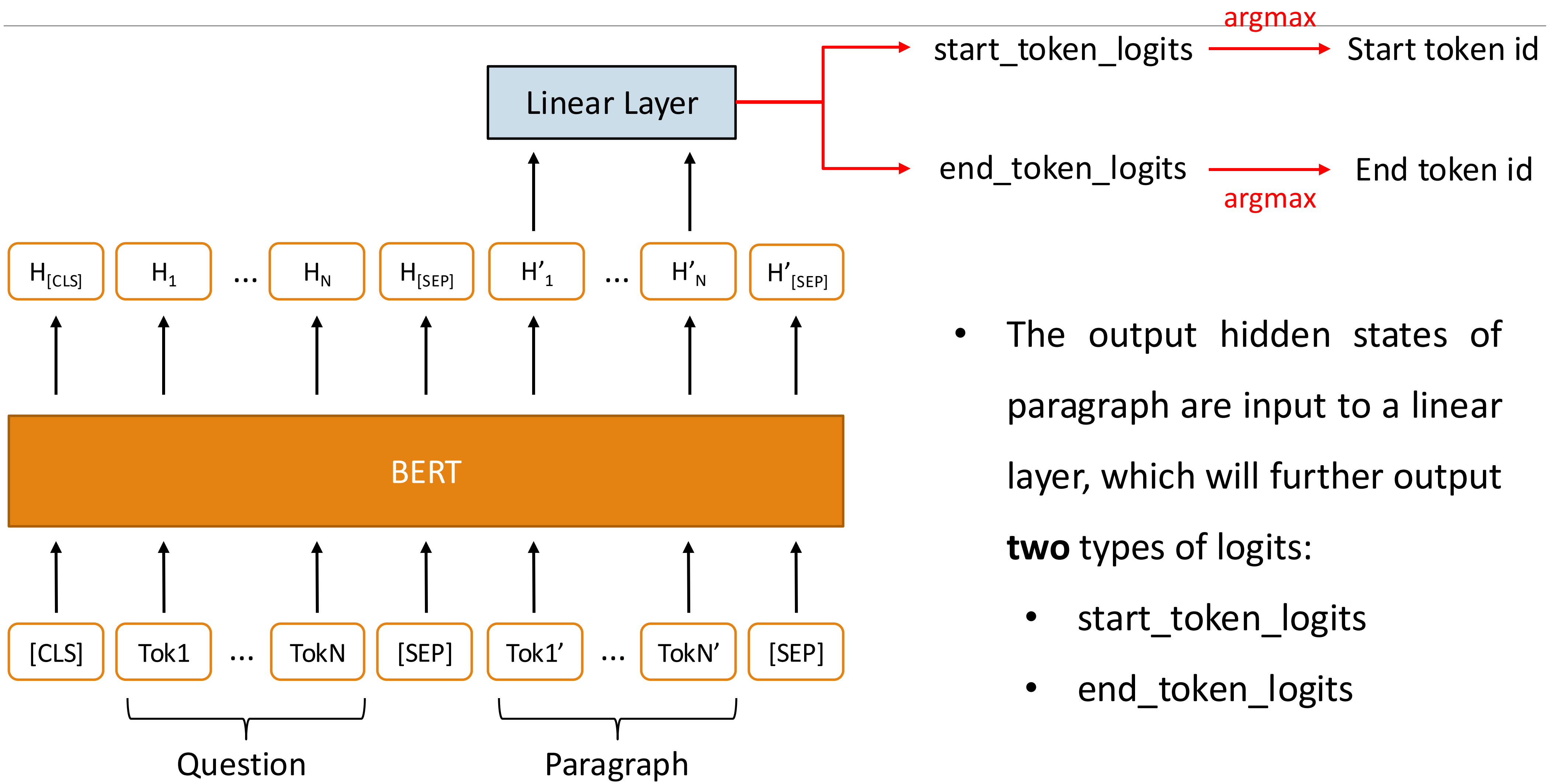
Open-domain QA with retrievers

做RAG最大的問題
=> Retrival檢索能力較難精進

- The QA model is an encoder (e.g., BERT).
  - ORQA
  - REALM
- The QA model is a generator (e.g., BART).

QA model is an encoder (e.g., BERT)

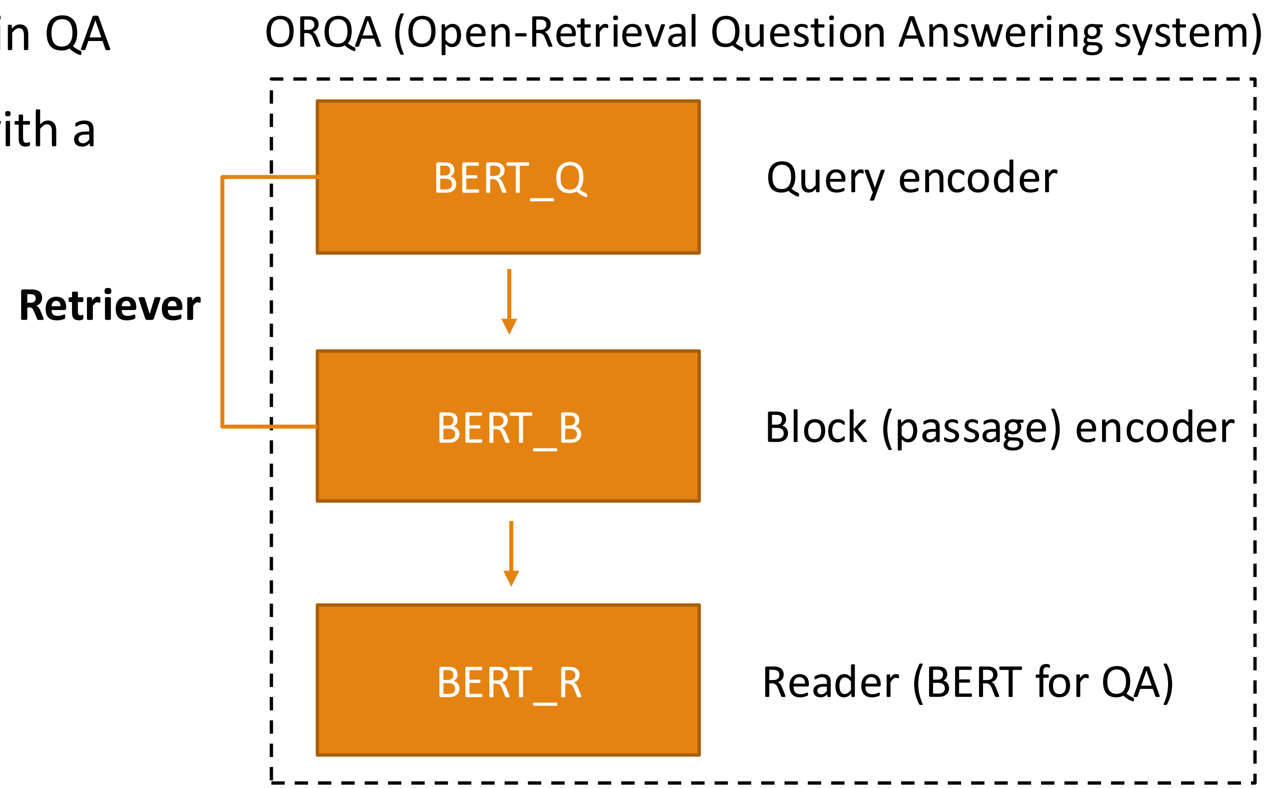
BERT for QA (Reading Comprehension)



- The output hidden states of paragraph are input to a linear layer, which will further output **two** types of logits:
  - start_token_logits
  - end_token_logits

ORQA (Open-Retrieval Question Answering system)

- Leverage **BERTs** for open-domain QA
- Extend BERT for QA to **ODQA** with a retriever model



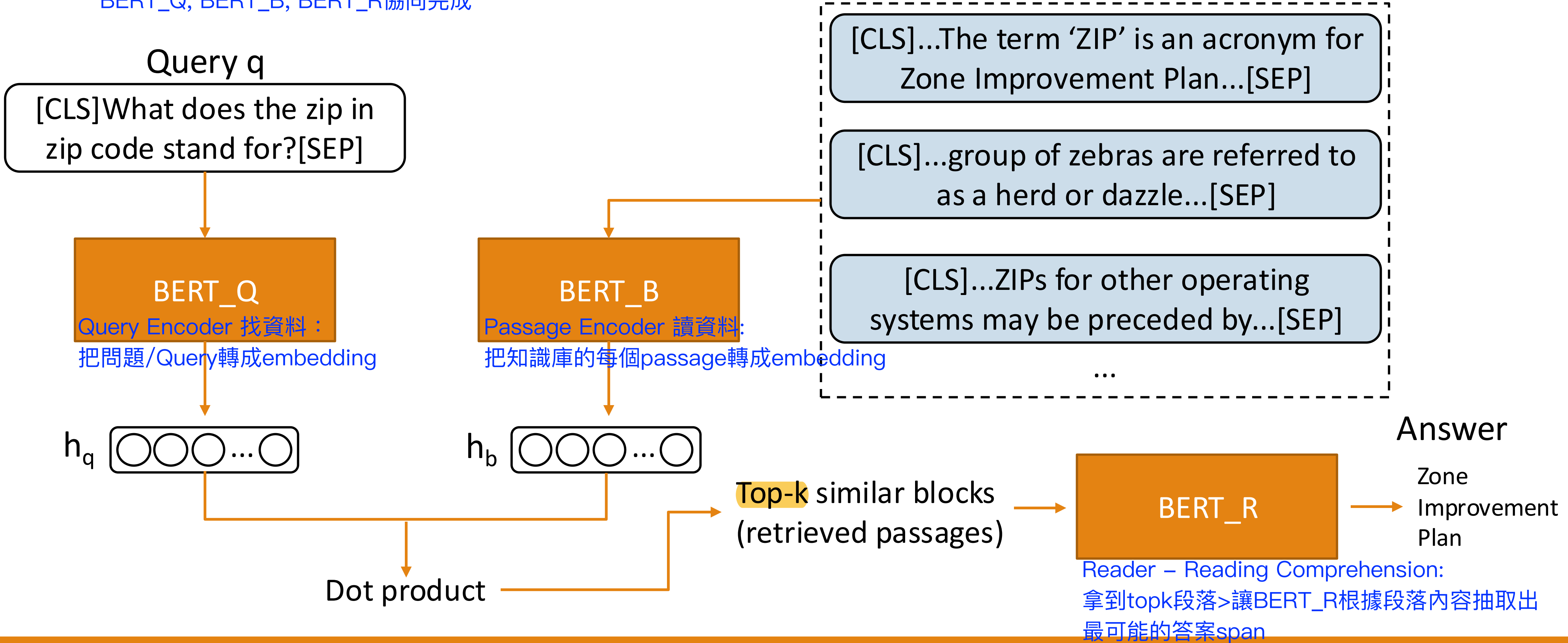
Lee, Kenton, Ming-Wei Chang, and Kristina Toutanova. "Latent Retrieval for Weakly Supervised Open Domain Question Answering." ACL 2019.

ORQA (Open-Retrieval Question Answering system)

Retriever+Reader結合在一起

- Leverage **BERTs** for open-domain QA
  BERT_Q, BERT_B, BERT_R協同完成

Evidence Block b (Passages to be retrieved)



＊Passage:從一篇文章裡面取出來的長字串段落
Ex:The term 'ZIP' is an acronym for Zone Improvement Plan.
ZIP codes were introduced by the United States Postal Service to improve mail delivery efficiency.

*Span: 在passage裡面被選出來的短字串
Ex: Passage = The term 'ZIP' is an acronym for Zone Improvement Plan.
Span = Zone Improvement Plan.

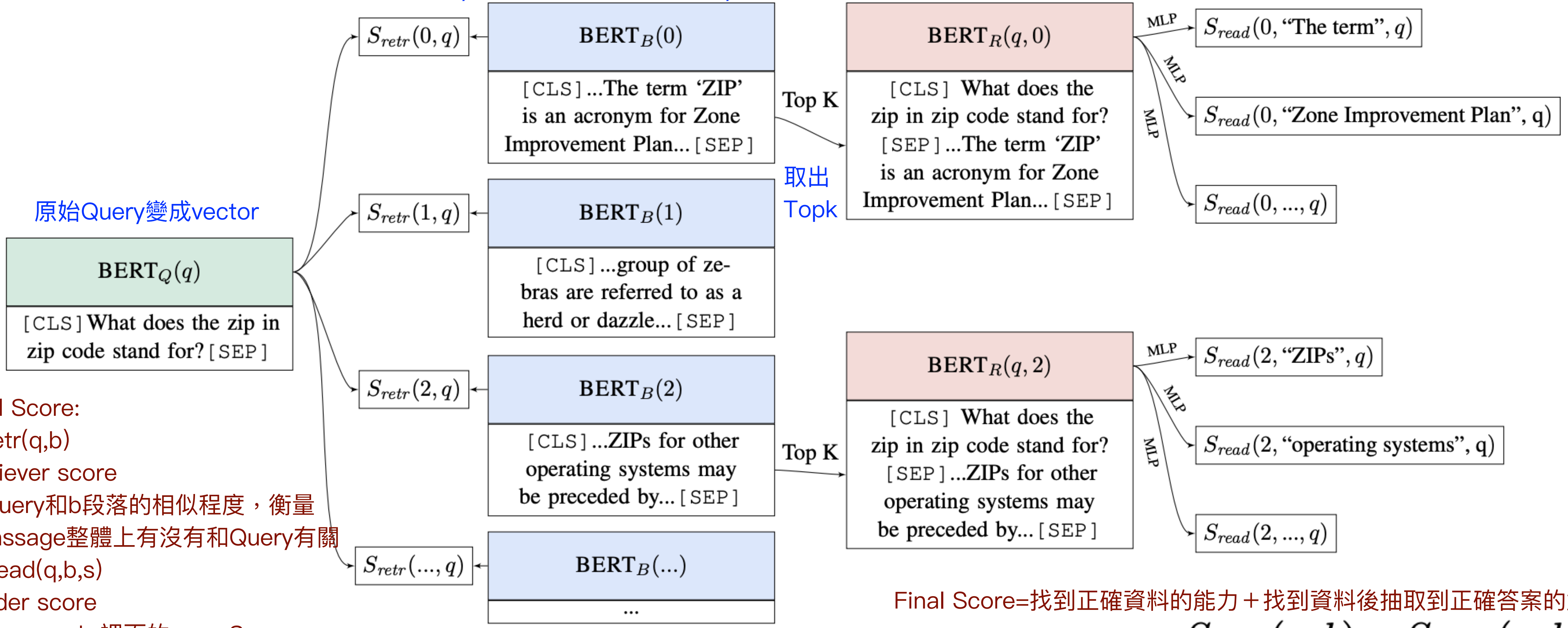
ORQA (Open-Retrieval Question Answering system)

將Query vector拿去跟所有passages比對
=>使用retriever計算出S_retr(i,q)
=>Query vs. 第i個段落的similarity

對每個Topk passage再做出
Bert_R,丟到reader model裡面
再做一次問答式判斷

問答模型會判對span s 是否為好答案？

原始Query變成vector



Final Score:
1.Sretr(q,b)
Retriever score
=>Query和b段落的相似程度，衡量此passage整體上有沒有和Query有關
2.Sread(q,b,s)
Reader score
=>passage b 裡面的span S 是否有直接回答Query了？

Final Score=找到正確資料的能力＋找到資料後抽取到正確答案的能力

Final score: $S_{retr}(q, b) + S_{read}(q, b, s)$

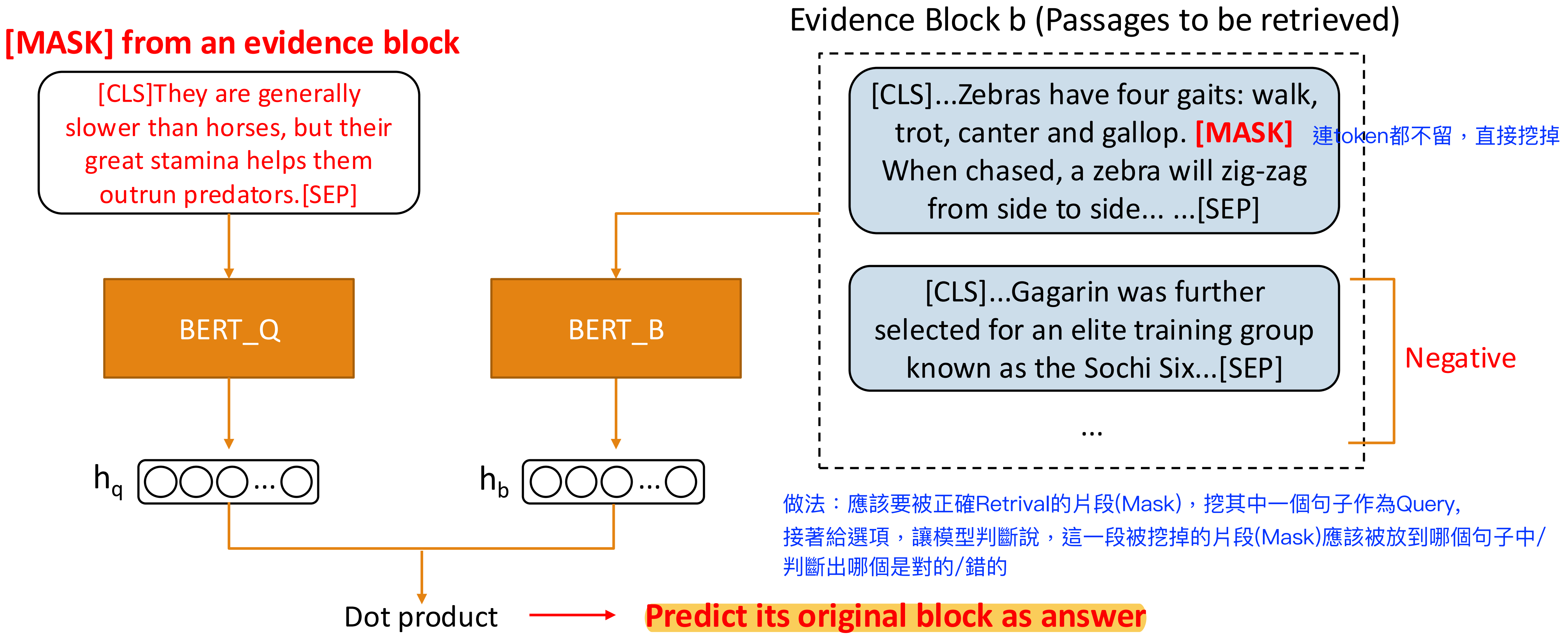
Inverse Cloze Task for Pre-training a Retriever

反向克漏字

- Cloze task: predict masked token from a corrupted context
- **Inverse cloze task (ICT): predict a corrupted context from a randomly sampled sentence** 屬於unsupervise
  - This is a **continual** pre-training (CP) task starting from the pre-trained BERT model in **ORQA**.

Lee, Kenton, Ming-Wei Chang, and Kristina Toutanova. "Latent Retrieval for Weakly Supervised Open Domain Question Answering." ACL 2019.

ICT for Pre-training a Retriever



ICT for Pre-training a Retriever

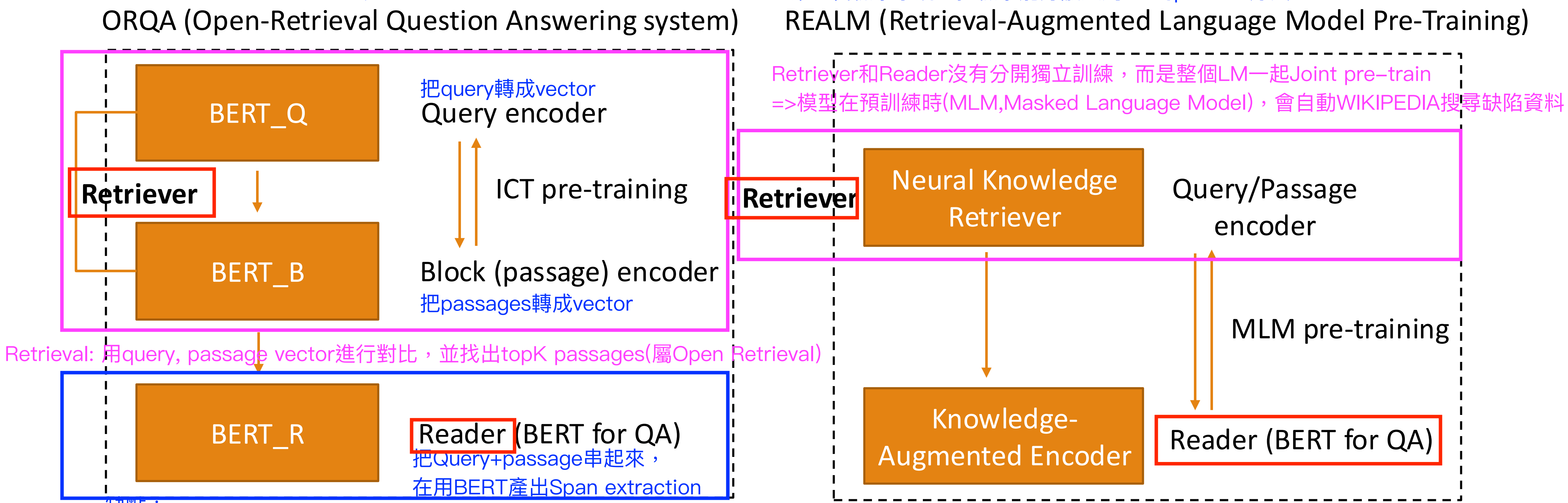
- Cloze task: predict masked token from a corrupted context
- **Inverse cloze task (ICT): predict a corrupted context from a randomly sampled sentence**
  - This is a **continual** pre-training (CP) task starting from the pre-trained BERT model in **ORQA**.
- Model to train:
  - Pre-training stage: BERT_Q and BERT_B
  - Fine-tuning stage: BERT_Q and BERT_R

Lee, Kenton, Ming-Wei Chang, and Kristina Toutanova. "Latent Retrieval for Weakly Supervised Open Domain Question Answering." ACL 2019.

REALM (Retrieval-Augmented Language Model Pre-Training)

第一代開放式檢索QA系統：
Retriever和Reader分開訓練,且為分割個體

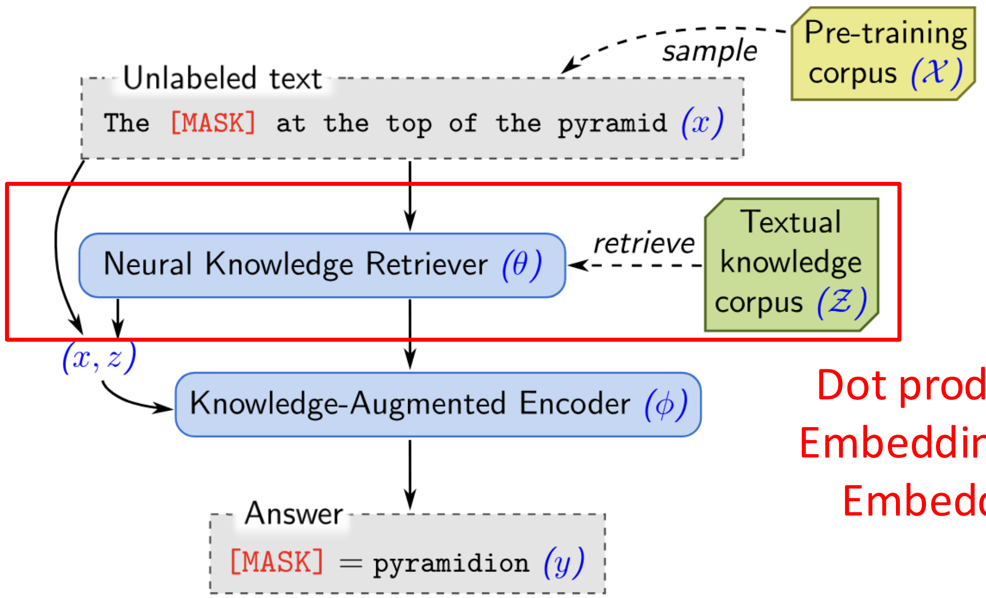
第二代檢索系統：把檢索能力融入到LM的pretrain方法



缺點：
retriever（BERT_Q、BERT_B）不是從閱讀任務學到的
reader（BERT_R）跟 retriever 是分開訓練
retriever 無法在訓練時看到“哪些 passage 真的重要”

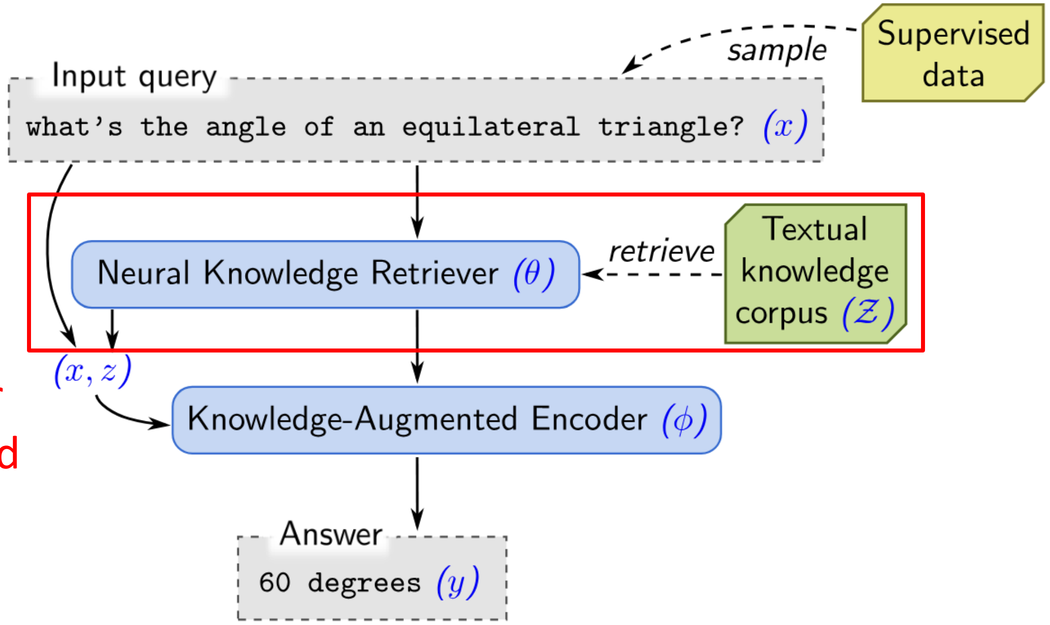
REALM (Retrieval-Augmented Language Model Pre-Training)

Unsupervised pre-training with masked language modeling (MLM)



Dot product for Embedding x and Embedding z

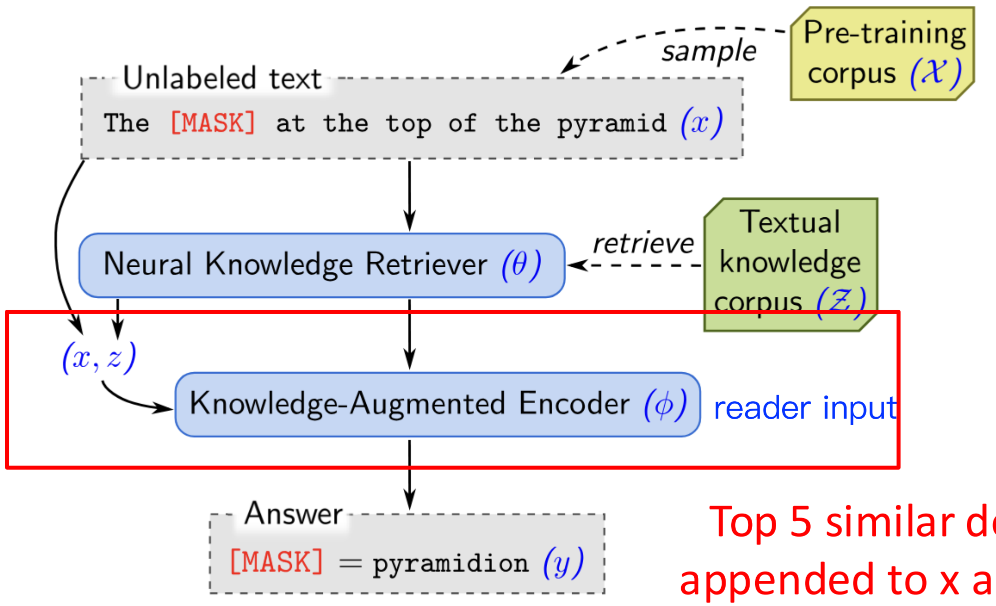
Fine-tuning on open-domain QA



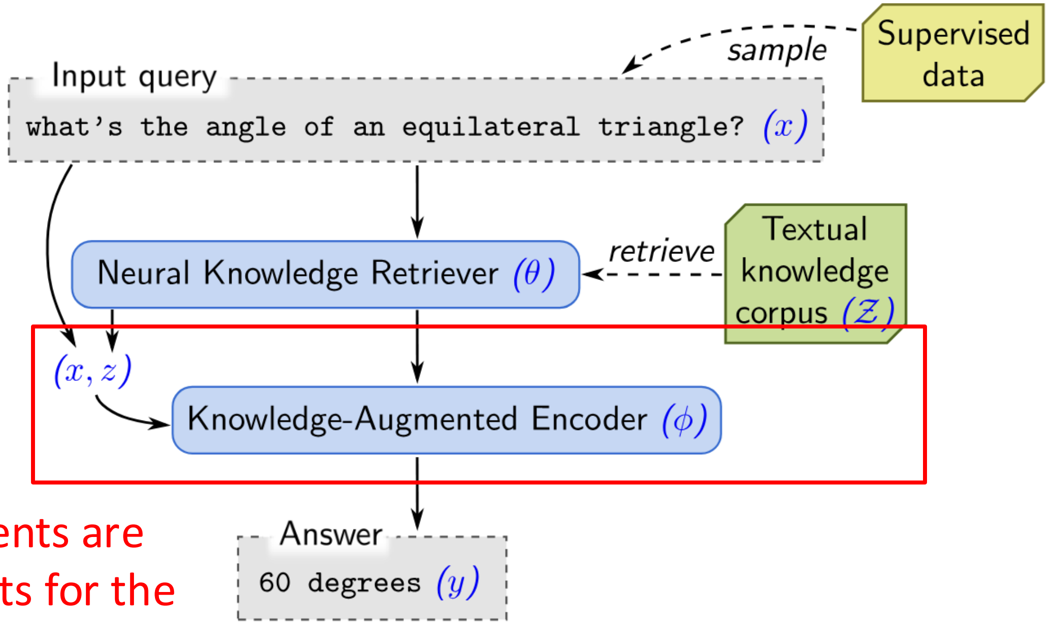
Guu, Kelvin, et al. "Retrieval augmented language model pre-training." ICML 2020.

REALM (Retrieval-Augmented Language Model Pre-Training)

Unsupervised pre-training with masked language modeling (MLM)



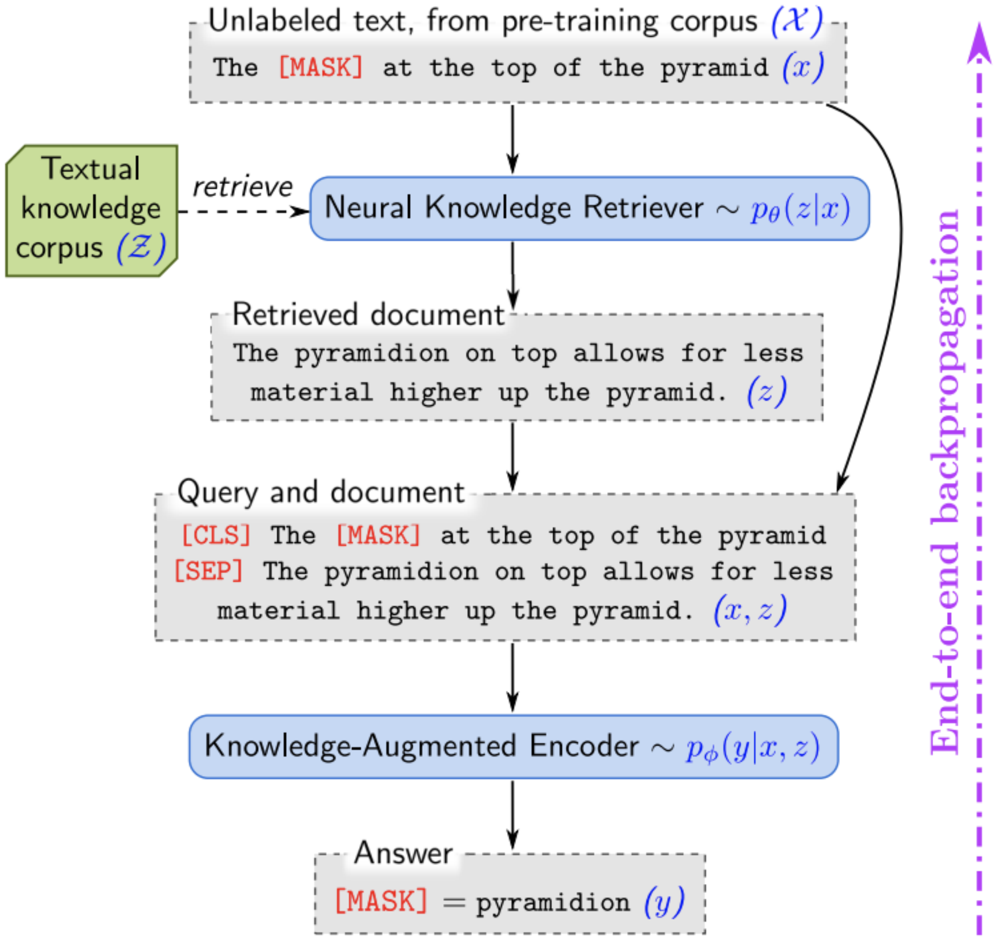
Fine-tuning on open-domain QA



Top 5 similar documents are appended to x as inputs for the Knowledge-Augmented Encoder

Guu, Kelvin, et al. "Retrieval augmented language model pre-training." ICML 2020.

REALM (Retrieval-Augmented Language Model Pre-Training)

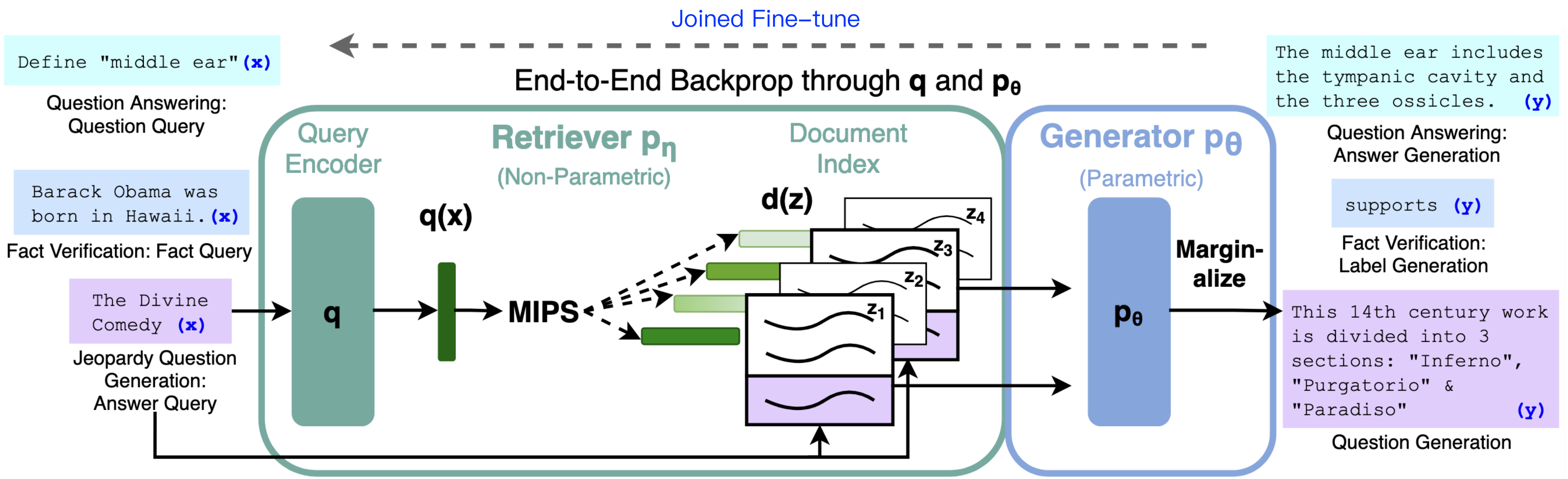


Some issues

- Corpus is big. (all of Wikipedia)
  - Index in embedding level
- $\mathcal{X}$ contain [MASK]
- End-to-end training

QA model is a generator (e.g., BART)

Retrieval-Augmented Generation (RAG)



FiD(Fusion–in–Decoder)

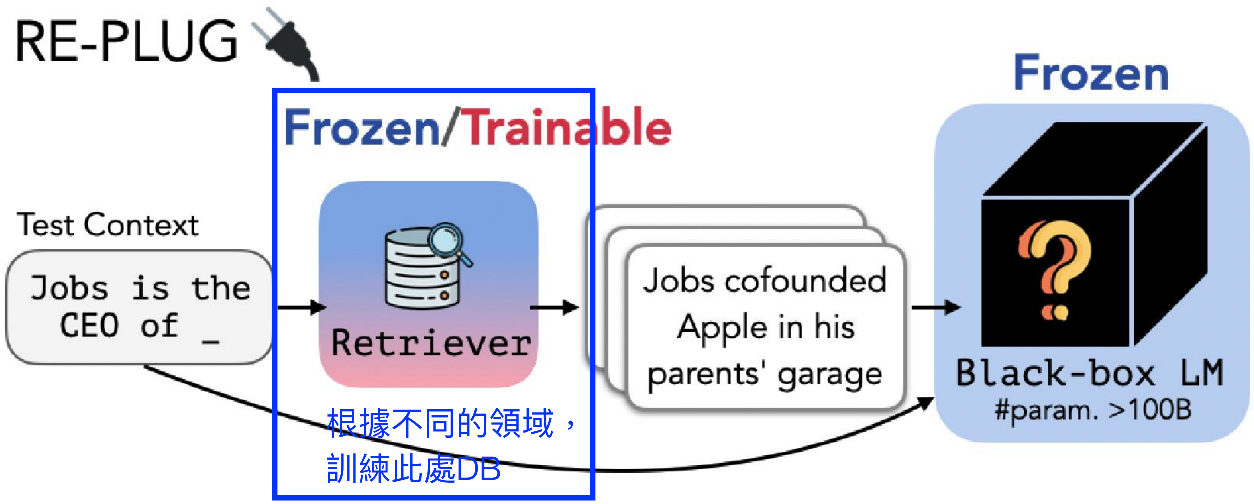
- This approach is the first RAG paper.
- No pre-training is involved. Only fine-tuning on open-domain QA.
- MIPS: Maximum Inner-Product Search (speed-up approach, supported by indexing packages like FAISS.)

Lewis, Patrick, et al. "Retrieval-augmented generation for knowledge-intensive nlp tasks." NeurIPS 2020.

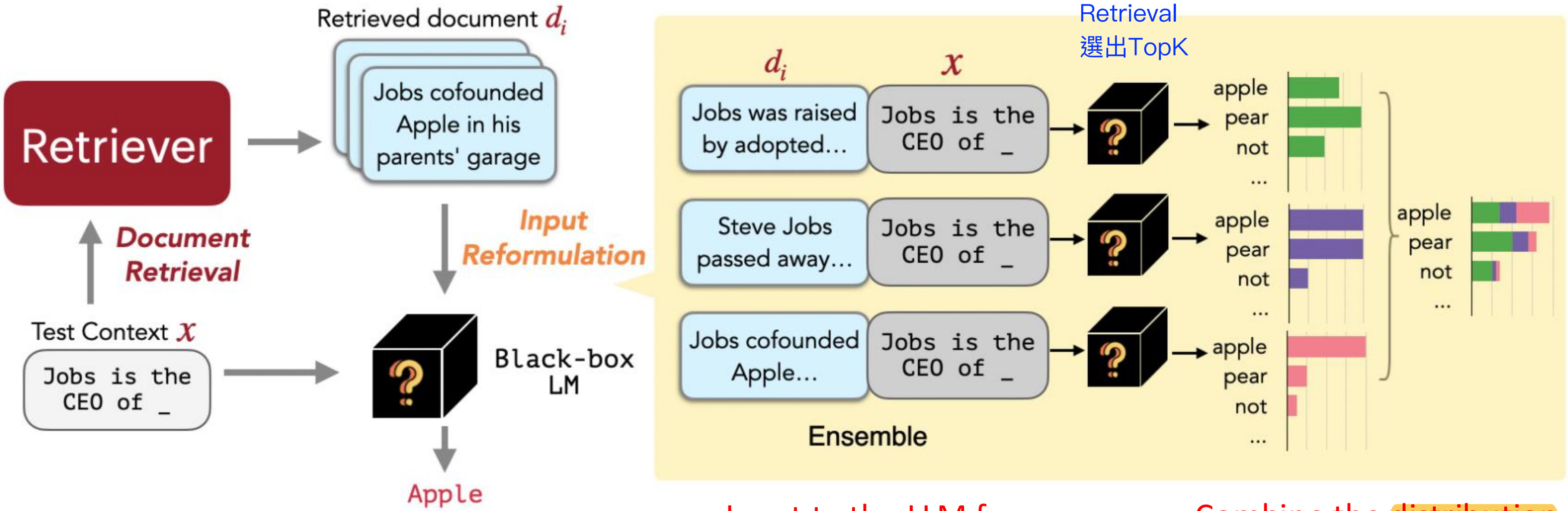
RAG with an LLM as the generator

Shi, Weijia, et al. "REPLUG: Retrieval-Augmented Black-Box Language Models." NAACL 2024.

- Due to a very large number of parameters, LLM is hard to train.
- **REPLUG (Retrieve and Plug)** proposes to train an LLM in the RAG framework.



REPLUG (at Inference Time)



Retrieval
選出TopK

Input to the LLM for each retrieved doc

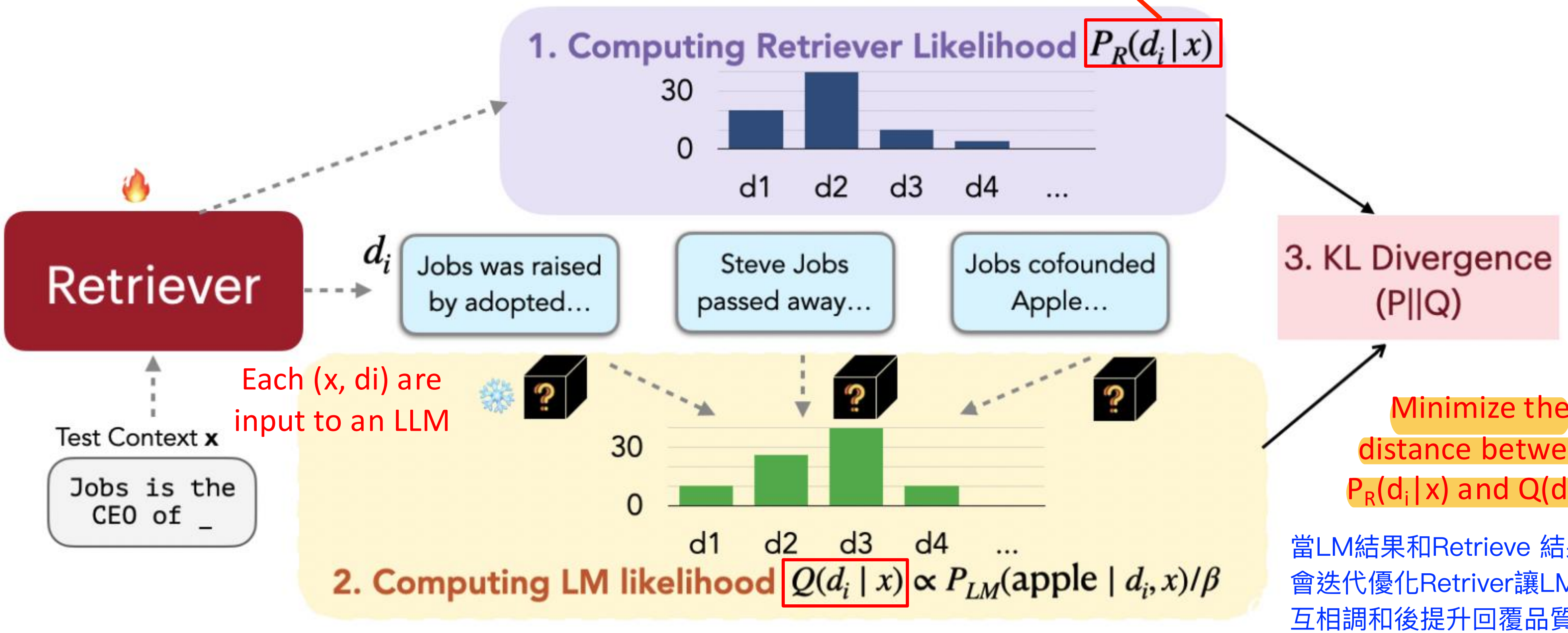
Combine the distribution for each retrieved doc

Shi, Weijia, et al. "REPLUG: Retrieval-Augmented Black-Box Language Models." NAACL 2024.

Training REPLUG with a Frozen LLM

REPLUG LSR (LM-Supervised Retrieval)

Train the retriever to retrieve the documents that are most helpful for the language modei to generate good answers.



Each (x, di) are input to an LLM

Minimize the distance between $P_R(d_i|x)$ and $Q(d_i|x)$

當LM結果和Retrieve 結果不一樣時，會迭代優化Retriver讓LM更加接近分佈，互相調和後提升回覆品質

Shi, Weijia, et al. "REPLUG: Retrieval-Augmented Black-Box Language Models." NAACL 2024.

Comparison between REPLUG and REPLUG LSR

Model		# Parameters	Original	+ REPLUG	Gain %	+ REPLUG LSR	Gain %
GPT-2	Small	117M	1.33	1.26	5.3	1.21	9.0
	Medium	345M	1.20	1.14	5.0	1.11	7.5
	Large	774M	1.19	1.15	3.4	1.09	8.4
	XL	1.5B	1.16	1.09	6.0	1.07	7.8
GPT-3 (black-box)	Ada	350M	1.05	0.98	6.7	0.96	8.6
	Babbage	1.3B	0.95	0.90	5.3	0.88	7.4
	Curie	6.7B	0.88	0.85	3.4	0.82	6.8
	Davinci	175B	0.80	0.77	3.8	0.75	6.3

Table 1: **Both REPLUG and REPLUG LSR consistently enhanced the performance of different language models.** Bits per byte (BPB) of the Pile using GPT-3 and GPT-2 family models (Original) and their retrieval-augmented versions (+REPLUG and +REPLUG LSR. The gain % shows the relative improvement of our models compared to the original language model.

壓縮率>>數字越小越好

Shi, Weijia, et al. "REPLUG: Retrieval-Augmented Black-Box Language Models." NAACL 2024.

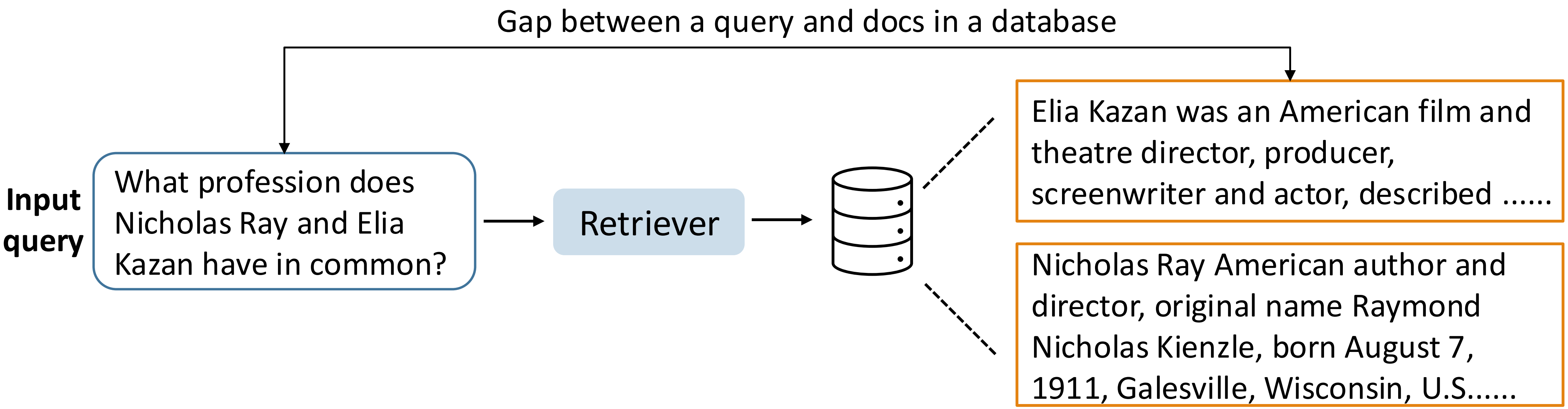
Recent RAG Developments

Type	Method	Paper	Venue Year
Enhancing Retrieval	Query Rewriting	Query Rewriting in Retrieval-Augmented Large Language Models	EMNLP 2023
	HyDE	Precise Zero-Shot Dense Retrieval without Relevance Labels	ACL 2023
Enhancing RAG	RetRobust	Making Retrieval-Augmented Language Models Robust to Irrelevant Context	ICLR 2024
	RAFT	RAFT: Adapting Language Model to Domain Specific RAG	COLM 2024
	Self-RAG	Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection	ICLR 2024
	RAAT	Enhancing Noise Robustness of Retrieval-Augmented Language Models with Adaptive Adversarial Training	ACL 2024
Enhancing RAG with Continual Retrieval	FLARE	Active Retrieval Augmented Generation	EMNLP 2023

Query Rewriting

Query 太短，會做Query expansion

- Motivation:
  - There is inevitably a gap between the input text and the knowledge that is really needed to query.



Ma, Xinbei, et al. "Query Rewriting in Retrieval-Augmented Large Language Models." EMNLP 2023.

Query Rewriting

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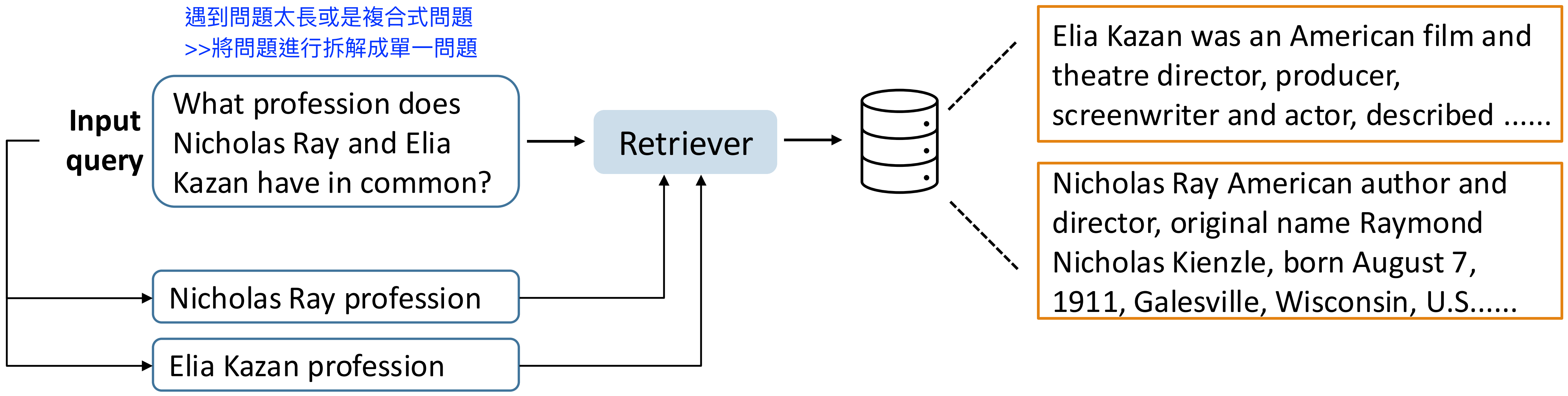
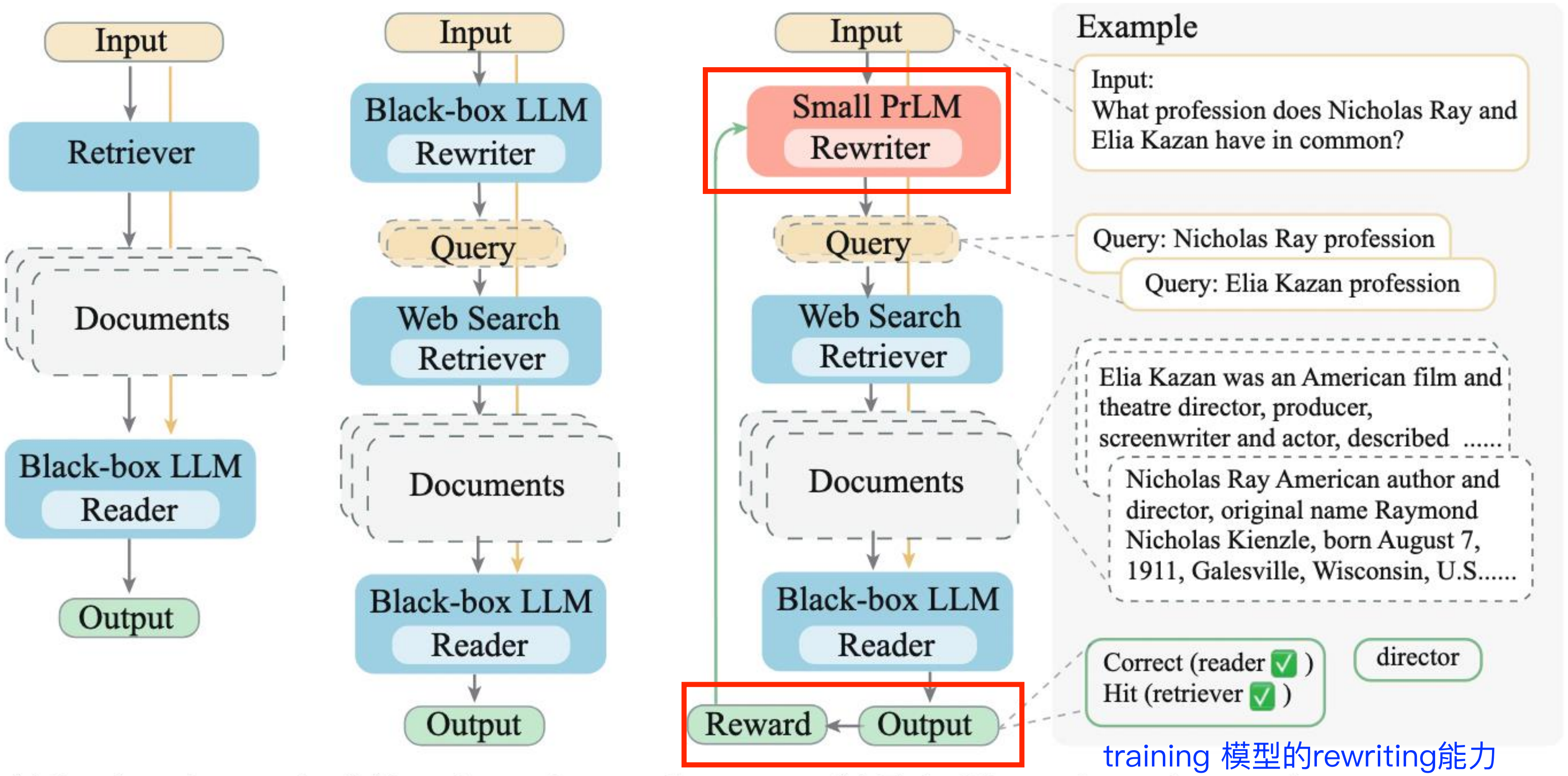


Query Rewriting – Approach



(a) Retrieve-then-read (b)Rewrite-retrieve-read (c) Trainable rewrite-retrieve-read

Ma, Xinbei, et al. "Query Rewriting in Retrieval-Augmented Large Language Models." EMNLP 2023.

Query Rewriting – Prompt for the LLMs

Direct prompt
Answer the question in the following format, end the answer with ’**’. {demonstration} Question: {x} Answer:
Reader prompt in retrieval-augment pipelines
Answer the question in the following format, end the answer with ’**’. {demonstration} Question: {*doc*} {x} Answer:
Prompts for LLM as a frozen rewriter 教LLM如何拆解問題，並從搜尋引擎找到需要的資料
Open-domain QA: Think step by step to answer this question, and provide search engine queries for knowledge that you need. Split the queries with ’;’ and end the queries with ’**’. {demonstration} Question: {x} Answer: *Multiple choice QA:* Provide a better search query for web search engine to answer the given question, end the queries with ’**’. {demonstration} Question: {x} Answer:

Table 1: Prompt lines used for the LLMs.

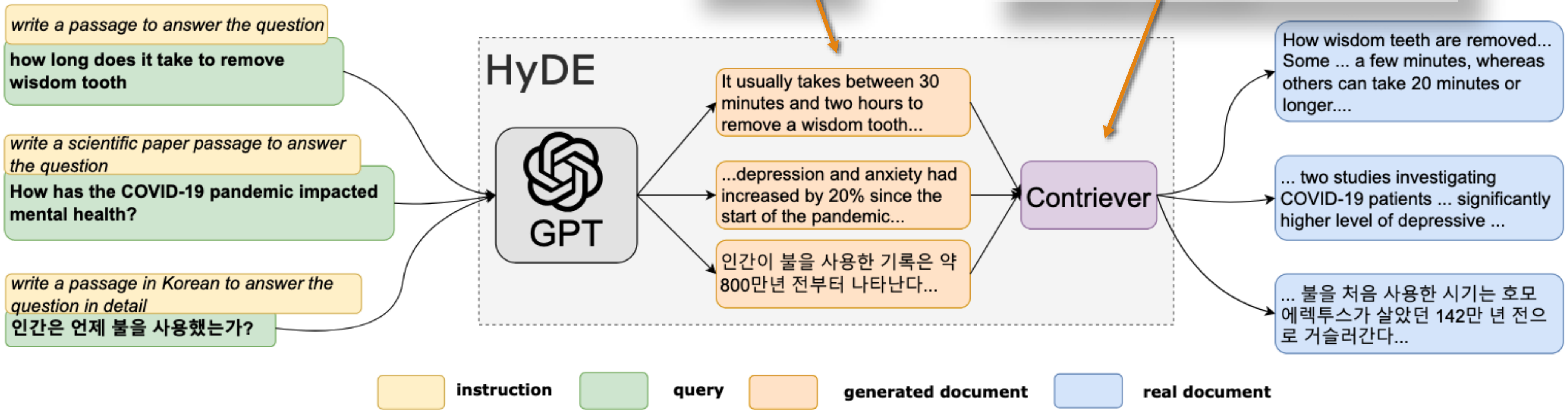
Ma, Xinbei, et al. "Query Rewriting in Retrieval-Augmented Large Language Models." EMNLP 2023.

Hypothetical Document Embeddings (HyDE)

- Use an LLM (InstructGPT) to perform query transformations by asking the LLM to write a passage (**similar to Query Rewriting**).

Not real

unsupervised contrastively learned encoder



Gao, Luyu, et al. "Precise Zero-Shot Dense Retrieval without Relevance Labels." ACL 2023.

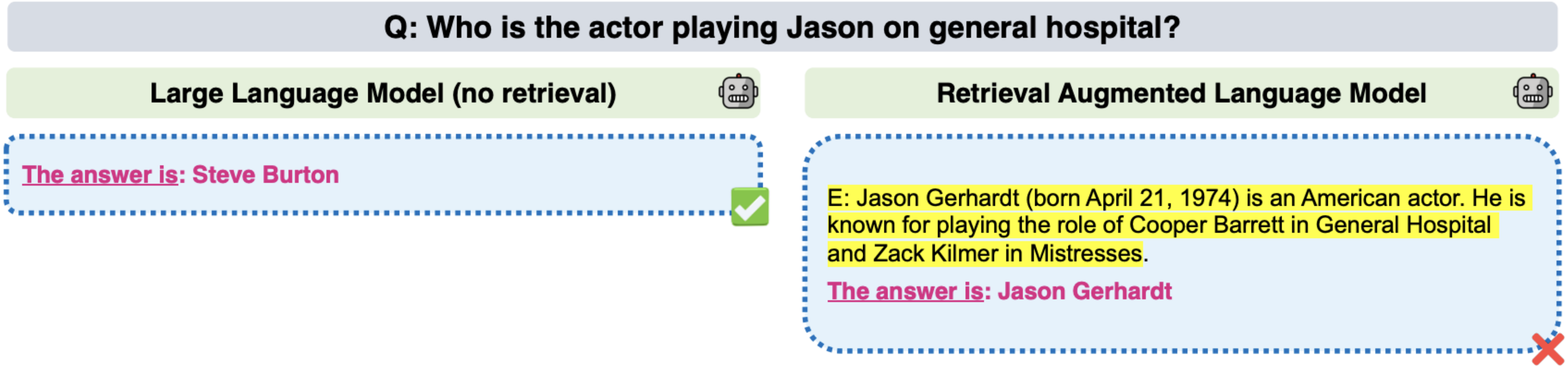
HyDE can be better than Query Rewriting for retrieval tasks

Method	TREC DL19					TREC DL20				
	mAP	nDCG@10	R@50	R@1k	Latency	mAP	nDCG@10	R@50	R@1k	Latency
unsupervised										
BM25	30.13	50.58	38.32	75.01	**0.07**	28.56	47.96	46.18	78.63	**0.29**
Contriever	23.99	44.54	37.54	74.59	3.06	23.98	42.13	43.81	75.39	0.98
supervised										
LLM-Embedder	44.66	70.20	49.06	84.48	<u>2.61</u>	45.60	68.76	61.36	84.41	<u>0.71</u>
+ Query Rewriting	44.56	67.89	51.45	85.35	7.80	45.16	65.62	59.63	83.45	2.06
+ Query Decomposition	41.93	66.10	48.66	82.62	14.98	43.30	64.95	57.74	84.18	2.01
+ HyDE	50.87	**75.44**	54.93	88.76	7.21	50.94	**73.94**	63.80	88.03	2.14
+ Hybrid Search	47.14	72.50	51.13	<u>89.08</u>	3.20	47.72	69.80	<u>64.32</u>	<u>88.04</u>	0.77
+ HyDE + Hybrid Search	**52.13**	<u>73.34</u>	**55.38**	**90.42**	11.16	**53.13**	<u>72.72</u>	**66.14**	**90.67**	2.95

Table 7: Results for different retrieval methods on TREC DL19/20. The best result for each method is made bold and the second is underlined.

Wang, Xiaohua, et al. "Searching for best practices in retrieval-augmented generation." EMNLP 2024.

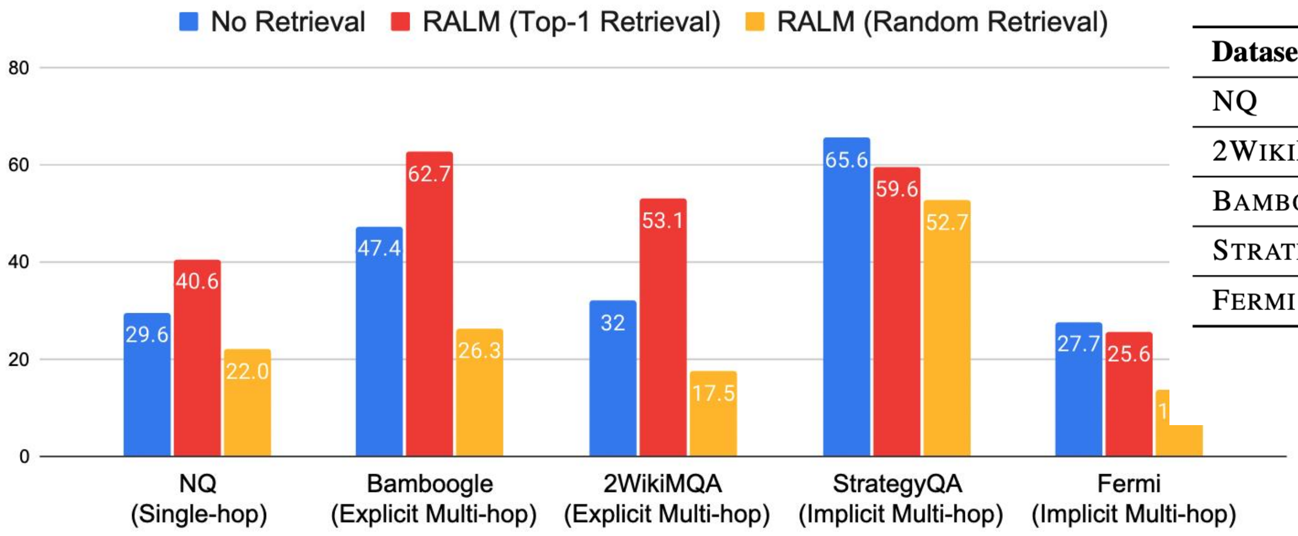
RetRobust: Making Retrieval-Augmented Language Models Robust to Irrelevant Context



RALM: Retrieval Augmented Language Model

Yoran, Ori, et al. "Making Retrieval-Augmented Language Models Robust to Irrelevant Context." ICLR 2024.

RetRobust: Making Retrieval-Augmented Language Models Robust to Irrelevant Context



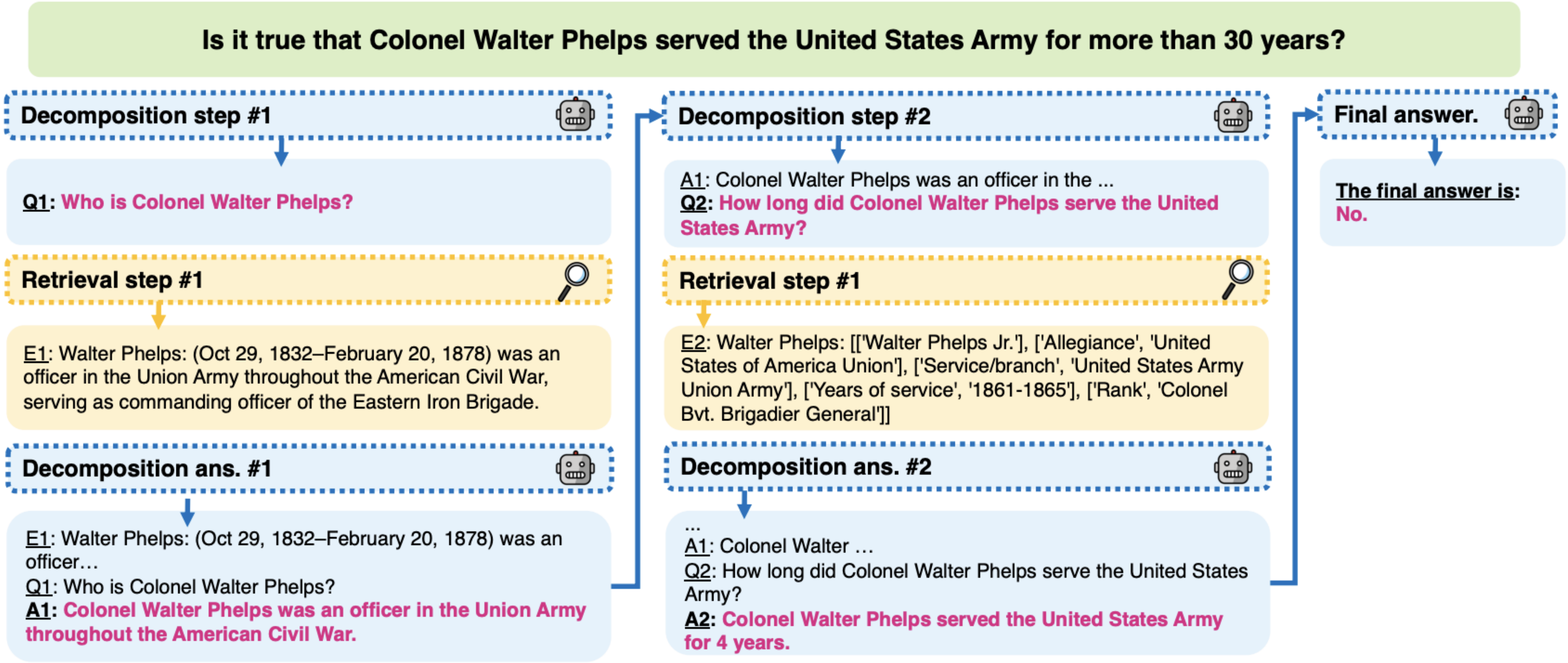
Dataset	Type	Example
NQ	Single-hop	What episode of law and order svu is mike tyson in?
2WIKIMQA	Explicit	Where was the place of death of Isabella Of Bourbon's father?
BAMBOOGLE	Explicit	What is the maximum airspeed (in km/h) of the third fastest bird?
STRATEGYQA	Implicit	Can Arnold Schwarzenegger deadlift an adult Black rhinoceros?
FERMI	Implicit	How many high fives has Lebron James given/received?

Table 1: The QA datasets in our experiments.

Retrieval augmentation can boost performance, but it also hurts performance on StrategyQA and Fermi, and random contexts reduce performance dramatically.

Yoran, Ori, et al. "Making Retrieval-Augmented Language Models Robust to Irrelevant Context." ICLR 2024.

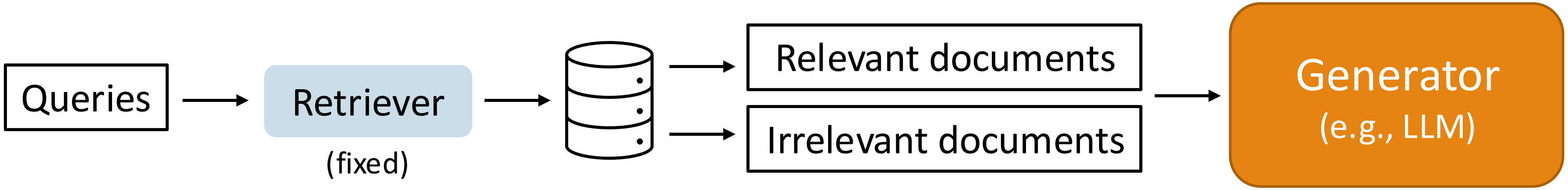
RetRobust: Making Retrieval-Augmented Language Models Robust to Irrelevant Context

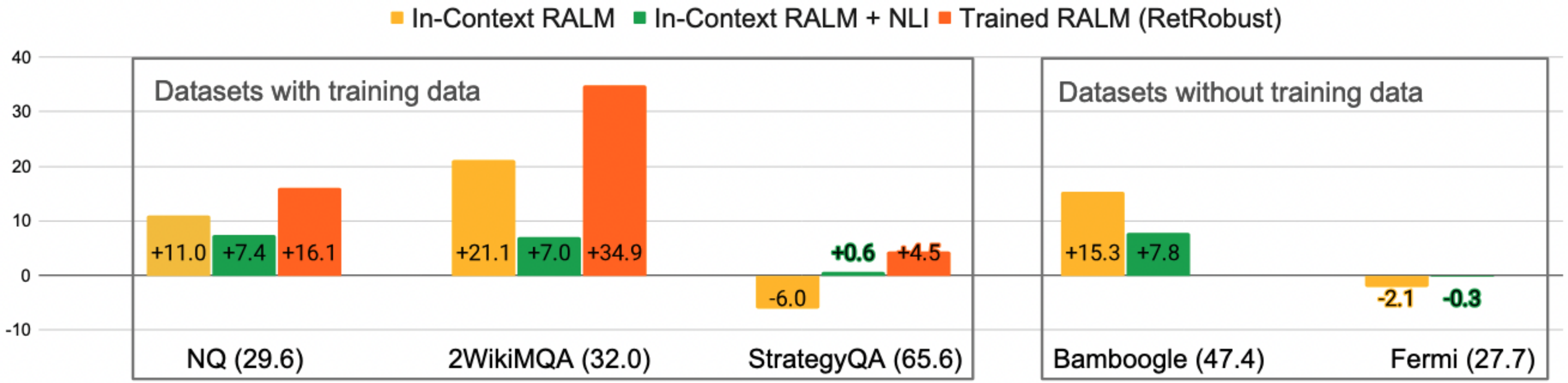


Yoran, Ori, et al. "Making Retrieval-Augmented Language Models Robust to Irrelevant Context." ICLR 2024.

RetRobust: Making Retrieval-Augmented Language Models Robust to Irrelevant Context

Train the generator on QA tasks with both relevant and irrelevant retrieved documents.

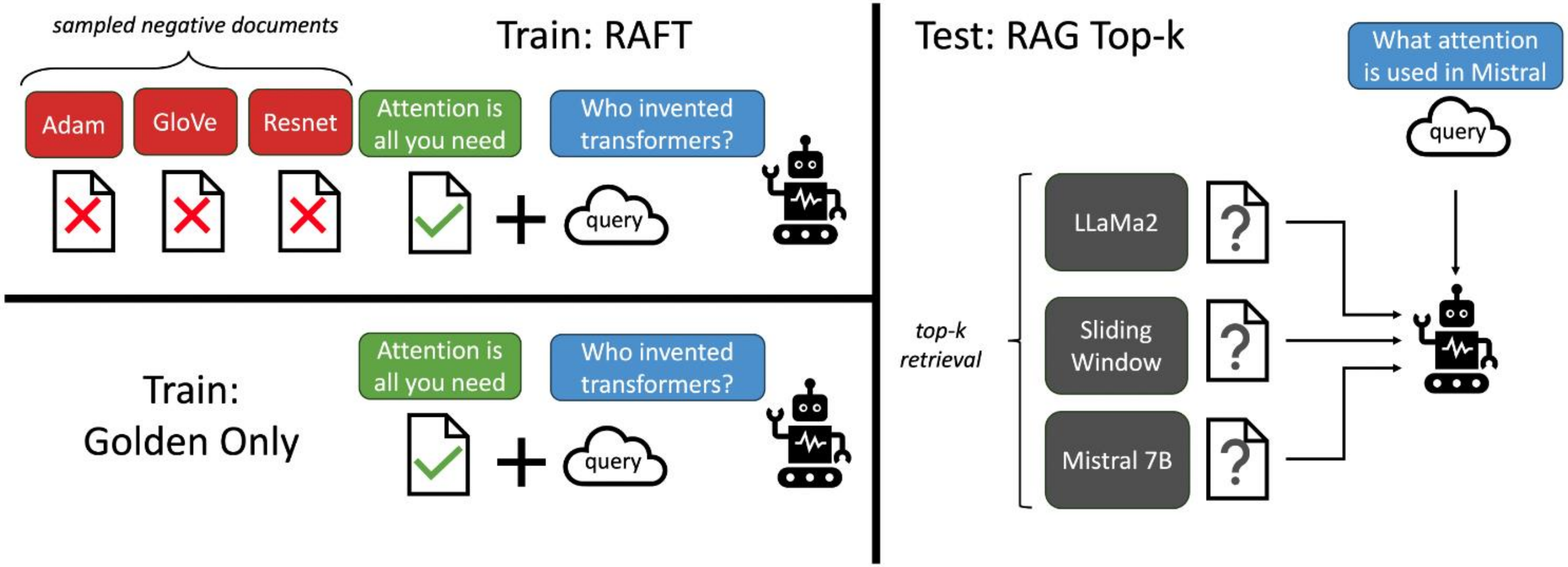




Yoran, Ori, et al. "Making Retrieval-Augmented Language Models Robust to Irrelevant Context." ICLR 2024.

RAFT: Adapting LLM to RAG

- In this work, the retriever is fixed. An LLM is trainable.
- Both correct and incorrect documents are included during training.



MateiZaharia, TianjunZhang ShishirG Patil NamanJain ShengShen, and IonStoica JosephE Gonzalez. "RAFT: Adapting Language Model to Domain Specific RAG." COLM 2024.

Difference between RetRobust and RAFT

RAFT uses **CoT Answer (answer with rationales).**

```
Question: The Oberoi family is part of a hotel company that has a head office
in what city?

context: [The Oberoi family is an Indian family that is famous for its
involvement in hotels, namely through The Oberoi Group]...[It is located in
city center of Jakarta, near Mega Kuningan, adjacent to the sister JW Marriott
Hotel. It is operated by The Ritz-Carlton Hotel Company. The complex has two
towers that comprises a hotel and the Airlangga Apartment respectively]...[The
Oberoi Group is a hotel company with its head office in Delhi.]

Instruction: Given the question, context and answer above, provide a logical
reasoning for that answer. Please use the format of: ##Reason: {reason}
##Answer: {answer}.

--------------------------------------------------------------------------------

CoT Answer: ##Reason: The document ##begin_quote## The Oberoi family is an
Indian family that is famous for its involvement in hotels, namely through The
Oberoi Group. ##end_quote## establishes that the Oberoi family is involved in
the Oberoi group, and the document ##begin_quote## The Oberoi Group is a hotel
company with its head office in Delhi. ##end_quote## establishes the head
office of The Oberoi Group. Therefore, the Oberoi family is part of a hotel
company whose head office is in Delhi. ##Answer: Delhi
```

MateiZaharia, TianjunZhang ShishirG Patil NamanJain ShengShen, and IonStoica JosephE Gonzalez. "RAFT: Adapting Language Model to Domain Specific RAG." COLM 2024.

EMNLP 2023

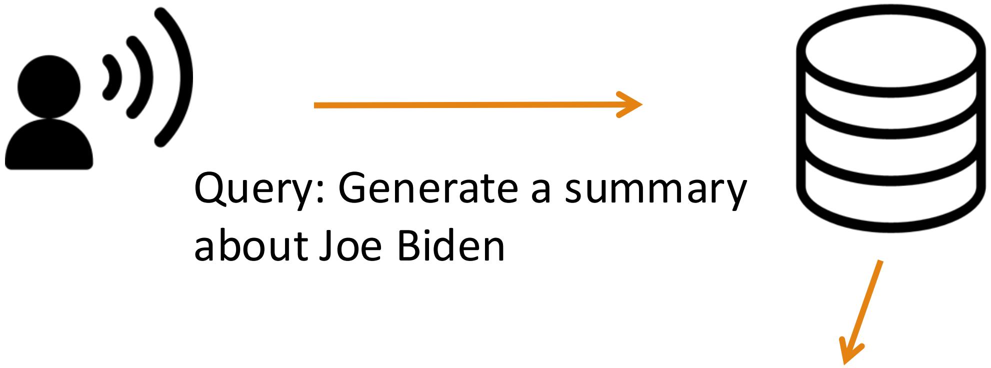
Carnegie Mellon University

Active Retrieval Augmented Generation

監督式的標注資料太少，讓模型自己標註自己train

Problem Formation

- For a naïve retrieval augmented generation:



Redundant information (Noise) may confuse the language model and make the generation wrong.

1. Joseph Robinette Biden Jr. was born on November 20, 1942
2. Neilia Hunter Biden, the first wife of Joe Biden, was born on July 28, 1942.
3. I started thinking as I was coming over here, why is it that Joe Biden is the first in his family ever to go to a university?

Noise

Methodology

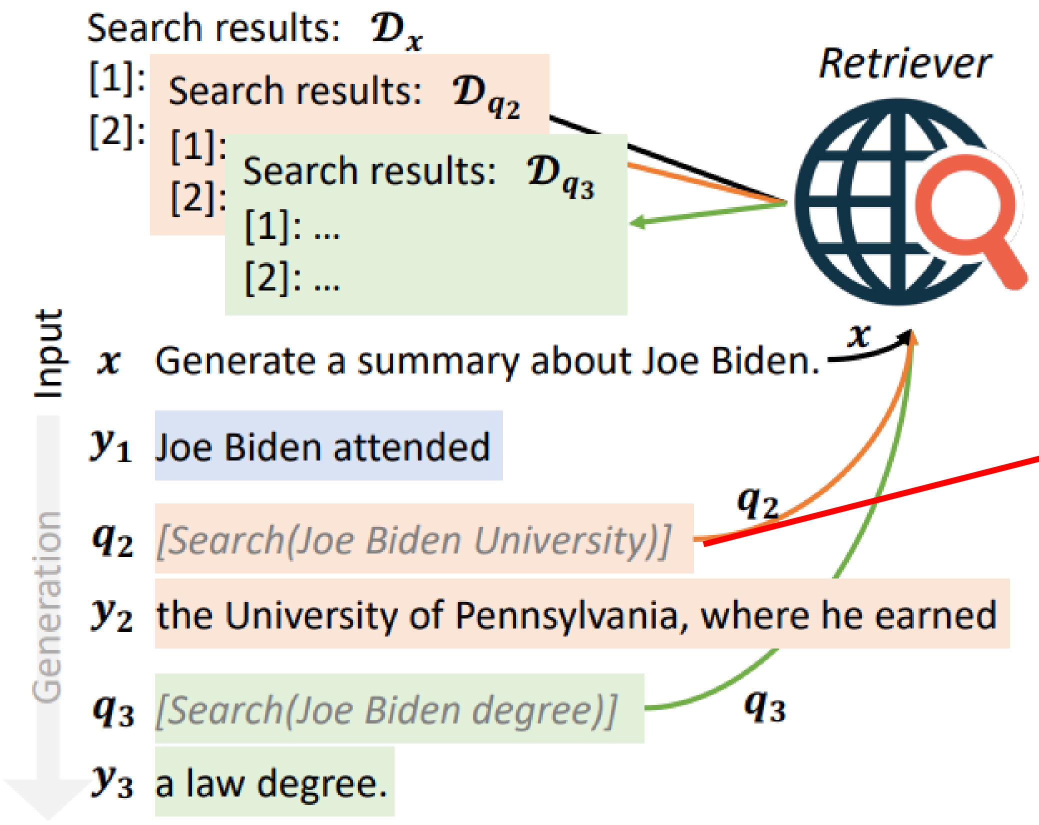
In retrieval-augmented generation, it is crucial to determine:

- **What to retrieve:** Selecting relevant and useful information.
- **When to retrieve:** Identifying the appropriate timing for retrieval.

The retrieved information should:

- Be essential for the model's generation process.
- Directly address the model's specific needs.

What to retrieve

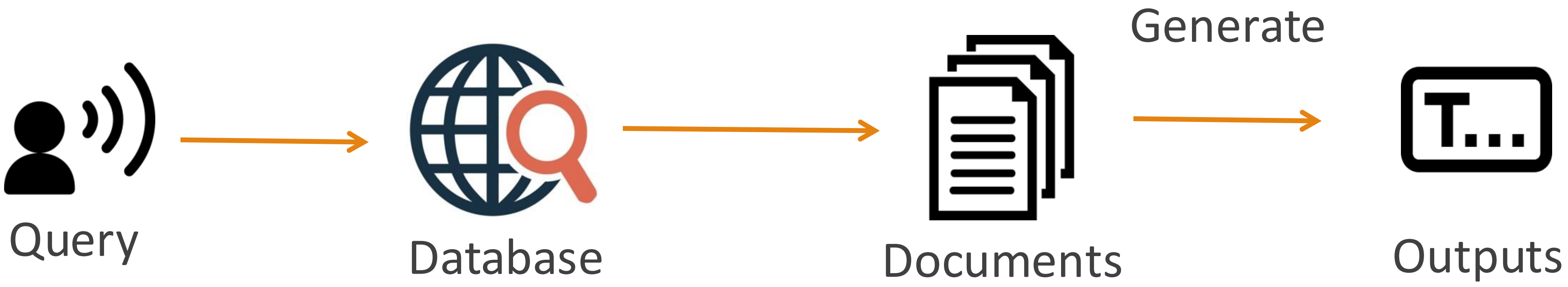


1. LMs should only retrieve information when they do not have the necessary knowledge to avoid unnecessary or inappropriate retrieval.
2. the retrieval queries should reflect the intents of future generations

Output = concat(y_1, y_2, y_3, …,)

When to retrieve

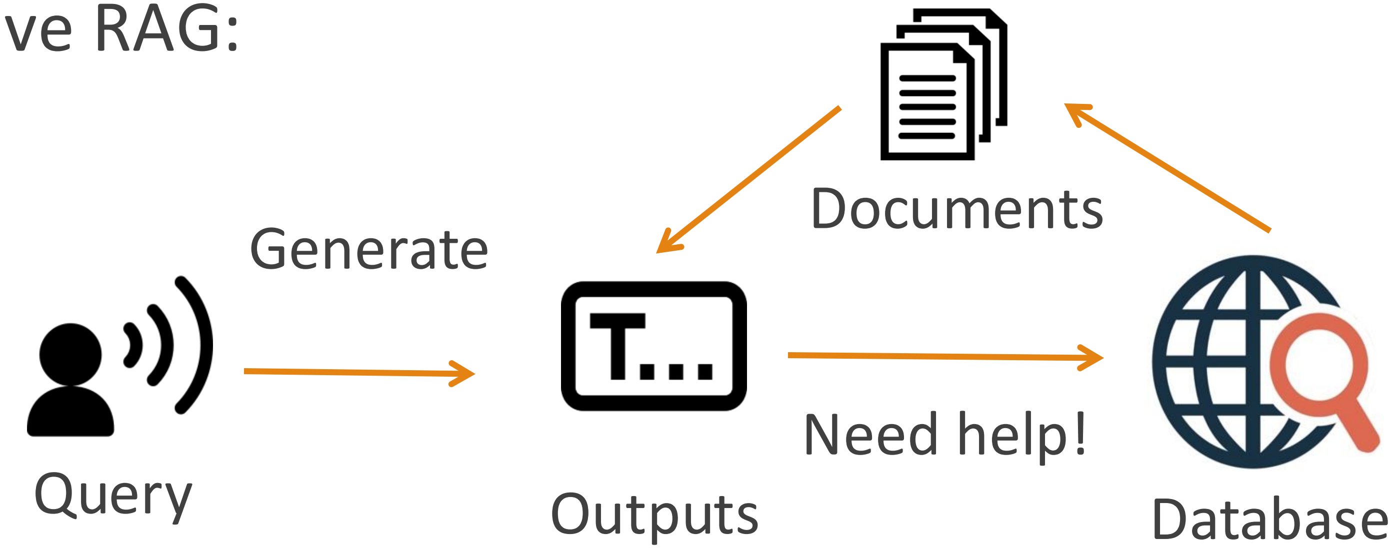
For traditional RAG:



The documents are retrieved based on the input query before generation.

When to retrieve

For Active RAG:



The documents are retrieved only when the language model needs help.

The definition of "need help"

Language models are often well-calibrated, meaning that low probability or confidence typically reflects a lack of knowledge.

This paper define a threshold $\theta \in [0, 1]$, when the probability of next token prediction is lower than θ, a retrieval is triggered.

There is no need to perform retrieval.

$$\boldsymbol{y}_t = \begin{cases} \hat{\boldsymbol{s}}_t & \boxed{\text{if all tokens of } \hat{\boldsymbol{s}}_t \text{ have probs} \geq \theta} \\ \boldsymbol{s}_t = \mathrm{LM}([\mathcal{D}_{\boldsymbol{q}_t}, \boldsymbol{x}, \boldsymbol{y}_{<t}]) & \text{otherwise} \end{cases}$$

Use $\widehat{S}_t$ to perform **a next** retrieval

Generation

Query: Generate a summary about Joe Biden

Outputs: Joe Biden attended *[Search(Joe Biden University)]*

Low Confidence!

Search & Generate

the University of Pennsylvania, where he earned *[Search(Joe Biden degree)]*

Low Confidence!

Search & Generate

a law degree.

RAAT

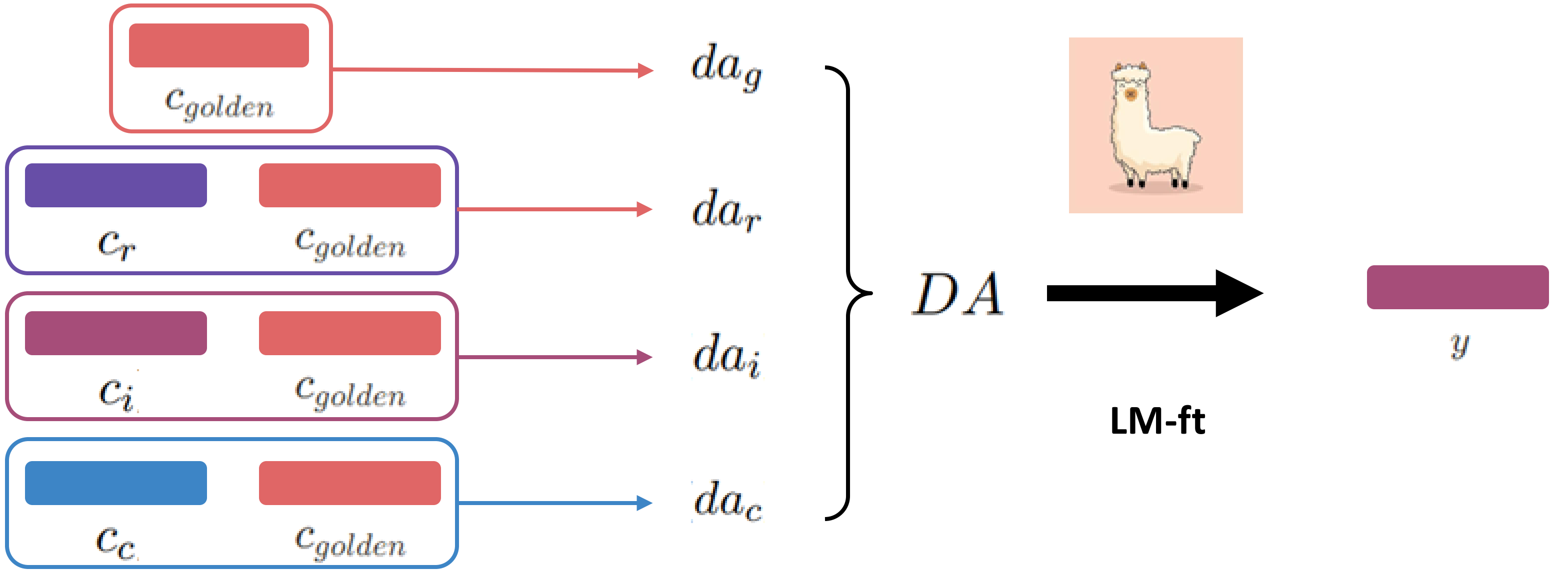
Adaptive Adversarial Training

- Recently, several studies have attempted to **enhance the noise robustness of RALMs through noisy training**, which involves incorporating retrieved noisy contexts into fine-tuning data.

Relevant noise
跟query相關但是對答案沒用

Irrelevant noise
跟query無關

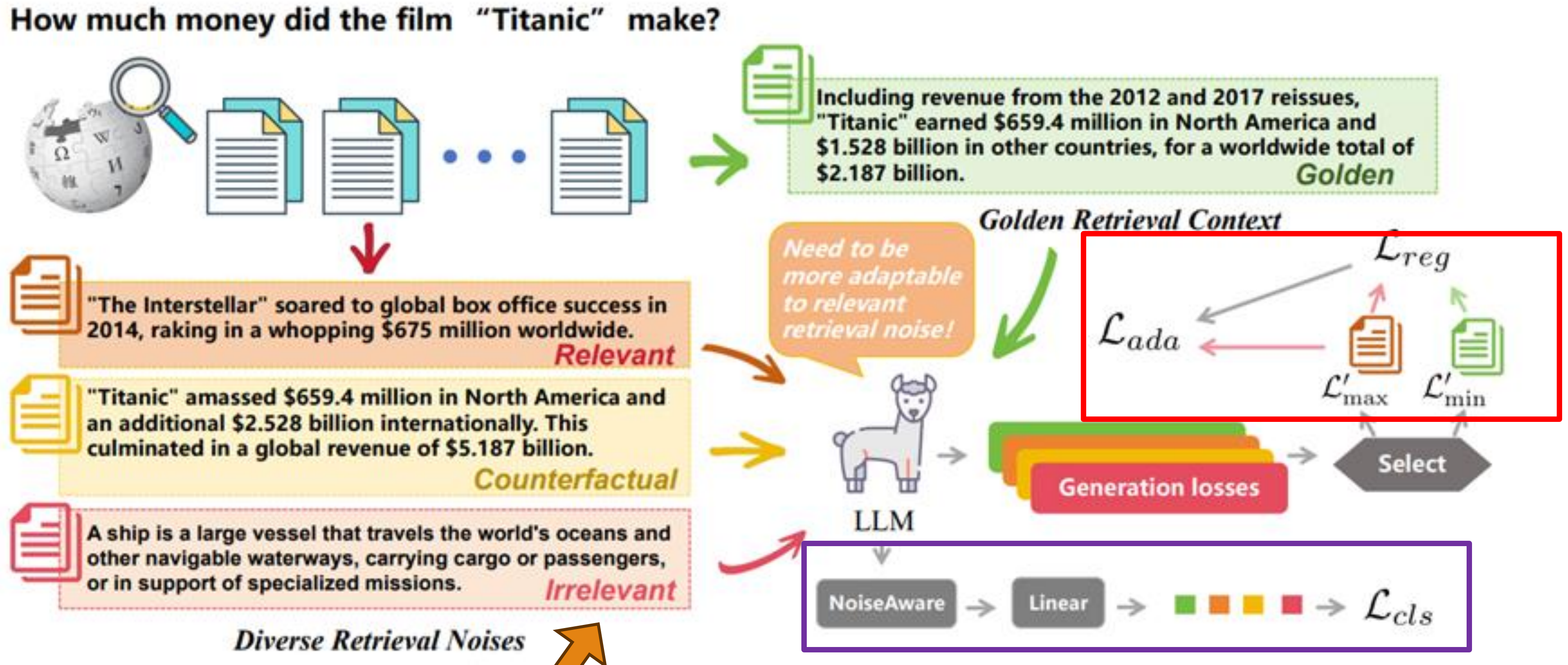
Counterfactual noise
假新聞



>>在做model training時就加入noise，增加他在未來的辨識能力

Fang, Feiteng, et al. "Enhancing Noise Robustness of Retrieval-Augmented Language Models with Adaptive Adversarial Training." ACL 2024.

RAAT



How to get these annotations?

Adaptive learning maximizes input manipulation and minimizes by fine-tuning to enhance model robustness.

An auxiliary task enhances RALMs' robustness by detecting and addressing retrieval noise.

Fang, Feiteng, et al. "Enhancing Noise Robustness of Retrieval-Augmented Language Models with Adaptive Adversarial Training." ACL 2024.

RAAT Conclusion

Method	Golden Only		Golden & c_i		Golden & c_r		Golden & c_c		Avg	
	F1	EM	F1	EM	F1	EM	F1	EM	F1	EM
LLaMA2$_{7B}$	65.56	51.80	56.14	42.87	53.10	39.73	51.81	38.37	56.68	43.19
Qwen$_{7B}$	62.57	47.07	61.48	46.06	55.50	40.50	53.26	36.90	58.20	42.63
LLaMA2$_{13B}$	69.27	55.00	63.25	49.47	62.27	47.97	62.07	47.17	64.22	49.90
Qwen$_{14B}$	67.45	51.43	66.71	51.20	61.88	46.16	58.65	41.30	63.67	47.52
LLaMA2$_{70B}$	71.43	56.56	70.05	55.13	65.97	51.33	63.91	48.27	67.84	52.82
ChatGPT$_{3.5}$	73.98	60.50	72.24	60.30	70.65	56.89	69.00	54.64	71.47	58.10
RALM$_{golden}$	80.31	74.03	79.33	72.73	73.26	66.33	73.08	65.40	76.50	69.62
RetRobust	80.10	73.80	79.25	72.97	74.81	68.30	75.46	68.43	77.41	70.88
RALM$_{retrieved}$	80.04	73.40	81.09	74.80	75.99	69.10	73.10	65.67	77.55	70.74
RALM$_{multiple}$	85.47	80.17	85.27	81.20	83.07	78.33	83.25	79.23	84.27	79.73
RAAT	**87.15**	**83.07**	**86.80**	**82.73**	**85.14**	**81.00**	**86.29**	**82.10**	**86.35**	**82.23**

Table 2: Experimental results on our RAG-Bench benchmark. “Golden Only” denotes a scenario where LLMs only consult the golden retrieval context. In “Golden & $c_i/c_r/c_c$”, LLMs consider both the golden retrieval context and *irrelevant retrieval noise/relevant retrieval noise/counterfactual retrieval noise*.

Fang, Feiteng, et al. “Enhancing Noise Robustness of Retrieval-Augmented Language Models with Adaptive Adversarial Training.” ACL 2024.

RAAT Conclusion

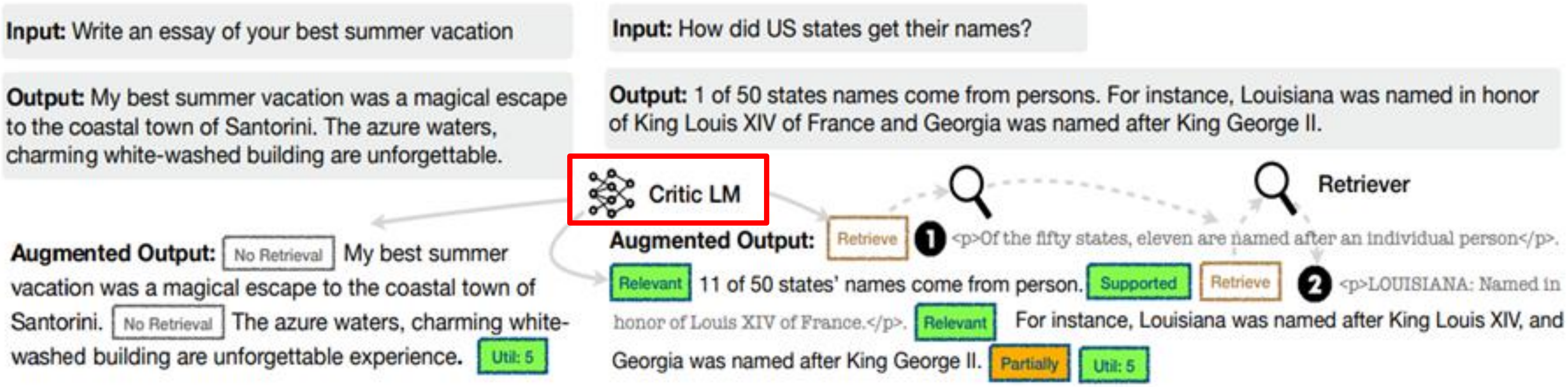
1. Investigated **retrieval noises** in RALMs and categorized them into **three distinct types**, reflecting real-world environments.
2. Introduced **RAAT** as a solution to address the noise robustness challenges faced by RALMs, which leveraged **adaptive adversarial learning** and **multi-task learning** to enhance the model's capability.
3. **Established a benchmark** to verify the effectiveness of RAAT based on three open-domain QA datasets.

Fang, Feiteng, et al. "Enhancing Noise Robustness of Retrieval-Augmented Language Models with Adaptive Adversarial Training." ACL 2024.

Self-RAG

Self-RAG (Training)

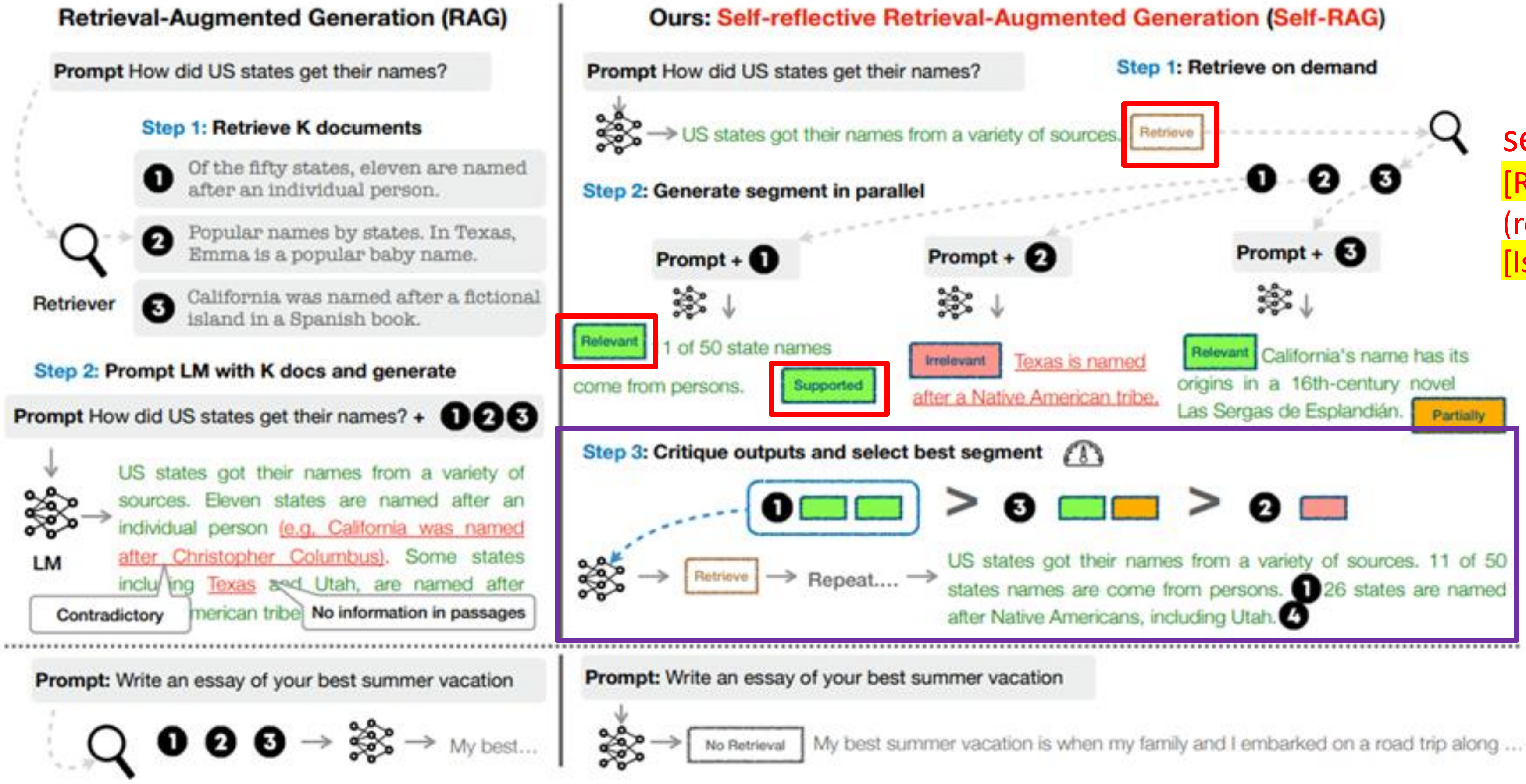
- Supervised data is created by prompting GPT-4 for reflection tokens, distilled into a critic model with token-specific definitions and inputs.
- Thus, they use critic model to collect training data for Generator.



Asai, Akari, et al. "Self-rag: Learning to retrieve, generate, and critique through self-reflection." ICLR 2024.

Self-RAG

Self-RAG (Inference)



self-reflection:
[Retrieve], [IsRel], [IsSup] (response supported by d), [IsUse] (useful response)

obtain the top-B segment continuations at each timestamp

Asai, Akari, et al. "Self-rag: Learning to retrieve, generate, and critique through self-reflection." ICLR 2024.

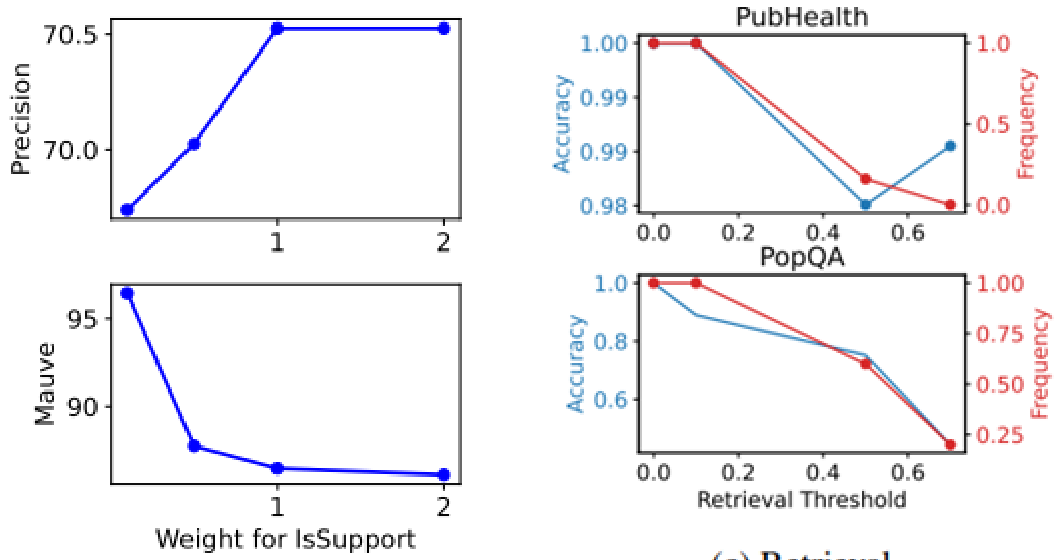
Self-RAG

	Short-form		Closed-set		Long-form generations (with citations)					
	PopQA	TQA	Pub	ARC	Bio			ASQA		
LM	(acc)	(acc)	(acc)	(acc)	(FS)	(em)	(rg)	(mau)	(pre)	(rec)
					LMs with proprietary data					
Llama2-c$_{13B}$	20.0	59.3	49.4	38.4	55.9	22.4	29.6	28.6	–	–
Ret-Llama2-c$_{13B}$	51.8	59.8	52.1	37.9	79.9	32.8	34.8	43.8	19.8	36.1
ChatGPT	29.3	**74.3**	70.1	**75.3**	71.8	35.3	36.2	68.8	–	–
Ret-ChatGPT	50.8	65.7	54.7	**75.3**	–	**40.7**	**39.9**	**79.7**	65.1	**76.6**
Perplexity.ai	–	–	–	–	71.2	–	–	–	–	–
					Baselines without retrieval					
Llama2$_{7B}$	14.7	30.5	34.2	21.8	44.5	7.9	15.3	19.0	–	–
Alpaca$_{7B}$	23.6	54.5	49.8	45.0	45.8	18.8	29.4	61.7	–	–
Llama2$_{13B}$	14.7	38.5	29.4	29.4	53.4	7.2	12.4	16.0	–	–
Alpaca$_{13B}$	24.4	61.3	55.5	54.9	50.2	22.9	32.0	70.6	–	–
CoVE$_{65B}$ *	–	–	–	–	71.2	–	–	–	–	–
					Baselines with retrieval					
Toolformer*$_{6B}$	–	48.8	–	–	–	–	–	–	–	–
Llama2$_{7B}$	38.2	42.5	30.0	48.0	78.0	15.2	22.1	32.0	2.9	4.0
Alpaca$_{7B}$	46.7	64.1	40.2	48.0	76.6	30.9	33.3	57.9	5.5	7.2
Llama2-FT$_{7B}$	48.7	57.3	64.3	65.8	78.2	31.0	35.8	51.2	5.0	7.5
SAIL*$_{7B}$	–	–	69.2	48.4	–	–	–	–	–	–
Llama2$_{13B}$	45.7	47.0	30.2	26.0	77.5	16.3	20.5	24.7	2.3	3.6
Alpaca$_{13B}$	46.1	66.9	51.1	57.6	77.7	**34.8**	36.7	56.6	2.0	3.8
Our SELF-RAG $_{7B}$	54.9	66.4	72.4	67.3	**81.2**	30.0	35.7	**74.3**	66.9	67.8
Our SELF-RAG $_{13B}$	**55.8**	**69.3**	**74.5**	**73.1**	80.2	31.7	**37.0**	71.6	**70.3**	**71.3**

Asai, Akari, et al. "Self-rag: Learning to retrieve, generate, and critique through self-reflection." ICLR 2024.

Self-RAG

- Reflection token settings balance precision versus fluency and accuracy versus retrieval frequency.



(b) Customization

(c) Retrieval

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Self-RAG Conclusion

1. It introduces **SELF-RAG**, a new framework to enhance the quality and factuality of LLMs through **retrieval on demand** and **self-reflection**.
2. SELF-RAG trains an LM to learn to **retrieve**, **generate**, and **critique text passages and its own generation** by **predicting the next tokens** from its original vocabulary as well as newly added special tokens, called **reflection tokens**.
3. SELF-RAG further enables the **tailoring of LM behaviors** at test time by leveraging reflection tokens.

Asai, Akari, et al. "Self-rag: Learning to retrieve, generate, and critique through self-reflection." ICLR 2024.

Challenges in Modern RAG

- A significant amount of noise information even fake news in the content available on the Internet.
- Currently, there is a lack of comprehensive understanding of how each model can improve performance through information retrieval.

Type of Noises

- Relevant (semantically similar) but not contain the answer
- Counterfactual information
- Irrelevant information
- Black box digestion
- …

Capabilities that LLMs Should Have in RAG

Noise Robustness

- LLMs must be able to extract the necessary information from documents despite there are noisy documents.

Question

Who was awarded the **2022** Nobel prize in literature?

External documents contain noises

The Nobel Prize in Literature for **2022** is awarded to the French author **Annie Ernaux**, “for the courage and clinical acuity …

The Nobel Prize in Literature for **2021** is awarded to the novelist **Abdulrazak Gurnah**, born in Zanzibar and active in …

Retrieval Augmented Generation

Annie Ernaux

Capabilities that LLMs Should Have in RAG

Negative Rejection

- In real-world situations, the search engine often fails to retrieve documents containing the answers.
- It is important for the model to have the capability to reject recognition and avoid generating misleading content.

Question

Who was awarded the **2022** Nobel prize in literature?

External documents contain noises

The Nobel Prize in Literature for **2021** is awarded to the novelist **Abdulrazak Gurnah**, born in Zanzibar and active in …

The **202**0 Nobel Laureate in Literature, poet **Louise Glück**, has written both poetry and essays about poetry. Since her…

Retrieval Augmented Generation

I can not answer the question because of the insufficient information in documents.

Capabilities that LLMs Should Have in RAG

Information Integration

- In many cases, the answer to a question may be contained in multiple documents.
- To provide better answers to complex questions, it is necessary for LLMs to have the ability to integrate information.

Question

When were the **ChatGPT app for iOS** and **ChatGPT api** launched?

External documents contain noises

On **May 18th**, 2023, OpenAI introduced its own **ChatGPT app for iOS**…

That changed **on March 1**, when OpenAI announced **the release of API access to ChatGPT** and Whisper,…

Retrieval Augmented Generation

May 18 and March 1.

Capabilities that LLMs Should Have in RAG

Counterfactual Robustness

- In the real world, there is an abundance of false information on the internet.
- LLMs should identify risks of known factual errors in the retrieved documents when the LLMs are given warnings about potential risks in the retrieved information through instruction.

Question

Which city hosted the Olympic games in **2004**?

External documents contain noises

The 2004 Olympic Games returned home to **New York**, birthplace of the …

After leading all voting rounds, **New York** easily defeated Rome in the fifth and final vote …

Retrieval Augmented Generation

There are factual errors in the provided documents. **The answer should be Athens.**